AI for Auto-Research: Roadmap & User Guide

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Awesome AI Auto-Research Team

AI research tools are evolving from isolated writing and coding assistants into long-horizon agents that propose ideas, run experiments, and manage research workflows. Producing these artifacts, however, does not establish that their claims are novel, reproducible, or scientifically meaningful. We survey developments through 11 October 2026 and organize the research lifecycle into four phases and eight stages: [1]**Creation** (idea generation, literature review, coding & experiments, tables & figures), [2]**Writing** (paper writing), [3]**Validation** (peer review, rebuttal & revision), and [4]**Dissemination** (communicating research through public-facing artifacts). We review stage-specific tools, end-to-end research agents, and benchmarks, emphasizing what their evaluations verify. Recent work extends assessment from output quality to research decision trajectories, execution-grounded evidence, independently tested model checkpoints, capability retention, and resource costs. These protocols reveal different capability boundaries: plausible ideas need not survive implementation, executable code need not implement the intended method, and stronger benchmark scores need not imply scientific understanding or reliable autonomy. We synthesize recurring design choices and failure modes across phase boundaries, including evidence traceability, independent verification, candidate selection, and human scientific judgment. A practitioner playbook translates this analysis into stage-specific assistance and verification checkpoints. The resulting roadmap identifies where automation can expand research capacity while researchers retain ownership of hypotheses, interpretation, and accountability. A continuously maintained reading list, tool inventory, and benchmark resources accompany the survey.

Project Page: https://worldbench.github.io/awesome-ai-auto-research

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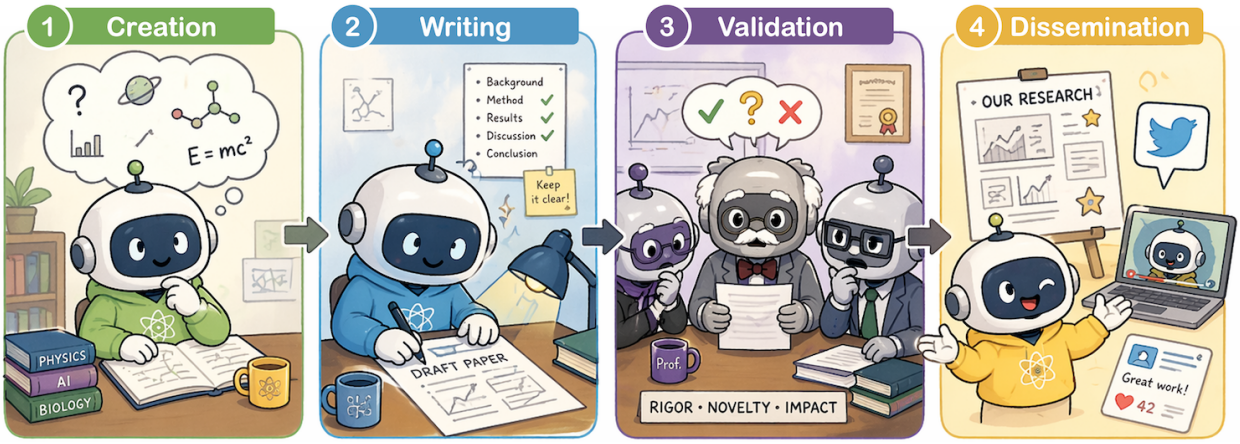


Figure 1 AI auto-research across the complete lifecycle. We organize AI assistance into four phases and eight stages: [1]**Creation** spans idea generation, literature review, coding & experiments, and tables & figures; [2]**Writing** centers on paper writing; [3]**Validation** includes peer review and rebuttal & revision; and [4]**Dissemination** transforms papers into posters, slides, videos, social media, project pages, and interactive paper agents.

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1 Introduction

AI-assisted research is crossing a threshold. Large language models (LLMs) and their agentic extensions are no longer limited to local writing or coding support; they are beginning to operate across the research lifecycle itself. Recent systems illustrate the scale of this shift: The AI Scientist generated complete research papers at roughly $15 per paper [223]; FARS ran continuously for 228 hours, consumed 11.4 billion tokens, and produced 100 papers, averaging one every 2.3 hours [20]; and ARIS reports an overnight workflow that ran 20+ GPU experiments, pruned unsupported claims, and improved a draft score from 5.0 to 7.5 through iterative review and revision [425]. These systems suggest a new paradigm: AI is moving from assisting individual research tasks to orchestrating multi-stage workflows that generate ideas, search literature, execute experiments, draft manuscripts, simulate critique, and prepare dissemination materials.

This rapid progress also exposes the defining tension of the field. AI systems are increasingly capable of producing research-like artifacts, yet remain far less reliable at verifying whether those artifacts are novel, faithful, executable, and scientifically meaningful. Generated ideas can appear promising but weaken after implementation [328]; generated code can run while implementing the wrong algorithm [134]; fluent manuscripts can conceal unsupported claims; automated reviews can be coherent yet lenient or vulnerable to manipulation [484]; rebuttals can promise revisions that are not later fulfilled [31]; and dissemination materials can simplify results beyond the evidence. The core challenge is therefore no longer whether AI can produce the *forms* of research, but whether it can preserve the *substance* of research: evidence, judgment, provenance, and accountability.

A lifecycle view is essential for understanding this challenge. Research is not a collection of independent tasks: ideas become experiments, experiments become claims, claims become manuscripts, reviews become revisions, and papers become public-facing summaries. Errors introduced early can be amplified downstream, especially when AI systems generate plausible outputs without preserving evidence or provenance. Despite the rapid emergence of research agents, writing assistants, scientific coding tools, automated reviewers, rebuttal systems, and Paper2X applications, the field still lacks a unified analysis of AI auto-research across the complete academic lifecycle. Without such a view, it is difficult to determine where AI reliably helps, where it fails systematically, and which deployment modes are scientifically credible.

Surveying developments through 11 October 2026, we present the first end-to-end analysis of AI auto-research across the complete academic research lifecycle. We organize the field into four epistemological phases and eight stages: [1]**Creation**, covering idea generation, literature review, coding & experiments, and tables & figures; [2]**Writing**, covering paper writing; [3]**Validation**, covering peer review and rebuttal & revision; and [4]**Dissemination**, covering posters, slides, videos, social media, project pages, and interactive paper agents. This structure follows the temporal sequence of research while making explicit the distinct AI capabilities, risks, and verification requirements introduced by each phase.

Our analysis yields five central findings. First, AI capability is strongest when tasks are structured, grounded, and externally checkable, but drops sharply for open-ended research tasks requiring novelty, implicit domain knowledge, long-horizon reasoning, or scientific judgment. Second, artifact generation consistently outpaces verification: across stages, AI can often produce plausible outputs faster than it can prove that they are correct, faithful, or meaningful. Third, the most reliable deployment mode is human-governed collaboration rather than full autonomy: AI can reduce mechanical friction in retrieval, drafting, coding, visualization, review support, and dissemination, but researchers must retain responsibility for judgment, interpretation, experimental design, argumentation, and accountability. Fourth, effective systems increasingly rely on layered architectures that combine exploration, tool-based execution, and verification, suggesting that orchestration, provenance, and feedback design are as important as model scale. Fifth, AI use in research is becoming a governance problem rather than a detection problem: as AI assistance becomes routine, the key questions are disclosure, attribution, responsibility, and whether scientific integrity is preserved.

This work makes three contributions to the emerging field of AI auto-research:

- We provide a unified taxonomy of AI auto-research across four phases and eight stages, covering both mature areas such as writing and coding, and underexplored areas such as rebuttal, scientific visualization, and research dissemination.

- We synthesize tools, benchmarks, and methodological families across the lifecycle, showing how systems have evolved from prompt-based assistance to retrieval-augmented, agentic, fine-tuned, and hybrid workflows.
- We identify cross-cutting capability boundaries and open challenges, including phase-boundary faithfulness, scientific judgment, reproducibility, citation provenance, governance, cross-domain generalization, and cognitive ownership.

The remainder of this paper is organized as follows. Section 2 introduces the lifecycle framework, methodological families, literature-collection scope, and development timeline. Section 3 to Section 6 build the roadmap of the four phases for AI-assisted research in temporal order. Section 7 synthesizes end-to-end systems, evaluation paradigms, cross-cutting insights, and open challenges. Section 8 concludes the paper.

2 Preliminaries

As AI-assisted research tools expand from isolated single stages (such as writing or coding aids) into multi-stage assistants, the field has become increasingly difficult to compare using a single vocabulary. Existing systems differ not only in their technical designs, but also in the research stages they target, the degree of autonomy they assume, and the forms of scientific risk they introduce.

To support a unified analysis, we first establish **four foundational elements**: (i) the high-level academic research lifecycle framework that organizes this survey (Section 2.1), (ii) the methodological families that recur across each stage (Section 2.2), (iii) the scope and methodology of our literature collection (Section 2.3), and (iv) a brief timeline of key developments (Section 2.4).

2.1 Research Lifecycle

We define the research lifecycle as **eight interconnected stages**, organized into **four phases**. Each phase groups stages that serve a shared function in the production, validation, and communication of scientific knowledge.

Phase 1: Creation. This phase covers the stages through which a research contribution is materially produced, including hypothesis formation, evidence gathering, experimentation, and scientific visualization.



S1 Idea Generation

Generating, refining, and evaluating research hypotheses. Techniques include direct LLM prompting, retrieval-augmented generation, knowledge-graph reasoning, and multi-agent collaboration for structured hypothesis formation.



S2 Literature Review

Retrieving, synthesizing, and organizing prior work into coherent research contexts. Modern systems span semantic retrieval, citation-graph traversal, survey generation, and deep research agents that iteratively explore the literature.



S3 Coding & Experiments

Translating ideas into executable code, running experiments, and analyzing empirical results. This stage includes code generation, paper-to-code translation, autonomous experiment orchestration, and result interpretation.



S4 Tables & Figures

Constructing method diagrams, result plots, comparison tables, mathematical formulas, and algorithmic illustrations. These artifacts transform raw outputs and conceptual designs into structured scientific representations.

Phase 2: Writing. This phase organizes the outputs of *Creation* into a formal scholarly manuscript for communication and external scrutiny.



S5 Paper Writing

Drafting, editing, polishing, and structuring academic manuscripts. AI assistance ranges from grammar correction and citation support to section-level drafting and full-paper generation.

Phase 3: Validation. This phase covers the stages through which the research community scrutinizes, critiques, and iteratively refines a manuscript.



S6 Peer Review

Generating structured reviews, matching reviewers to manuscripts, assessing review quality, and supporting meta-review decisions. These systems aim to assist, rather than replace, the community's evaluative process.



S7 Rebuttal & Revision

Analyzing reviewer comments, identifying required evidence, drafting responses, and supporting manuscript revision. This stage connects external critique with additional analysis, clarification, and experimental follow-up.

Phase 4: Dissemination. This phase converts the manuscript and its supporting materials into formats accessible to broader research and public audiences.



S8 Paper2X

Converting papers into posters, slides, videos, project pages, demos, and social media content. Each output format targets a different audience and requires distinct design choices, fidelity constraints, and communication strategies.

Although presented in temporal order, the lifecycle is not strictly linear. Reviewer critiques in Phase 3 (*Validation*) may require returning to Phase 1 (*Creation*) for additional experiments, while dissemination outputs in Phase 4 (*Dissemination*) may expose ambiguities or errors that trigger revisions in Phase 2 (*Writing*). These feedback loops are central to research practice and are especially important for AI-assisted workflows, where errors can propagate across stages if not explicitly checked.

This four-phase grouping reflects the functional structure of research. Evidence and artifacts are produced in P1 *Creation*, organized into a manuscript in P2 *Writing*, externally scrutinized in P3 *Validation*, and communicated to broader audiences in P4 *Dissemination*.

We separate *Writing* from *Creation* because manuscript construction is not merely a formatting step: it is a rhetorical and evidential organization process that requires different AI capabilities from those used to produce code, experiments, or figures. We group Peer Review and Rebuttal under *Validation* because together they form the community-facing mechanism through which claims are challenged, defended, and revised. Finally, we treat *Dissemination* as a full phase because posters, slides, videos, project pages, and social media summaries are increasingly important knowledge artifacts with their own fidelity and trust requirements.

2.2 Methodological Families

Across the research lifecycle, AI-assisted research systems reuse a small set of methodological patterns. We group them into five broad families: [1]*prompt engineering*, [2]*retrieval-augmented generation (RAG)*, [3]*training-free agentic methods*, [4]*training-based methods*, and [5]*hybrid approaches*. These families are not mutually exclusive or strictly chronological; rather, they describe how current systems elicit, ground, specialize, and orchestrate LLM behavior. Many practical systems combine several of them, for example using prompts for decomposition, RAG for grounding, tools for execution, and trained modules for scoring or ranking.

Table 1 Dominant methodological families, representative systems, and research maturity across each stage of the four-phase research lifecycle. Notations: ✓ = "primary approach", ○ = "secondary/emerging", — = "not used".

Stage	Prompt Eng.	RAG	Agentic	Training	Hybrid	Representative works	Maturity
Phase 1: Creation							
S1: Idea Generation	✓	○	✓	—	○	AI Scientist [223], VirSci [344], Spark [305]	★★★★☆
S2: Literature Review	○	✓	✓	—	✓	PaperQA2 [335], AutoSurvey [384], STORM [318]	★★★☆☆
S3: Coding & Exp.	○	○	✓	—	✓	AIDE [150], PaperCoder [314], R&D-Agent [426]	★★★★☆
S4: Tables & Fig.	✓	—	✓	○	—	MatPlotAgent [431], AutoFigure [205], DeTikZify [17]	★☆☆☆☆
Phase 2: Writing							
S5: Paper Writing	✓	✓	○	✓	○	CycleResearcher [394], ScholarCopilot [385], XtraGPT [39]	★★★★☆
Phase 3: Validation							
S6: Peer Review	○	○	✓	✓	✓	DeepReviewer [486], MARG [60], ReviewAgents [84]	★★★☆☆
S7: Rebuttal	○	✓	✓	—	○	RebuttalAgent [119], Paper2Rebuttal [234]	★☆☆☆☆
Phase 4: Dissemination							
S8: Dissemination	✓	○	✓	—	—	Paper2Poster [267], PPTAgent [477], SlideGen [199]	★☆☆☆☆

Prompt engineering provides the simplest interface for adapting general-purpose LLMs to research tasks [390, 438]. It includes direct prompting, chain-of-thought reasoning, role assignment, structured templates, rubric-based instructions, and output constraints. Because it requires no additional training, it remains widely used for lightweight tasks such as brainstorming, editing, review drafting, rebuttal outlining, and social media generation, but it is sensitive to prompt wording and usually lacks persistent grounding.

Retrieval-augmented generation (RAG) grounds model outputs in external sources, including paper corpora, citation graphs, code repositories, benchmark records, and experimental logs [175]. It is especially important for literature review, citation support, evidence checking, rebuttal generation, and stages where source attribution is required. RAG reduces hallucination by exposing models to evidence at inference time, but does not ensure that selected sources are correct, version-consistent, or faithfully represented.

Training-free agentic methods extend LLMs with planning, tool use, memory, self-reflection, and iterative execution, enabling multi-step workflows without updating model parameters [306, 326, 438]. These methods are central to deep literature exploration, code debugging, experiment orchestration, review-response planning, and Paper2X workflows. Their strength lies in orchestration, while their main risk is error propagation when retrieval, tool use, or self-critique fails.

Training-based methods specialize models for stage-specific distributions, such as peer reviews, scientific manuscripts, code repositories, citation contexts, rebuttal traces, or benchmark trajectories [265, 383]. They include supervised fine-tuning, instruction tuning, preference optimization, reinforcement learning, and domain-specific adaptation. They can improve consistency, format adherence, domain vocabulary, and task-specific judgment, but depend heavily on data quality and may overfit to narrow benchmark or venue distributions.

Hybrid approaches combine multiple families into integrated research systems, for example by coupling RAG with agentic planning, fine-tuning domain-specific submodules, or embedding prompt-based controllers inside larger workflows [11, 223, 232, 318]. Hybrid systems are increasingly dominant because research workflows require generation and grounding, autonomy and verification, and flexible reasoning with stage-specific specialization.

Table 1 maps these methodological families to the eight lifecycle stages, using primary and secondary markers to indicate common design patterns in recent systems.

2.3 Scope & Literature Collection

This survey focuses on AI tools, methods, and benchmarks that support *human-driven academic research*, with an emphasis on computer science and machine learning. We cover work published or publicly released between 2023 and 11 October 2026, while also referencing earlier foundational methods when they define

recurring technical paradigms. Cross-disciplinary systems are included when they demonstrate capabilities relevant to the research lifecycle, such as autonomous experimentation, literature synthesis, scientific coding, or evidence-grounded writing. We exclude general-purpose LLM capabilities that are not explicitly connected to research workflows, as well as closed systems for which insufficient technical or evaluative information is available.

To construct the survey corpus, we combined three complementary collection strategies:

- **Systematic keyword search** across Google Scholar, Semantic Scholar, arXiv, and DBLP, using queries related to AI-assisted research, automated research agents, literature review, scientific coding, paper writing, peer review, rebuttal generation, and research dissemination.
- **Snowball citation tracing** from representative seed papers in each lifecycle stage, including both backward tracing to foundational work and forward tracing to recent systems and benchmarks.
- **Community and repository monitoring**, including open-source projects, curated reading lists, and benchmark leaderboards that document emerging tools not yet covered by formal publications.

A paper, system, or benchmark was included only if it satisfied all three criteria: (i) it targets at least one stage of the research lifecycle defined in Section 2.1; (ii) it is publicly accessible through a publication, preprint, open-source repository, benchmark page, or technical report; and (iii) it provides sufficient methodological or evaluative detail to support critical analysis. When multiple versions of the same system exist, we prioritize the most recent or most technically complete version, while noting earlier versions when they mark important historical milestones.

The resulting corpus spans all four phases of the lifecycle, but the distribution is uneven. Most documented systems concentrate on P1 (*Creation*), especially literature review, coding, and experiment automation, followed by P2 (*Writing*), P3 (*Validation*), and P4 (*Dissemination*). This imbalance reflects both research maturity and publication availability: creation-stage tools are more frequently benchmarked and open-sourced, whereas dissemination-oriented tools are often commercial, workflow-specific, or evaluated through less standardized criteria. The benchmark landscape across stages is summarized in Table 2.

2.4 Development Timeline

The development of AI-assisted research can be understood as a shift from *stage-specific assistance* toward *multi-stage research automation*. Before 2024, most systems targeted isolated research tasks, such as literature search, scientific question answering, code generation, or domain-specific experiment planning. Early demonstrations, including Coscientist [21], showed that LLM-based agents could plan and execute scientific workflows in constrained laboratory settings, while domain foundation models such as AlphaFold 3 [1] illustrated the broader potential of AI systems to transform specialized scientific discovery.

In 2024, the field began moving from isolated tools toward end-to-end research agents. The AI Scientist [223] provided an early demonstration of an automated pipeline spanning idea generation, experiment execution, paper writing, and review-style evaluation. Around the same period, general coding agents, retrieval-augmented literature systems, and scientific reasoning benchmarks matured rapidly, making it possible to evaluate individual components of the research lifecycle more systematically. This transition marked an important change in emphasis: AI systems were no longer viewed only as assistants for local tasks, but increasingly as orchestrators of multi-step research workflows.

By 2025 and early 2026, the field entered a stage of rapid specialization and benchmarking. Dedicated systems emerged for nearly every lifecycle stage, including literature synthesis, paper-to-code translation, autonomous experiment orchestration, manuscript writing, peer review, rebuttal support, figure generation, and research dissemination. For example, OpenScholar [11] advanced retrieval-augmented scientific synthesis, AI Scientist v2 [419] explored stronger forms of end-to-end automated research, and FARS [20] demonstrated large-scale autonomous paper generation. At the same time, previously underexplored stages began receiving dedicated attention, including rebuttal writing (*e.g.*, RebuttalAgent [119]) and scientific visualization (*e.g.*, AutoFigure-Edit [205]).

Table 2 Summary of datasets and benchmarks for AI-assisted research, organized by phases and stages.

#	Stage	Benchmark	Ref.	Year	GitHub	HF	Evaluation Focus	Scale	Link
Phase 1: Creation									
1	S1: Idea Gen.	IdeaBench	[109]	2024	-	-	Novelty, feasibility	Multiple LLMs	
2	S1: Idea Gen.	LiveIdeaBench	[294]	2024	-	-	Real-time model comparison	40+ models	
3	S1: Idea Gen.	AI Idea Bench 2025	[282]	2025	✓	-	Multi-dimensional assessment	3,495 papers	
4	S1: Idea Gen.	ResearchBench	[218]	2025	-	-	Inspiration-based task decomp.	-	
5	S1: Idea Gen.	Scientist-Bench	[356]	2025	-	-	Guided & open-ended AI research	Multi-domain	
6	S1: Idea Gen.	HindSight	[147]	2026	-	-	Impact-based idea evaluation	-	
7	S1: Idea Gen.	HeurekaBench	[268]	2026	✓	-	Open-ended data-driven science	Multi-domain	
8	S1: Idea Gen.	MechHypoBench	[407]	2026	-	-	Controlled consequence agreement	300 tasks/14 fields	
9	S2: Lit. Rev.	LitSearch	[6]	2024	✓	✓	Literature retrieval	-	
10	S2: Lit. Rev.	DeepScholar-Bench	[270]	2025	✓	-	Research synthesis quality	-	
11	S2: Lit. Rev.	ReportBench	[185]	2025	✓	-	Deep research report quality	100 prompts	
12	S2: Lit. Rev.	ScholarGym	[321]	2026	-	-	Information-gathering evaluation	2,536 queries	
13	S2: Lit. Rev.	SciNetBench	[316]	2026	-	-	Relation-aware retrieval	18M papers	
14	S2: Lit. Rev.	IDRBench	[76]	2026	-	-	Interactive deep research	100 tasks	
15	S3: Coding	SWE-bench	[151]	2024	✓	✓	GitHub issue resolution	500 problems	
16	S3: Coding	MLAgentBench	[137]	2024	✓	-	ML experimentation	13 tasks	
17	S3: Coding	LAB-Bench	[172]	2024	✓	✓	Biology research tasks	Multi-domain	
18	S3: Coding	DiscoveryBench	[237]	2024	✓	✓	Data-driven discovery	-	
19	S3: Coding	DiscoveryWorld	[144]	2024	✓	-	Virtual discovery environment	120 tasks	
20	S3: Coding	MLE-Bench	[30]	2024	✓	✓	Kaggle ML competitions	75 tasks	
21	S3: Coding	ScienceAgentBench	[49]	2024	✓	✓	Scientific data analysis	-	
22	S3: Coding	KernelBench	[264]	2025	✓	✓	GPU kernel generation	-	
23	S3: Coding	TritonBench	[181]	2025	✓	-	Triton operator generation	-	
24	S3: Coding	ResearchCodeBench	[134]	2025	-	-	Novel ML code implementation	212 tasks	
25	S3: Coding	SciReplicate-Bench	[402]	2025	✓	-	Algorithm reproduction	100 tasks	
26	S3: Coding	MLR-Bench	[36]	2025	-	✓	Open-ended ML research	201 tasks	
27	S3: Coding	MLGym	[251]	2025	-	-	AI research agent framework	-	
28	S3: Coding	CURIE	[168]	2025	✓	-	Rigorous experimentation	-	
29	S3: Coding	PaperBench	[342]	2025	✓	✓	Paper replication	20 ICML papers	
30	S3: Coding	AstaBench	[24]	2025	✓	✓	Scientific research suite	2,400+ problems	
31	S3: Coding	ResearchClawBench	[413]	2025	✓	-	Scientist-aligned workflows	Multi-domain	
32	S3: Coding	EXP-Bench	[169]	2026	✓	✓	AI conducting experiments	461 tasks/51 papers	
33	S3: Coding	FrontierScience	[375]	2026	-	-	Expert-level scientific tasks	Olympiad + PhD	
34	S3: Coding	PostTrainBench	[287]	2026	✓	✓	LLM post-training automation	-	
35	S3: Coding	MMPostTrainBench	[217]	2026	✓	-	Checkpoint tests; non-target retention	8 multimodal tasks	
36	S3: Coding	ST-Bench	[52]	2026	-	-	Earth-science output coverage; cost	100 tasks/2,067 queries	
37	S4: Tab. & Fig.	MatPlotBench	[431]	2024	-	-	Data visualization	-	
38	S4: Tab. & Fig.	PlotCraft	[457]	2025	-	-	Complex visualization	1K tasks	
39	S4: Tab. & Fig.	TeXpert	[155]	2025	-	-	LaTeX code generation	3 difficulty levels	
40	S4: Tab. & Fig.	PaperBananaBench	[485]	2026	-	-	Scientific illustration quality	292 test cases	
41	S4: Tab. & Fig.	SciFlow-Bench	[462]	2026	-	-	Framework figure evaluation	500 figures	
42	S4: Tab. & Fig.	Figure-Bench	[487]	2026	✓	✓	Text-to-illustration generation	3,300 pairs	
Phase 2: Writing									
43	S5: Writing	ScholarCopilot	[385]	2025	-	-	Citation accuracy	40.1% top-1 acc.	
44	S5: Writing	SciIG	[87]	2025	-	-	Introduction writing quality	NAACL/ICLR papers	
45	S5: Writing	PaperWritingBench	[338]	2026	-	-	AI paper writing quality	200 papers	
Phase 3: Validation									
46	S6: Peer Rev.	ClaimCheck	[263]	2025	-	-	Grounded LLM critiques	-	
47	S6: Peer Rev.	Review-CoT	[84]	2025	-	-	Review reasoning chains	142K reviews	
48	S6: Peer Rev.	AI Detection Bench	[445]	2025	-	-	AI review detection	788K reviews	
49	S7: Rebuttal	ReviewMT	[353]	2024	-	-	Multi-turn review dialogue	26,841 papers	
50	S7: Rebuttal	Re^2	[454]	2025	-	-	Full-stage review + rebuttal	19,926 papers	
51	S7: Rebuttal	Commitment Checklist	[31]	2026	-	-	Unfulfilled rebuttal commitments	ICLR 2025	
Phase 4: Dissemination									
52	S8: P2Slides	PPTEval	[477]	2025	✓	-	Slide content, design, coherence	10K+ presentations	
53	S8: P2Video	PresentQuiz	[490]	2025	✓	-	Video faithfulness	101 paper-video pairs	
Cross-Phase									
54	Cross-Phase	RE-Bench	[395]	2024	✓	-	Open-ended ML R&D	7 environments	
55	Cross-Phase	PaperBench	[342]	2025	✓	✓	End-to-end paper replication	20 ICML papers	

These developments suggest that the field is no longer bottlenecked by model capability alone, but also by orchestration, evaluation, reliability, and governance across the full research lifecycle.

3 Phase 1: Creation

This phase covers the stages through which a research contribution is materially produced: generating an idea (S1), situating it within prior work (S2), producing empirical or analytical evidence (S3), and constructing visual representations of methods and results (S4). Together, these stages address two foundational questions: *what is the contribution, and what evidence supports it?*

Among the four phases, *Creation* currently has the richest tool ecosystem and broadest benchmark coverage, but its maturity remains uneven. S1 (*Idea Generation*) has attracted extensive tooling, yet suffers from an ideation–execution gap in which seemingly novel ideas often weaken after implementation. S2 (*Literature Review*) is rapidly improving through retrieval-augmented and agentic synthesis, but citation fidelity, coverage completeness, and multi-paper relational reasoning remain difficult. S3 (*Coding and Experiments*) has progressed through code generation, paper-to-code translation, and autonomous experiment orchestration, but performance still drops sharply on genuinely novel research code. S4 (*Tables and Figures*) remains comparatively underdeveloped despite its importance in daily research practice. We discuss these four stages in order below.

3.1 Idea Generation

Idea generation is the entry point of the research lifecycle, where candidate hypotheses, research questions, and experimental directions are proposed and refined. Existing approaches range from direct LLM prompting to externally grounded generation, multi-agent collaboration, and dedicated evaluation of novelty, feasibility, diversity, and downstream impact. Across these directions, the central challenge is that LLMs can produce ideas that appear novel and well-motivated, yet often struggle to generate ideas that remain feasible, distinctive, and impactful after execution. What makes this stage distinctive is that, unlike later stages whose outputs can be checked against code, data, or reviewer feedback, the value of an idea is only weakly observable at the moment it is proposed: it depends on experiments not yet run and on a research context that no single prompt fully captures. Idea generation is therefore simultaneously the most heavily tooled and the least verifiable stage of *Creation*, which is why much of the recent progress lies less in producing more ideas than in grounding and assessing them.

We organize methods into four groups that trace this progression: generation from the model's internal parametric knowledge (Section 3.1.1), generation anchored in external signals such as knowledge graphs, retrieved literature, and research trends (Section 3.1.2), multi-agent collaborative generation that simulates community-style critique and debate (Section 3.1.3), and the dedicated assessment of idea quality along novelty and feasibility (Section 3.1.4). A comprehensive inventory of ideation systems is provided in Table 4 (Appendix).

3.1.1 LLM Internal Knowledge-Based Generation

The simplest form of AI-assisted ideation prompts an LLM directly with a research domain, problem description, or literature context. Si *et al.* [327] established an influential baseline through a large-scale human study involving 100+ NLP researchers, finding that LLM-generated ideas were rated significantly higher in novelty than human ideas ($p < 0.05$). This result demonstrates the surface-level generative capacity of LLMs, but it also raises a central question for this stage: whether apparent novelty corresponds to executable and impactful research. Subsequent work has explored three ways to strengthen direct generation. First, *iterative refinement* uses feedback loops to improve idea specificity and reduce shallow novelty. ResearchAgent [13] incorporates academic graph feedback to refine generated ideas, SciMON [376] iteratively compares candidate ideas against prior work to mitigate the tendency of direct LLM prompting toward shallow contributions, and Chain of Ideas [183] organizes literature into progressive reasoning chains that outperform simple prompting baselines.

Second, *learned quality signals* introduce explicit scoring or optimization objectives. Spark [305] combines retrieval-augmented generation with a judge model trained on 600K OpenReview reviews to estimate creativity, DeepInnovator [70] trains a 14B model under a "Next Idea Prediction" paradigm and reports 80–94% win rates against frontier models on ideation tasks, and Goel *et al.* [95] optimize AI co-scientist plans using rubric rewards extracted from existing papers, with RL-optimized plans preferred by human experts 70% of the time.

Third, *adaptive test-time compute* treats reasoning effort as a controllable resource. IRIS [88] uses MCTS in a human-in-the-loop ideation platform to allocate search as ideas converge, while FlowPIE [377] evolves scientific ideas at test time through flow-guided literature exploration. Recent systems push this search view further by making the space being searched explicit: Tree-of-Ideas [192] reasons across trajectories of scholarly evolution rather than over a static snapshot of the literature, and IDEAgent [105] frames ideation as quality-diversity search so that candidate ideas are selected for coverage of the space and not for score alone.

A recent creativity-centered survey [315] further categorizes these methods into knowledge augmentation, prompt steering, inference-time scaling, multi-agent collaboration, and parameter adaptation.

3.1.2 External Signal-Driven Generation

Direct LLM generation is limited by the model's parametric knowledge and by its tendency to produce plausible but weakly grounded ideas. External signal-driven methods address this limitation by anchoring ideation in structured knowledge, retrieved literature, or temporal research trends. Three signal sources are especially common, each grounding ideas from a different perspective: relational structure, textual evidence, and temporal opportunity.

Knowledge graphs provide relational structure for hypothesis formation. SciAgents [91] performs multi-agent reasoning over scientific knowledge graphs, while MOOSE-Chem [433] decomposes chemistry hypothesis generation into inspiration retrieval, hypothesis composition, and ranking, rediscovering hypotheses from 51 high-impact papers. MOOSE-Chem2 [432] extends this direction toward fine-grained, experimentally actionable hypotheses. *Paper retrieval* grounds ideas in unstructured literature. SciPIP [380] proposes ideas anchored to retrieved papers, and IdeaSynth [277] represents idea facets as nodes on an interactive canvas for literature-grounded refinement; in a user study with 20 participants, IdeaSynth encouraged users to explore more alternatives than LLM-only baselines. *Trend analysis* targets the temporal dimension of research opportunity. Nova [131] uses iterative planning and search to identify emerging research directions with improved diversity. External grounding is also being pushed toward the reasoning form that ideation actually requires, namely abduction: HALO [146] supports interactive co-abductive reasoning with a human collaborator, a complementary analysis asks what representational grounding the abduction loop presupposes and whether a disembodied model can supply it [73], and inspiration retrieval can be learned against target-conditioned abstractions rather than surface similarity [349]. Domain instantiations show the same grounding logic at work in causally structured settings, both in empirical software engineering [368] and in materials science, where mechanism recovery can be made visually inspectable [340]. Together, these methods suggest that external grounding is not merely an auxiliary feature, but a key mechanism for connecting generated ideas to the research frontier.

IdeaAnchor [50] makes literature-grounded synthesis a training objective by mining the roles of prior papers, their relationships, and contribution-specific criteria as privileged signals. Demonstration learning, self-distillation, and reinforcement learning improve criteria satisfaction and expert preferences on temporally held-out proposals, while retrieval mainly adds methodological detail. These measures assess ideation quality; they do not establish the success of subsequently executed research.

3.1.3 Multi-Agent Collaborative Generation

Multi-agent ideation systems attempt to improve idea quality by simulating aspects of research-community interaction, such as role specialization, critique, revision, and debate. VirSci [344] constructs a virtual scientific community in which multiple LLM agents participate in structured discussions, reporting higher novelty scores than a single-agent AI Scientist baseline (5.24 vs. 4.94). Its analysis suggests that agent diversity and discussion structure matter, with the best configuration using 8 members over 5 rounds with 50% diversity.

However, multi-agent scaling is not uniformly beneficial. A SIGDIAL 2025 study [364] finds that three critique–revision rounds are often sufficient, while additional rounds produce diminishing returns. Other systems explore richer collaboration mechanisms beyond discussion alone: Gu *et al.* [104] study *combinatorial creativity* by composing ideas across domains, and Deep Ideation [472] designs agents that navigate scientific concept networks through structured graph exploration.

Yet recent evidence also points to a deeper limitation: the "Artificial Hivemind" study [148] reports that LLM-generated ideas tend to cluster in narrow regions of the idea space, suggesting that diversity collapse may be a structural property of current models rather than a problem solved simply by adding more agents. In response, recent methods target diversity explicitly: *Towards Diverse Scientific Hypothesis Search* [372] casts hypothesis generation as a diversity-aware search problem, using population-based sampling to escape these narrow modes across molecular, equation, and algorithm discovery. Complementing diversity-oriented search, auditable hypothesis-evolution protocols [352] make the iterative proposal and refinement of hypotheses by LLM agents transparent and inspectable. The concern has also been reframed at the level of the research community rather than the model: when many researchers query similar assistants, ideation can homogenize across users, motivating methods that deliberately diversify personalized ideation against this AI-induced convergence [412].

Skill Reuse, Local Operation, and Precommitted Falsifiability. Three further shifts are visible in this window. First, ideation systems are beginning to accumulate reusable competence rather than re-deriving strategy each run: HypoForge [280] learns reusable scientific skills spanning both hypothesis generation and hypothesis testing, and MAIL [12] replaces static inspiration corpora and predefined heuristics with a memory-driven, incremental, literature-grounded loop. Second, the assumption that problem discovery requires a frontier proprietary model is being tested: SGHA [92] runs a corpus-first structural-gap pipeline entirely on a local LLM. Third, and most consequential for evaluation, benchmarks are starting to define idea quality operationally rather than by preference judgement. RATIO [320] declares relevance through three typed ideation moves — address, broaden, specify — so that retrieval for ideation has a defined target, and Lit2Test [389] requires every proposal to precommit the observation that would falsify it, which makes proposal quality decidable rather than merely arguable. That last move is a direct answer to the verification gap this survey identifies as the field's central weakness.

3.1.4 Assessment: Novelty and Feasibility

Evaluating generated ideas is difficult because strong research ideas must satisfy multiple criteria simultaneously: novelty, feasibility, clarity, significance, and eventual impact. Early benchmarks quantify parts of this space, but the central question is whether an idea remains valuable after it is implemented, tested, and situated against prior work.

IdeaBench [109] evaluates idea generation against 2,374 influential papers across eight research domains, while LiveIdeaBench [294] probes scientific creativity using 1,180 keyword prompts across 22 domains. Both suggest that scientific creativity is not well predicted by general-purpose benchmarks, with reasoning-focused models often performing better. ResearchBench [218] extends evaluation through inspiration-based task decomposition, and AI Idea Bench 2025 [282] scales assessment to 3,495 papers across two evaluation axes.

A recurring pattern across these benchmarks is the gap between apparent novelty and practical feasibility. IdeaBench reports that many LLMs score above 0.6 on novelty but below 0.5 on feasibility [109], indicating that generating plausible-sounding ideas remains easier than generating ideas that can be executed and validated. HindSight [147] sharpens this concern by introducing a time-split, impact-based evaluation, showing that LLM-as-Judge can overvalue novel-sounding ideas that do not later materialize into impactful work. This finding suggests that current evaluation protocols may reward apparent novelty rather than genuine research potential, reinforcing the need for execution-grounded and temporally robust assessment. Two recent benchmarks question whether models can perform this assessment at all: SoundnessBench [123] tests whether AI-scientist systems can distinguish methodologically sound proposals from flawed ones, while a complementary analysis [334] characterizes the limits of LLM-as-Judge for novelty assessment; both report a pervasive optimism bias that undermines the use of LLMs as first-pass idea evaluators. Two further benchmarks tighten the protocol itself: LigBench [371] unifies ideation evaluation against human-aligned judgments, and a blind formulation asks models to recover the ideas of papers from pre-publication bibliographies alone [42], which removes the leakage that makes retrospective novelty scoring unreliable.

Recent benchmarks further separate literature grounding from implementation readiness. AgentIdeaBench [243] compares static observation with active exploration across five disciplines, finding improvements in feasibility, clarity, and specificity without corresponding gains in measured originality under its critics. IdeaAMBIG [236] examines a different boundary: whether an apparently plausible research idea specifies enough methodological

detail for faithful implementation. Its real and synthetic specification gaps distinguish defect localization from clarification once a defect is known. Together, these benchmarks suggest that idea quality requires both grounding in prior work and sufficient specification for execution; neither is captured by a single novelty score.

Judging Novelty Without a Comparative Oracle. Novelty assessment has depended on comparison against retrieved prior work, which inherits the retriever's blind spots. Two contributions probe the model itself instead. Rajwal *et al.* [286] score hypotheses by a logit-based energy derived from the model's intrinsic confidence rather than by comparative judgement, and Think-Probe-Respond [310] reads latent novelty judgements out of hidden states during reasoning and feeds them back into the final verdict. Both are cheap enough to run at generation scale, and both raise the same caution: an internal signal cannot distinguish a genuinely novel idea from one merely absent from the model's training distribution. Set against this, Patel *et al.* [271] report that across three years of models, performance on open-ended tasks where originality and diversity matter has not tracked the gains seen on verifiable-answer benchmarks — a caution against assuming ideation quality improves with general capability.

MechHypoBench [407] shifts hypothesis assessment from plausibility to consequences under withheld conditions. Its controlled tasks instantiate paper-derived mechanisms on empirical dataset structure and score whether open-form hypotheses imply the expected response changes. A fixed LLM interprets these implications, so the results probe consequence agreement within constructed mechanisms rather than establish causal identification in the original experiments.

3.1.5 Findings and Observations

Stage 1: Idea Generation



State & Progress

Maturity | Progression | Grounding | Collaboration | Training | Benchmarks

- Idea generation is one of the most tool-rich stages in Phase 1 (*Creation*), with systems spanning prompting, retrieval, multi-agent collaboration, learned scoring, and test-time search.
- Clear capability progression: prompting → RAG → multi-agent → RL-trained, each generation addressing the weaknesses of its predecessor.
- External grounding is increasingly central: retrieval- and knowledge-graph-based methods better connect generated ideas to the research frontier than LLM-only prompting [380, 433].



Gaps & Limitations

Execution | Feasibility | Diversity | Mis-Evaluation | Shallow | Closed-Loop

- Ideas that score well before implementation can degrade substantially after execution ($\Delta = -1.98$ vs. -0.63 for human ideas [328]), exposing a gap between surface novelty and executable substance.
- Persistent novelty-feasibility tradeoff (> 0.6 *vs.* < 0.5 [109]) remains unresolved, and diversity collapse is structural, not solvable by scaling [148].
- LLM-as-Judge evaluation can reward apparent rather than genuine innovation, with reported novelty judgments negatively correlating with later real-world impact ($\rho = -0.29$ [147]).

3.2 Literature Review

Literature review anchors research in prior knowledge by retrieving relevant work, synthesizing evidence, and organizing existing findings into a coherent intellectual context. Compared with idea generation, this stage is more grounded and externally verifiable, making it one of the fastest-maturing areas in AI-assisted research. Existing systems have moved from semantic paper retrieval to citation-aware synthesis and long-horizon deep research agents. Yet two limitations remain central: systems can retrieve and summarize individual papers increasingly well, but still struggle with faithful citation, coverage completeness, and multi-paper relational reasoning. These limitations carry outsized consequences because literature review sits upstream of the rest of the lifecycle: a missed competing method, a misattributed result, or an incomplete account of prior work propagates silently into ideation, writing, and even the author's later response to reviewers. For this

stage, then, the binding constraint is not fluency but *faithfulness*—whether each surfaced claim is correctly attributed to a source that genuinely supports it.

We follow the natural pipeline of AI-assisted reviewing: retrieving candidate papers from large scientific corpora (Section 3.2.1), synthesizing them into survey or related-work narratives (Section 3.2.2), and the long-horizon deep research agents (Section 3.2.3) that increasingly fold retrieval and synthesis into a single iterative loop; we then examine how retrieval and synthesis quality are assessed (Section 3.2.4). A comprehensive inventory of literature review systems is provided in Table 5 (Appendix).

3.2.1 Literature Retrieval

Retrieval is the foundation of AI-assisted literature review: every downstream synthesis depends on whether the system can surface the right papers from scientific corpora that now contain tens of millions of entries. Existing methods can be grouped into three modes. *Semantic retrieval* forms the baseline, using dense representations and LLM-based query understanding to move beyond keyword matching. LitLLM [4] integrates LLMs with academic databases for dense retrieval, while PaperQA2 [335] extends this direction with citation verification and reports strong performance on scientific literature search.

Citation-graph-augmented retrieval adds structural signals beyond embeddings. Instead of treating papers as isolated documents, these methods use citation links, paper relations, and graph traversal to improve contextual coverage. OpenResearcher [481] combines RAG with graph traversal for accelerated literature exploration. *Agentic multi-step retrieval* further shifts retrieval from a one-shot ranking problem to an iterative search process. PaSa [116] deploys an LLM agent that issues follow-up queries and refines candidate sets, approximating how human researchers probe an unfamiliar topic. Pushing this agentic view further, self-evolving retrieval agents [66] continually adapt their own search policy rather than relying on a fixed retrieval strategy. Recent work has also revisited the retrieval substrate itself rather than only the agent wrapped around it: structure-aware Boolean retrieval exploits corpus structure instead of relying solely on dense similarity [378], rerankers can be trained directly against search rubrics for research agents [212], and query refinement can be personalized through graph-scaffolded evidence grounding [442]. Alongside these methods, dedicated benchmarks have emerged to audit retrieval quality: LitSearch [6] targets retrieval precision, CiteME [276] focuses on citation fidelity, and MasterSet [289] provides a large-scale benchmark for must-cite citation recommendation in the AI/ML literature. Together, these efforts show that finding relevant papers is becoming easier, but ensuring that retrieved papers are used faithfully remains difficult.

3.2.2 Survey and Related Work Generation

Synthesis transforms retrieved papers into structured narratives. This marks a shift from retrieval-oriented systems, which optimize paper ranking and coverage, to generation-oriented systems, which must identify themes, compare methods, expose contradictions, and articulate research gaps. The subfield has developed through several increasingly structured designs.

Single-pass systems established the feasibility of automated survey drafting. AutoSurvey [384] demonstrated that LLMs can generate surveys of reasonable quality end-to-end, while SurveyX [198] improved content quality and approached human-expert performance on selected dimensions. *Structure-aware systems* then elevated outline planning from a formatting step to a core synthesis artifact. STORM [318] introduces multi-perspective question-asking to build comprehensive topic outlines, and SurveyForge [421] learns outline heuristics from human-written surveys together with memory-driven content generation, outperforming AutoSurvey on outline quality.

Multi-agent decomposition separates retrieval, verification, organization, and narrative writing into specialized subtasks. LiRA [94] and Agentic AutoSurvey [214] employ dedicated agents for different roles, while IterSurvey [455] treats outline generation as an iterative planning problem with stability checks. InteractiveSurvey [393] further introduces user customization, allowing researchers to refine reference categorization and outline structure through an interactive interface. STRUCTSURVEY [272] pushes the agentic direction further, coupling structured retrieval to outline planning and evidence gathering for automated survey-paper generation.

Citation- and editor-aware systems close the loop between synthesis and the writing environment. SurveyG [253] constructs a three-layer citation graph (Foundation/Development/Frontier) with hierarchical traversal, Citegeist [15] builds a dynamic RAG pipeline on the arXiv corpus, and CiteLLM [126] embeds hallucination-free reference discovery directly inside a LaTeX editor. DeepSurvey [436] pushes this direction further, coupling full-text key-point extraction with evidence-constrained citation assignment to improve both analytical depth and citation reliability. Open-source systems such as GPT Researcher [67], PaperQA [232], and ChatPaper [227] further illustrate the growing practical adoption of literature synthesis tools beyond controlled research prototypes. However, citation fidelity remains a bottleneck: ScholarCopilot [385] reports only 40.1% top-1 citation accuracy, suggesting that generating plausible related-work text is still easier than grounding each claim in the correct source.

3.2.3 Deep Research Agents

Deep research agents differ from single-pass retrieval or survey-generation systems by treating literature exploration as an *iterative, agentic* process. Given an open-ended query, they plan sub-queries, retrieve and read sources, update their internal state, and continue until a report can be synthesized with sufficient confidence. This loop makes deep research agents closer to a workflow for long-horizon information seeking than a single retrieval model.

Commercial systems have popularized this paradigm for broad information synthesis. OpenAI Deep Research, Google Deep Research, Perplexity, and Elicit all support multi-source retrieval and report generation, though they differ in latency, citation style, interactivity, and target use cases. *Open-source literature-specific systems* adapt this paradigm to scientific research. OpenScholar [11], published in Nature, is a retrieval-augmented LM that searches large-scale open-access scientific corpora and outperforms PaperQA2 and Perplexity Pro on scientific literature benchmarks. Tongyi DeepResearch [9] from Alibaba is an agentic LLM specialized for long-horizon deep information seeking, achieving strong results on deep research benchmarks.

Training-era approaches target the data and optimization bottlenecks that limit long-horizon research agents. O-Researcher [439] combines multi-agent distillation with agentic reinforcement learning to improve benchmark performance, while OpenResearcher [196] addresses the trajectory-data bottleneck by constructing an offline trajectory synthesis pipeline over large document collections. These synthesized trajectories provide long-horizon tool-use supervision for training research agents. Two further directions have since appeared: pretraining objectives designed specifically for research navigation rather than generic language modeling [482], and agents that recursively improve their own research procedure across episodes [225].

Cost- and control-aware designs have emerged as a distinct concern, reflecting that an agent which reads indefinitely is not deployable. Work in this direction estimates the marginal value of each additional token or tool call so that search can stop when further reading is unlikely to change the answer [167], diagnoses and repairs the stagnation that causes agents to loop without new evidence [339], exposes the interpretable execution plan so users can steer and optimize a query [297], and pushes the whole loop onto small on-device models, where the authors report that exposure bounds faithfulness while retrieval bounds coverage [28]. A complementary line diagnoses *how* search fails rather than only whether it succeeded, characterizing failure modes of long-horizon search agents [211]. Retrieval strategy itself is also being broadened, for instance by retrieving historical analogies to support foresight-style analysis [47].

Domain-focused variants remain important for specialized synthesis tasks: CHIME [129] provides LLM-assisted hierarchical organization of scientific studies, and ASReview [365], published in Nature Machine Intelligence, uses active-learning-based screening to reduce manual effort in systematic reviews while maintaining recall. This specialization continues: Plato-Bio [55] performs verification-first biological novelty screening with temporal rediscovery checks, Albilich [96] orchestrates proof states with computer-algebra integration for mathematical research, and discipline-level studies now assess how far such assistance actually reaches, both in physics and astrophysics literature work [120] and across scientific knowledge discovery more broadly [2]. Collectively, deep research agents span a spectrum from lightweight factual lookup to long-horizon autonomous synthesis, but increasingly converge on the same iterative architecture: plan, retrieve, read, update, and synthesize.

State, Bounds, and Auditability in Deep Research. A reaction against unconstrained agentic search is now visible, on three distinct grounds. On *state*: DAS [418] makes survey generation a stateful closed loop in which

agents maintain explicit literature, organization, writing, and finalization states with candidate-grounded taxonomy planning and reverse paper-to-section routing, and AgentR [304] persists research intent, generated queries, candidate assessments, and gap analyses as durable, recoverable workflow artifacts rather than as transient context. On *bounds*: Crase [115] offers structurally bounded, inspectable graph exploration as a deliberate alternative to deep-research agents, reporting up to three times the recall@50 of proprietary-model agents at roughly a third of the cost — evidence that unbounded agency is not required for scholarly search. On *synthesis*: DeepWeaver [382] argues retrieval alone is insufficient and targets the organization of noisy, fragmented evidence into well-cited answers, while ITER [37] trains the retriever itself on agent interaction trajectories so that retrieval is aware of the multi-step process consuming it.

3.2.4 Assessment: Retrieval and Synthesis Quality

Evaluation has shifted from *retrieval accuracy* alone ("did the system find the right papers?") toward broader *synthesis quality* ("did it produce a useful, accurate, and well-organized review?").

At the output level, DeepScholar-Bench [270] establishes a dedicated benchmark for research synthesis across coverage, coherence, and factual accuracy. ReportBench [185] scales this direction to deep research reports derived from survey-style prompts. At the process level, ScholarGym [321] isolates the information-gathering stage of deep research by decomposing it into query planning, tool invocation, and relevance assessment. This is an early step toward evaluating *how* a system reaches its answer, rather than judging only the final output. Benchmarks have also begun probing structural and interactive dimensions of literature competence. SciNetBench [316] introduces a relation-aware benchmark for literature retrieval agents over large-scale AI literature, revealing that relation-aware retrieval accuracy often remains low. IDRBench [76] addresses the human-in-the-loop dimension through interactive deep research tasks with on-demand user interaction. AutoResearchBench [409] complements these by benchmarking agents on complex, multi-hop scientific literature discovery. The benchmark surface has continued to widen along two axes: breadth, with suites spanning wide as well as deep research queries [274] and verifiable tasks grown by iterative evolution from simple questions [370]; and scientific specificity, with evaluations that trace an agent from scientific navigation through to information integration [358].

Retrieval evaluation is also moving closer to the open-ended decisions that precede a research contribution. ScholarCatalyst [161] uses authors' firsthand judgments about papers that could advance their projects, while INSPIRE [64] evaluates search against the cited antecedents of later work and decomposes failure into exposure, selection, and ranking. These supervision sources capture different notions of relevance: author-judged inspiration can extend beyond published citations, whereas citation-derived research lineages offer a more structured but incomplete proxy. Both make temporal restrictions important, since access to the eventual solution would change the task from finding useful prior work to recovering a known lineage.

A distinct and pointed evaluation cluster has formed around *reliability under corrupted evidence*, which speaks directly to the capability-versus-integrity tension. Rather than asking whether an agent can find good sources, these benchmarks ask what it does when the sources are bad: DRNOISE [256] places deep research agents in deliberately misleading evidence environments, HiEviDR-Bench [346] tests whether evidence can be aggregated hierarchically rather than accepted flat, and a systematic study finds that misleading knowledge propagates into false conclusions [488]. The implication is uncomfortable and worth stating plainly: an agent that retrieves and synthesizes fluently can also launder unreliable evidence into a confident report, and no coverage- or coherence-oriented metric will notice.

Even accurate individual citations can support a selectively assembled conclusion. CESS [405] treats deep-research search traces as adaptive evidence samples and audits their direction against an explicit candidate pool. Its corrections require logged selection probabilities and adequate coverage; estimating the pool's evidence differs from establishing how a search-policy intervention changes the evidence read. This distinction adds a process-level requirement to synthesis evaluation: sources must not only support the claims attributed to them, but their selection must also be examined for systematic omissions.

Across these efforts, four evaluation dimensions have crystallized:

- *Citation Accuracy*, whether the references in the paper are correctly attributed and faithfully support the associated claims.

- *Coverage Completeness*, whether the review captures the relevant landscape without major omissions.
- *Narrative Coherence*, whether the synthesis has logical flow, thematic organization, and readability.
- *Factual Grounding*, whether claims are supported by cited evidence rather than hallucinated.

SurveyX [198] exemplifies this multi-dimensional view by evaluating content quality, structure quality, and citation accuracy as separate axes. The main open challenge is to develop automated metrics that correlate with expert judgment on synthesis quality while remaining robust across domains, venues, and writing styles.

Self-Confirmation and Attributable Citation Error. Two failure modes specific to agentic literature work were characterized this window, and both are measurement contributions rather than system contributions. Zhang *et al.* [463] identify an objectivity failure: once an agent has produced a query, plan, or intermediate conclusion, it becomes systematically less objective when later judging the consequences of that same action, and they operationalize this as the IBIS benchmark. Hirsch *et al.* [121] attack the attribution problem, localizing faithfulness and citation mistakes to the specific agent in a deep-research trace that introduced them — which converts an aggregate hallucination rate into a debuggable signal. Alongside these, Tree-of-Concerns [241] inverts the usual synthesis target by deploying skeptic personas to surface the limitations papers do *not* state, and a multi-aspect retrieval benchmark [392] asks whether retrievers can recover the same paper when queried from different aspects. Together they mark a shift from measuring whether an agent finds enough to measuring whether what it reports can be trusted and traced.

3.2.5 Findings and Observations

Stage 2: Literature Review



State & Progress

Fastest | Convergence | Open-Source | Retrieval | Synthesis | DeepResearch

- Fastest-maturing stage: four generations in two years (single-pass → structure-aware → multi-agent → editor-aware), with 35 systems spanning retrieval, synthesis, and deep research.
- Commercial and open-source systems increasingly converge on an iterative architecture: plan → retrieve → read → update → synthesize.
- Recent evidence suggests that trajectory data and long-horizon tool-use supervision can be as important as model scale for improving the performance of deep research systems [196].



Gaps & Limitations

Relations | Hallucination | Citation | Cross-Domain | Coherence | Scalability

- Multi-paper relational reasoning remains a core bottleneck: the citation top-1 accuracy metric remains largely limited [385], and relation-aware retrieval often remains weak [316].
- Hallucination has shifted from obvious fabrication to subtle misgrounding: generated claims may appear well-cited while not being faithfully supported.
- Nearly all benchmarks and systems target ML/NLP literature; cross-domain synthesis (chemistry, biology, physics) remains largely untested and likely requires domain-specific retrieval infrastructure.

3.3 Coding and Experiments

This stage translates research ideas into executable implementations, runs experiments, and analyzes the resulting evidence. Compared with literature review, coding and experimentation require AI systems to interact with external environments: repositories, dependencies, datasets, compute resources, test suites, and evaluation scripts. Existing work spans general code generation, paper-to-code translation, experiment orchestration, and result analysis. Across these directions, the central challenge is not whether LLMs can write plausible code, but whether they can produce semantically correct research implementations, execute meaningful experiments, and interpret results reliably.

A comprehensive inventory of coding and experiment systems is provided in Table 6 (Appendix).

3.3.1 Code Generation

General-purpose code generation has become one of the most mature capabilities of current LLMs. On SWE-bench Verified [350], which evaluates real-world GitHub issue resolution, frontier systems now exceed 76%. Agent frameworks have played a central role in this progress. SWE-agent [423] established the *agent–computer interface* paradigm, giving LLMs structured access to files, tests, and tool calls rather than relying on unstructured shell interaction. OpenHands [381] extends this direction into a general open platform for software engineering agents and has become a common backbone for coding-oriented workflows.

However, high performance on standard software benchmarks does not directly imply readiness for research coding. SWE-bench Verified has been questioned for potential contamination, and more challenging variants expose a sharper limitation: performance drops to 23% on SWE-bench Pro [62] and 25% on SWE-EVO [359]. These results suggest that standard benchmarks may overestimate robustness when tasks are familiar, well-scaffolded, or pattern-matchable. This distinction becomes more pronounced in research settings, where the target is not only to fix existing software but to implement underspecified algorithms, reproduce implicit design choices, and validate scientific claims. Discipline-level accounts corroborate this from practice rather than from benchmarks: a review of generative-AI use in statistical research, paired with a code-generation case study, finds the same divide between fluent code and methodologically defensible analysis [247].

3.3.2 Paper-to-Code

Paper-to-code translation is a research-specific form of code generation. It is harder than conventional software engineering because research papers often mix natural-language descriptions, equations, pseudocode, ablation details, and domain conventions, while leaving key implementation choices implicit. PaperCoder [314] addresses this setting with a three-stage multi-agent framework for planning, analysis, and code generation, transforming ML papers into executable repositories.

Dedicated benchmarks quantify how difficult this setting remains. ResearchCodeBench [134] evaluates LLMs on 212 novel ML implementation tasks, where the best reported model achieves only 37.3% accuracy; notably, 58.6% of errors are semantic, meaning that the generated code runs but implements the wrong algorithm or behavior. SciReplicate-Bench [402] reports a similar ceiling of 39% across 100 tasks from 36 NLP papers. SciCode [361] extends research-level coding evaluation to mathematics, physics, and chemistry, while PaperBench [342] decomposes 20 ICML 2024 papers into individually gradable subtasks covering environment setup, experiment execution, and result reproduction. Together, these benchmarks reveal a substantial gap between general software issue resolution and faithful research implementation.

At the high end, FunSearch [293] demonstrates that LLM-generated programs can contribute to genuine mathematical discovery when embedded inside an evolutionary search loop. This result is important, but it also clarifies the boundary of current capability: success comes not from raw one-shot code generation alone, but from coupling generation with aggressive search, evaluation, and selection. The resulting contrast between strong performance on familiar software benchmarks and much lower performance on novel research code defines the capability cliff of this stage.

Recent work sharpens paper-to-code into a specification-recovery problem. ReproAgent [133] observes that the specification is split — explicit paper content such as algorithms, metrics, and artifacts is lost across long agent trajectories, while implicit requirements are never written down — and introduces contract-guided reproduction to hold both halves. DeepRepro [337] adds state-aware subplanning for repositories that evolve during reproduction. PaperCompiler [216] makes the intermediate representation itself enforceable, compiling paper-grounded evidence into repository-level specifications with provenance, cross-file dependencies, and explicit distinctions between supported, inferred, and unresolved requirements. Its reported fidelity gains concern correspondence to the source method, which should be distinguished from reproducing the original empirical results. SciCoQA [14] approaches the same alignment question from the assurance side, treating paper–code correspondence as a quality-assurance target in its own right rather than a byproduct of successful execution.

3.3.3 Experiment Execution and Orchestration

Once the code is available, the next challenge is to run experiments systematically and efficiently. Experiment orchestration systems provide infrastructure for planning runs, modifying code, launching jobs, monitoring results, and iterating over failures. MLAgentBench [137] evaluates language agents on ML experimentation; MLR-Copilot [188] separates autonomous research into idea and experiment agents; DS-Agent [108] targets end-to-end data-science workflows; EvoDS [434] extends data-science agents with self-evolving skill learning and explicit context management; and AIDE [150] frames ML engineering as tree search in code space. Broader evaluation environments, including MLR-Bench [36], MLE-Bench [30], MLGym [251], and CURIE [168], provide increasingly standardized testbeds for measuring autonomous experimentation.

Recent systems push this infrastructure toward higher-throughput and closed-loop research workflows. R&D-Agent [426] uses a Researcher-Developer dual-agent design for ML experimentation, while Karpathy's autoresearch [157] demonstrates high-throughput experiment iteration. Closed-loop systems such as CodeScientist [145], Dolphin [448], and NovelSeek [453] attempt to connect hypothesis generation, implementation, execution, and verification. EvoScientist [231] further illustrates the ambition of this direction by reporting accepted papers generated through a self-evolving research pipeline. AutoScientists [83] organizes self-organizing agent teams for long-running experimentation, while EurekAgent [408] argues that engineering the agent's environment is central to autonomous scientific discovery. These systems show that experimental throughput and workflow automation are improving rapidly, but their reliability still depends heavily on task scaffolding, benchmark design, and verification quality.

A complementary line of work couples execution with search and learning signals. AlphaEvolve [259] improves algorithms through LLM-generated mutations and automated evaluation. Si *et al.* [329] use execution-grounded search with large-scale parallel GPU experiments, outperforming GRPO baselines. SciNav [460] uses pairwise tree-search judgments to select promising branches, while Yuksekgonul *et al.* [450] combine test-time training and reinforcement learning for continuous improvement across mathematics, GPU kernel optimization, and computational biology. AutoReproduce [474] addresses a different but related problem: reproducing cited experiments by extracting implicit knowledge from paper lineages.

Domain-specific systems illustrate how orchestration changes when the environment is scientific rather than purely computational. In chemistry, Coscientist [21] and ChemCrow [25] use LLM-driven tools to support autonomous research workflows. In biology, AlphaFold 3 [1] extends protein structure prediction to biomolecular complexes, while CRISPR-GPT [284], BioPlanner [261], and LAB-Bench [172] target gene-editing design, protocol planning, and biology research evaluation. For systems-level optimization, KernelBench [264] and TritonBench [181] evaluate whether LLMs can generate efficient GPU kernels and Triton operators. Cross-domain suites such as AstaBench [24] and EXP-Bench [169] broaden evaluation to multi-domain scientific tasks and autonomous experiment execution.

Overall, the execution layer has advanced quickly, especially when tasks are well specified and feedback is automated. The harder problem is experiment *planning*: deciding which experiments are worth running, in what order, and how to interpret failures. Many current systems perform well on prescribed task pools, but remain less reliable when asked to choose genuinely novel research directions. In this sense, coding and experimentation expose the same broader pattern as idea generation: execution capability is improving faster than the scientific judgment needed to decide what should be executed.

3.3.4 Assessment: Code Correctness and Reproducibility

Assessing coding and experiment systems requires more than checking whether the generated code runs. Research code must implement the intended algorithm, reproduce reported results, support meaningful ablations, and generate evidence that can be interpreted correctly. This makes semantic correctness and reproducibility central evaluation criteria. Several benchmarks expose the difficulty of this interpretive layer. DiscoveryBench [237] and ScienceAgentBench [49] evaluate scientific reasoning over experimental data, showing that LLMs still struggle with multi-step analysis over complex result sets. DiscoveryWorld [144] provides a virtual environment with 120 challenge tasks for automated scientific discovery agents. InfiAgent-DABench [132] benchmarks end-to-end data-analysis workflows, including data cleaning, statistical testing, and visualization generation across diverse domains. At the whole-system level, MLReplicate [81] benchmarks

autonomous research systems on end-to-end machine-learning reproducibility, directly probing whether agents can regenerate the results reported in a paper.

Two newer benchmarks isolate complementary parts of experimental competence. AgentActionBench [125] records reproduction actions and scores them with paper-specific rubrics across 150 ML and AI4Science papers, extending assessment beyond the final repository. Its scalable rubric construction uses model-assisted expansion, with a human-annotated subset covering 10% of the papers. AutoDataBench [449] instead holds non-data factors fixed across three controlled tasks to evaluate data interventions and agents' predictions about their effects. Process adherence and data-effect reasoning thus become explicit targets, while neither alone establishes that a complete research claim has been reproduced.

The core bottleneck is moving from raw outputs to trustworthy claims. Current systems can often produce plots, summary statistics, and local interpretations, but they are less reliable at identifying statistically meaningful trends, diagnosing failure modes, designing decisive ablations, and synthesizing results into a coherent empirical argument. This limitation is particularly consequential because coding errors and experimental misinterpretations can propagate into later writing and review stages, where polished narratives may obscure weak or incorrect evidence.

Execution Success Is Not Experimental Fidelity. The most pointed result in this window is that passing execution says little about whether the intended experiment was run. Yu *et al.* [444] audit experimental fidelity directly and document *methodological hallucinations*: agents silently reduce datasets or training budgets, or design experiments that do not test the paper's claims, while still producing runnable code and plausible numbers. This is precisely the generation-versus-verification asymmetry this survey argues is the field's defining weakness, now measured inside the coding stage rather than inferred. Two responses point in different directions. Praxist [182] makes provenance the unit of account, organizing autonomous R&D into solution lineages so that a gain can be traced to a cause rather than attributed to a run. A skill-based agentic workflow [246] instead constrains the task, generating competitive baselines for medical-imaging challenges under literature-guided reasoning — a narrower claim that is correspondingly easier to verify.

3.3.5 Findings and Observations

Stage 3: Coding & Experiments



State & Progress

37% Ceiling | Closed-Loop | Search | Throughput | Orchestration | Cross-Domain

- Sharpest capability boundary across all stages: 76% on pattern-matching *vs.* 37–39% on novel research code, consistently reproduced across 4+ independent benchmarks [134, 402].
- Execution infrastructure is no longer the bottleneck: systems sustain ∼12 experiments/hour in closed-loop, with generated papers accepted at academic venues [157, 231].
- Coupling generation with search (evolutionary, tree, RL) consistently outperforms raw code generation [150, 259, 293], suggesting that search strategy matters more than model capability alone.



Gaps & Limitations

Semantic | Planning | Fabrication | Insight Gap | Benchmark Leak | Verification

- Semantic failures are especially problematic: generated code may execute successfully while implementing the wrong algorithm or producing misleading results [134].
- Current systems execute prescribed tasks more reliably than they choose meaningful experiments; experiment planning remains strongly dependent on human scientific judgment.
- 80% of fully autonomous results are fabricated [36], and downstream review catches only half of methodological issues [228], creating a compounding verification deficit.

3.4 Tables and Figures

Tables and figures transform experimental outputs, statistical summaries, algorithms, and conceptual designs into publication-ready research artifacts. Existing systems cover scientific figure generation, data visualization, table construction, formula generation, and algorithmic illustration. Compared with coding and experimentation, this stage is less about producing new evidence than about representing evidence faithfully. Across these artifact types, the central challenge is the gap between *visual plausibility* and *scientific correctness*: AI-generated outputs may look professional while containing incorrect labels, misleading layouts, invalid numerical relationships, or domain-specific notation errors.

A comprehensive inventory of figure and table generation systems is provided in Table 7 (Appendix).

3.4.1 Scientific Figure Generation

Scientific figure generation spans method diagrams, architecture illustrations, result plots, data visualizations, and pipeline figures. Standard result plots are comparatively tractable because they can often be grounded in structured data and executable plotting code. In contrast, method diagrams and framework figures are harder because they require faithful spatial organization, correct information flow, domain-specific symbols, and paper-specific visual conventions.

For *method and architecture diagrams*, AutoFigure-Edit [205] generates editable text-to-SVG scientific illustrations from long-form text, enabling users to revise generated figures rather than treating them as fixed images. Its companion system AutoFigure [487] introduces FigureBench for generating and refining publication-oriented scientific illustrations. PaperBanana [485] employs multiple specialized agents for retrieval, planning, styling, visualization, and critique, while StarVector [292] focuses on scalable vector graphics from textual descriptions. More recent systems broaden the input modalities and grounding signals: Crafter [471] provides a multi-agent harness that produces editable scientific figures from heterogeneous inputs, DiagramRAG [447] retrieves reference diagrams to ground figure generation, and GeoSVG-RL [190] applies geometry-aware reinforcement learning to layout-constrained text-to-SVG diagram generation. FigTree [215] further treats a paper-grounded figure as a hierarchy of short SVG programs, combining recursive construction with critique of both the rendering and its source. This makes local visual revisions traceable to program changes. Together, these systems show a shift from static image generation toward editable, structured, and critique-aware figure construction.

For *result plots and data visualization*, MatPlotAgent [431] uses VLM-based visual feedback to improve data visualization quality, while PlotGen [98] and PlotCraft [457] study chart generation across diverse plot types and task difficulties. CoDA [48] explores multi-agent collaboration for visualization, and ChartGPT [362] decomposes chart generation into sequential reasoning steps for handling abstract natural-language inputs. More recent systems broaden the scope of generation and evaluation: SciFig [138] introduces rubric-based evaluation for pipeline figures, VisCoder [255] studies code-based visualization generation at scale, DiagramAgent [391] targets multiple diagram categories with specialized agents, and SciFlow-Bench [462] evaluates scientific framework figures through structure-first analysis. These efforts indicate that standard data plots are increasingly tractable, while complex framework figures remain difficult because they require structural consistency rather than only visual appeal.

For *figure editing and optimization*, VIS-Shepherd [266] provides constructive feedback for LLM-based data visualization, emphasizing critique and revision rather than direct generation alone. [32] surveys publisher policies on AI-generated figures and proposes best-practice guidelines for responsible use. The SAIL framework [311] separates domain logic from code syntax, allowing researchers to retain scientific oversight while delegating implementation details to AI. A newer framing treats editing as the primary task and learns it from evidence of real revisions: SciDiagramEdit [347] learns diagram edits from paper revision histories, Edit2TikZ [467] benchmarks instruction-conditioned figure editing in TikZ source, and GenGA [158] generates editable, data-grounded graphical abstracts rather than one-shot images. EdiTikZ [103] extends revision-based supervision by mining TikZ edit pairs and learning instruction-conditioned editing alongside reconstruction. Its render and edit evaluations address whether a requested change is realized, rather than certifying the scientific content depicted. Across these systems, the emerging design principle is human-guided refinement: AI can accelerate layout, rendering, styling, and accessibility improvements, but researchers must verify

whether the figure faithfully represents the underlying method or data.

3.4.2 Table Understanding and Generation

Table generation spans two complementary tasks: understanding existing tables and creating new ones. On the understanding side, Chain-of-Table [388] performs table reasoning through multi-step table transformations, reflecting the fact that many table tasks require sequential operations rather than single-pass extraction. On the generation side, ArxivDIGESTables [252] synthesizes scientific literature into structured comparison tables, ShowTable [219] introduces collaborative reflection and refinement for creative table visualization, and Table2LaTeX-RL [206] converts table images into LaTeX code using reinforced multimodal language models. CSPO [430] similarly targets structured table-to-LaTeX generation, introducing component-specific optimization to alleviate reward ambiguity during training.

Compared with standard figure generation, table generation remains less mature because scientific tables must satisfy stricter semantic constraints. Comparison tables require consistent axes, fair grouping of methods, complete citation coverage, and correct numerical transcription. Ablation tables are even more demanding because they encode experimental design choices, not only final results. AbGen [475] evaluates LLMs on ablation study design using expert-annotated examples from NLP papers, revealing a significant gap between LLM-generated table plans and human expert judgments. This suggests that table generation is not merely a formatting problem; it requires understanding which comparisons are scientifically meaningful and how evidence should be organized.

3.4.3 Mathematical Formulas and Algorithm Pseudocode

Mathematical formulas, TikZ diagrams, and algorithm pseudocode are compact representations of scientific reasoning, making them particularly sensitive to small errors. Unlike ordinary prose or standard charts, these artifacts require exact syntax and exact semantics simultaneously: a misplaced symbol, index, operator, arrow, or dependency can change the meaning of the method. As a result, formula and pseudocode generation remain less robust than natural-language polishing or standard visualization.

Recent systems address this challenge through specialized datasets, multimodal inputs, and iterative refinement. AutomaTikZ [16] introduces DaTikZ, a large-scale TikZ dataset, and shows that fine-tuned models can outperform general-purpose LLMs on scientific vector graphics. DeTikZify [17] extends this line with multimodal input and MCTS-based iterative refinement over a larger collection of TikZ graphics. TikZilla [102] further suggests that domain-specific training with supervised fine-tuning and reinforcement learning can make smaller open-source models competitive with larger general-purpose models on TikZ generation. TeXpert [155] highlights the remaining difficulty: accuracy drops sharply as LaTeX tasks become more complex, especially for tables with merged cells, nested environments, and nontrivial formatting constraints. These results reinforce the broader pattern of table and figure generation: specialized training and iterative refinement help, but human verification remains necessary when visual or symbolic artifacts carry scientific meaning.

3.4.4 Assessment: Visual Fidelity and Scientific Accuracy

Evaluation for table and figure generation must assess both *visual fidelity* and *scientific accuracy*. Visual fidelity asks whether an artifact is readable, aesthetically coherent, and consistent with publication conventions. Scientific accuracy asks whether the artifact faithfully represents the underlying data, method, comparison, or mathematical relation. The distinction is crucial: an AI-generated figure may look professional while containing misaligned arrows, incorrect labels, invalid quantitative relationships, or domain-specific notation errors.

Recent benchmarks increasingly target this gap. SciFlow-Bench [462] uses inverse-parsing evaluation to detect structurally incorrect but visually plausible framework figures. FigureBench [487] evaluates scientific illustration generation and refinement. PlotCraft [457] studies chart generation across diverse chart types, while SciFig [138] provides a rubric-based evaluation for pipeline figures. TeXpert [155] evaluates LaTeX generation across difficulty levels, exposing steep performance degradation on hard cases. AbGen [475] extends evaluation to ablation study design, where the challenge is not only formatting a table but selecting scientifically meaningful comparisons. *Can AI Draw Science?* [33] broadens figure-generation evaluation to

text-to-image and multimodal models, probing whether they render scientifically faithful figures rather than merely plausible images.

A further wave of benchmarks evaluates figures *against the manuscript they belong to*, which is the setting that matters for research use. SciFigQual-Bench [63] scores figure quality with full-manuscript context, and SciFigAlign [410] scores figures by fine-tuned alignment between the visual and the manuscript evidence it is supposed to depict; both make consistency with the paper, rather than standalone appearance, the object of measurement. Two adjacent capabilities are also now probed directly: whether vision-language models actually read a figure or merely pattern-match, tested by observing their behaviour when blinded or misled [260] and by broader multimodal comprehension of scientific images [65]; and whether figure provenance can be traced at all, addressed by provenance-aware plagiarism detection for scientific figures [57], which extends integrity concerns from text into the visual record.

Across artifact types, maturity remains uneven. Standard result plots are the most tractable because they can be generated from structured data and validated through executable plotting code. Method diagrams and framework figures remain harder because they require spatial organization and semantic consistency. Tables are difficult when they encode comparison logic or ablation design rather than simple formatting. Mathematical formulas, TikZ diagrams, and pseudocode exhibit steep accuracy cliffs because small syntactic errors can alter scientific meaning. Together, these benchmarks show that S4 evaluation is moving from appearance-based judgment toward structure-, semantics-, and task-aware assessment.

3.4.5 Findings and Observations

Stage 4: Tables & Figures



State & Progress

Emerging | Visualization | Small Models | Multi-Agent | Benchmarks | Editing

- Fastest-growing stage from zero: first dedicated tools appeared in late 2025, yet 20+ systems already span figures, tables, formulas, and editing.
- Standard data visualization becomes increasingly tractable: 90%+ execution pass rate on Matplotlib/Seaborn [255], with multi-agent approaches boosting quality $> 40\%$ over baselines [48].
- Domain-specific training and iterative refinement can make smaller specialized models competitive for structured visual languages such as TikZ [17, 102].



Gaps & Limitations

Correctness | Uneven | Tables | Formulas | Spatial | Symbols

- Visual plausibility $\neq$ scientific correctness: generated artifacts may look polished while misrepresenting data, structure, notation, or information flow.
- Maturity is sharply uneven: figures are most advanced, table generation lags with no high-traction LaTeX tool, and formula accuracy drops from 78.8% to 15% with complexity [155].
- Tools remain assistants, not producers: AI-generated figures frequently require human modification for domain-specific symbols, spatial relationships, and paper-specific visual languages.

3.5 Summary and Transition: Creation

The four Creation stages are tightly coupled in practice. S1 (*Idea Generation*) generates candidate hypotheses, S2 (*Literature Review*) situates them within prior work, S3 (*Coding and Experiments*) turns them into executable implementations and empirical evidence, and S4 (*Tables and Figures*) converts the resulting outputs into visual and structured artifacts for communication. Progress across these stages shows a consistent pattern: AI systems are increasingly effective at producing the artifacts of research, including ideas, literature summaries, code, experiments, figures, and tables, but they remain less reliable at verifying whether these artifacts are novel, faithful, executable, and scientifically meaningful.

This gap appears differently at each stage. In S1, plausible novelty often weakens after implementation. In

S2, fluent synthesis can conceal citation errors or incomplete coverage. In S3, executable code can still be semantically wrong, and automated runs do not guarantee meaningful experimental design. In S4, polished visual artifacts may misrepresent data, notation, or methodological structure. These failure modes suggest that Creation-stage automation is most credible when coupled with grounding, execution feedback, explicit verification, and human scientific judgment.

The outputs of Phase 1 (*Creation*) constitute the raw material for Phase 2 (*Writing*). Ideas, retrieved literature, validated experiments, statistical summaries, comparison tables, and scientific figures must be organized into a coherent manuscript that explains the contribution, justifies its significance, and prepares it for external scrutiny. We therefore next turn to *Writing*, where AI assistance shifts from producing research artifacts to structuring evidence into a scholarly argument.

4 Phase 2: Writing

This phase consists of a single stage: *Paper Writing* (S5). Writing merits its own phase because it transforms the artifacts produced in Phase 1 into a scholarly argument. It is not merely a formatting step: a manuscript must select evidence, structure claims, situate contributions in the literature, explain methods with sufficient detail for reproducibility, and anticipate objections before external scrutiny in Phase 3 (*Validation*). Compared with Phase 1 (*Creation*), which emphasizes artifact production, Phase 2 (*Writing*) emphasizes rhetorical organization and evidential justification.

This distinction matters for AI-assisted research. Writing tools are among the most widely adopted systems in the AI-for-research ecosystem, spanning grammar correction, sentence polishing, section drafting, citation support, and full-paper generation. At the same time, Writing is one of the most ethically sensitive phases because questions of authorship, attribution, disclosure, and the boundary between assistance and generation remain unresolved. The central challenge is therefore not whether AI can produce fluent academic prose, but whether it can preserve factual grounding, argumentative depth, citation fidelity, and human accountability.

4.1 Paper Writing

AI-assisted writing has moved from occasional support to mainstream research practice. Large-scale corpus analyses estimate detectable AI modification in up to 17.5% of computer science abstracts [197] and 13.5% of biomedical abstracts [166], while self-reported adoption is higher: a 2025 Nature survey found that more than half of researchers report seeking AI writing help [249]. These measurements are imperfect, but together they indicate a clear shift: AI writing assistance is now embedded in everyday scientific workflows. This makes the quality, transparency, and governance of AI-assisted writing increasingly important.

We organize this stage along a spectrum of increasing autonomy: *semi-automated assistance* (Section 4.1.1), in which AI handles local operations such as polishing, citation formatting, and drafting while the researcher retains control of the argument; *fully automated generation* (Section 4.1.2), in which systems attempt end-to-end manuscript production; and the *assessment* of writing quality and AI detection (Section 4.1.3). What separates these regimes is not surface fluency, which current models already supply, but the degree to which they preserve the author's evidential and argumentative judgment. As assistance shades into generation, the unresolved boundary between acceptable support and unattributed authorship turns paper writing from a largely technical task into a question of disclosure and accountability, a theme we revisit when discussing governance in Section 7.

A comprehensive inventory of AI-assisted writing systems is provided in Table 8 (Appendix).

4.1.1 Semi-Automated Writing Assistance

Semi-automated writing assistance supports different parts of the manuscript workflow, from planning and drafting to polishing and revision. At the planning stage, systems help generate titles, outlines, section structures, and citation suggestions. ScholarCopilot [385], for example, trains LLMs for academic writing with integrated citation recommendation, reflecting a broader trend toward tools that combine text generation with literature grounding.

During drafting, commercial tools such as Grammarly, Writefull, Paperpal, and GPT-based editors support paragraph generation, sentence polishing, citation insertion, and style refinement. Open-source prompt templates [193] provide lightweight alternatives, while CoAuthor [174] studies human–AI collaborative writing workflows. The dominant paradigm is increasingly shifting from *"AI writes for you"* to *"AI writes with you"*: AI handles mechanical or local operations, such as polishing, citation formatting, and initial drafting, while researchers retain responsibility for novelty, argumentation, experimental interpretation, and scientific judgment.

Editor-integrated systems make this collaboration more explicit. PaperDebugger [128] embeds a multi-agent system into Overleaf, running Reviewer, Enhancer, Scoring, and Researcher agents within the writing environment. In a similar vein, PaperMentor [210] acts as a human-centered multi-agent writing tutor that delivers Overleaf-native inline comments on research drafts while leaving authoring to the writer. A complementary line of work emphasizes cognitive engagement and transparency. Script&Shift [330] structures AI-assisted writing around *source transformation* rather than direct text generation, aiming to preserve the writer's active reasoning. DraftMarks [332] provides visual traces of revision intensity and AI-generated content, making the human–AI writing process more transparent to readers and reviewers. Empirical evidence [331] further suggests that purposeful AI support can assist student writing without fully displacing cognitive effort.

Post-writing systems focus on revision, consistency, and style. XtraGPT [39] provides an open-source LLM suite for instruction-guided scientific paper revision, SciIG [87] benchmarks introduction writing using recent NAACL and ICLR papers, OpenDraft [61] uses specialized agents to generate long research drafts with citation support, and LimAgents [8] integrates OpenReview comments and citation networks to generate research-limitation statements. A further step closes the loop between a suggested revision and evidence that it is warranted: AutoSupervision [180] verifies proposed revisions against grounded signals from the scientific workflow instead of accepting them on fluency, and domain assistants such as ReasFlow [117] organize reasoning-centric work in applied mathematics around a knowledge structure rather than free text. Together, these systems show that semi-automated writing assistance is most credible when it augments researcher control rather than replaces the intellectual work of framing, interpreting, and defending a contribution.

4.1.2 Fully Automated Paper Generation

Fully automated paper generation attempts to move beyond local assistance toward end-to-end manuscript production. Existing systems can be grouped into three directions. First, end-to-end research systems such as The AI Scientist [223, 224] and Agent Laboratory [308] generate full papers as part of broader automated research pipelines. These systems demonstrate the feasibility of producing complete paper-like artifacts, but their outputs often remain limited by shallow argumentation, weak experimental validation, or insufficient novelty.

Second, benchmarked paper-generation systems aim to approach human review standards. CycleResearcher [394] reports generated papers scoring 5.36 on the ICLR scale, approaching but still below the reported accepted-paper average of 5.69. This gap is important because it suggests that the main bottleneck is no longer surface fluency alone. Rather, near-threshold papers often lack the argumentative depth, experimental rigor, and reviewer anticipation that distinguish publishable work from plausible drafts.

Third, rubric-guided and section-specific systems improve parts of the manuscript rather than generating the entire paper from scratch. APRES [469] discovers rubrics predictive of citation counts and revises papers accordingly, with human experts preferring revised papers 79% of the time. FutureGen [7] targets "Future Work" section generation, and LECTOR [403] targets introduction generation, jointly optimizing a scientific reasoning graph with the generated text to ground citations and curb hallucinated references. PaperWritingBench [338], introduced as part of the PaperOrchestra framework, provides a dedicated benchmark for automated paper writing by evaluating multi-agent systems against reverse-engineered top-tier conference papers. RWGBench [406] narrows the lens to related-work generation, evaluating whether generated related-work sections correctly position a paper against prior scholarship. These systems indicate that automated writing is increasingly measurable, but also reinforce that high-quality papers require more than fluent text: they require evidence-grounded reasoning and coherent scientific contribution.

4.1.3 Assessment: Writing Quality and AI Detection

Assessment of AI-assisted writing involves two related but distinct questions: whether AI use can be detected, and whether the resulting manuscript is scientifically strong. Detection remains unreliable as a governance mechanism. Current detectors can produce unacceptable false positives, especially for formal, non-native, or highly edited academic prose, motivating a shift at major venues from attempting to *detect* AI use toward requiring authors to *declare* AI use. Watermarking offers a more principled route under controlled settings [288], but it requires model-provider cooperation and remains vulnerable to paraphrasing, translation, and post-editing. A related concern is the integrity of AI-generated citations rather than AI authorship per se: multi-agent detectors such as CiteTracer [186] classify each reference as genuine, potential, or hallucinated, targeting fabricated citations directly.

Quality evaluation is more important, but also harder. Good academic writing must be assessed along multiple dimensions: factual correctness, citation accuracy, argumentative coherence, methodological completeness, novelty of framing, and stylistic appropriateness. LLM-as-Judge frameworks are increasingly used to approximate parts of this evaluation. CycleReviewer [394] reports a 26.89% reduction in Proxy MAE relative to individual human reviewers for score prediction, while the Stanford Agentic Reviewer [149] achieves review-score correlations comparable to human inter-rater agreement ($\rho = 0.42$ vs. human $\rho = 0.41$). These results suggest that automated evaluators can provide useful review-style signals, but they should not be treated as substitutes for expert assessment: score prediction and agreement metrics only partially capture factual grounding, evidential rigor, novelty, and scientific contribution. Complementing such outcome-based evaluation, process-oriented analyses [333] study how researchers actually revise AI-generated versus human-written drafts, finding that AI assistance helps most at the sentence level while struggling with global coherence.

SciSlopBench [262] targets failures in the connections between a manuscript's structure, argument, and artifacts, using 390 matched AI-generated and human-written paper pairs. Its companion SciSlopHarness constrains revisions by the available experimental records, distinguishing evidence-supported repair from direct optimization of a writing-quality score. This extends evaluation from local fluency to the consistency of the scientific argument. However, the corpus is predominantly computer science and covers specific generators, so its measures should not be treated as universal authorship detectors or evidence of misconduct.

The central failure mode of AI writing is therefore not ungrammatical prose, but unsupported persuasion: text that is fluent, well-structured, and citation-like, yet insufficiently grounded in evidence or scientific judgment. This issue is amplified by the productivity–quality divergence observed in recent studies: AI use can increase publication output, but AI-assisted papers with complex language may be less likely to be accepted [170]. As in Phase 1 (*Creation*), greater artifact production does not necessarily imply stronger research.

From Detection to Accountability. A counter-position to the detection framing appeared this window. Feng *et al.* [75] argue that treating LLM-assisted writing primarily as a detection problem misframes it, because the observable stylistic patterns do not establish whether the underlying science is better or worse; they propose evaluating manuscripts on scholarly quality and accountability rather than on inferred tool use. This aligns with the governance direction developed in Sec. 7.4: as assistance becomes ubiquitous, disclosure and responsibility are more tractable levers than classification, whose error rates fall hardest on non-native speakers.

4.1.4 Findings and Observations

Stage 5: Paper Writing



State & Progress

Commercial | Near Threshold | Cognitive | Collaboration | Rubric-Guided | Transparency

- Paper writing is among the most widely adopted AI-assisted research stages, with tools supporting planning, drafting, polishing, citation assistance, and manuscript revision [197, 249].
- Strong automated systems score 5.36 on the ICLR scale (*vs.* 5.69 accepted [394]); rubric-guided revision achieves 79% expert preference [469]. The gap to acceptance is argumentative depth, not fluency.
- Cognitive engagement and transparency are emerging design principles, aiming to preserve human understanding rather than merely producing polished text [330, 332].



Gaps & Limitations

Paradox | Mediocrity | Detect Failing | Shallow Args | Attribution | Deskilling

- Productivity and quality can diverge: AI-assisted workflows may increase output, but more fluent or complex language does not necessarily translate into stronger acceptance outcomes [170].
- Valley of mediocrity: papers are fluent enough to look real but lack argumentative depth, experimental rigor, and reviewer-anticipation, the skills that separate publishable from near-publishable work.
- Detection tools have unacceptable false-positive rates, forcing a shift from "detect" to "declare" policies, while 17.5% of CS papers already carry detectable AI modification [197].

4.2 Summary and Transition: Writing

This phase shifts the focus from producing research artifacts to organizing them into a scholarly argument. S5 (*Paper Writing*) takes the outputs of Phase 1, including ideas, retrieved literature, experiments, figures, and tables, and converts them into a manuscript that explains what was done, why it matters, and how the evidence supports the claims. Progress in this phase shows that AI systems are increasingly effective at assisting the writing workflow, from planning and drafting to polishing, citation support, revision, and even full-paper generation.

The central limitation is that fluent writing can conceal weak reasoning. AI-generated or AI-assisted text may improve readability and productivity while leaving deeper scientific requirements unresolved: whether claims are grounded, whether citations faithfully support them, whether experiments are sufficient, and whether the contribution is argued with appropriate nuance. This limitation appears both in semi-automated writing assistance and in fully automated paper generation. The former is most credible when it preserves researcher control over framing, interpretation, and final responsibility; the latter increasingly approaches reviewable quality in selected settings, but still struggles with argumentative depth, evidential rigor, and reviewer anticipation.

The output of this phase is a manuscript ready for external scrutiny. We therefore next turn to Phase 3 (*Validation*), where the manuscript is evaluated through peer review and revised through author rebuttal. This transition shifts the role of AI from structuring evidence into a coherent argument to assessing whether that argument is sound, fair, and sufficiently supported.

5 Phase 3: Validation

This phase encompasses the stages through which a manuscript produced in Phase 2 (*Writing*) is externally scrutinized and iteratively refined: peer review and rebuttal with revision. Together, these stages address a different question from *Creation* or *Writing*: *does this contribution meet the epistemic standards of the field?*

Validation is distinct because it introduces adversarial evaluation by reviewers who are expected to identify unsupported claims, methodological flaws, missing comparisons, unclear writing, and insufficient novelty. This

makes Phase 3 a high-stakes setting for AI assistance. Automated systems can help summarize manuscripts, draft reviews, synthesize reviewer opinions, identify weaknesses, and support rebuttal preparation, but they also risk amplifying leniency, bias, adversarial manipulation, and weakly grounded criticism. The central challenge is therefore not whether AI can produce review-like text, but whether it can support fair, critical, and evidence-grounded evaluation without replacing independent expert judgment. The two stages also form a feedback loop: reviewer critiques in S6 (*Peer Review*) may require additional experiments in S3 (Coding and Experiments), revised figures in S4 (*Tables and Figures*), or manuscript rewrites in S5 (*Paper Writing*), while rebuttal and revision in S7 (*Rebuttal and Revision*) determine how those critiques are addressed.

5.1 Peer Review

Peer review is the gateway to validation. It evaluates whether a manuscript is technically sound, sufficiently novel, clearly presented, and supported by appropriate evidence. Existing systems span automated review generation, meta-review drafting, reviewer–paper matching, and review quality assessment. Across these directions, the central limitation is that LLMs can often produce structured and plausible critiques, but may under-detect methodological flaws, over-score weak submissions, and remain vulnerable to adversarial manipulation.

A comprehensive inventory of automated review systems is provided in Table 9 (Appendix).

5.1.1 Automated Review Generation

Automated review generation aims to produce structured critiques of manuscripts, including summaries, strengths, weaknesses, questions, and rating recommendations. Existing approaches can be grouped into four broad families. *Fine-tuned reviewer models* specialize LLMs on expert review data to improve domain alignment and review format. DeepReviewer-14B [486] reports strong performance against GPT-o1 and on ICLR 2024 accept/reject prediction, while OpenReviewer [142] fine-tunes Llama-8B on 79K expert reviews. These systems show that supervised review data can improve review-style generation, but acceptance prediction remains a narrow proxy for review quality.

Multi-agent review systems decompose reviewing into specialized roles. MARG [60] uses multi-agent collaboration to generate multiple substantive comments per manuscript, the open-source ai-peer-review tool [273] uses multiple LLMs to produce independent reviews followed by meta-review synthesis, and ScholarPeer [101] extends this paradigm with literature search and claim verification. Such decomposition is useful because high-quality reviewing requires several distinct operations: understanding the manuscript, checking related work, evaluating claims, identifying weaknesses, and producing actionable feedback.

RL-optimized review systems attempt to improve review quality through more explicit training signals. REMOR [351] optimizes review generation with multi-objective reinforcement learning, while ReviewRL [451] combines retrieval-augmented context with RL to produce more comprehensive and grounded reviews. *Prompt-based systems* provide a lighter-weight alternative. Reviewer2 [86] introduces a two-stage framework that models the distribution of review aspects, while ChatReviewer [254] provides a deployed ChatGPT-based tool for analyzing strengths, weaknesses, and possible improvements. Moving beyond passive generation, ProReviewer [71] frames reviewing as a proactive investigation, with the agent actively probing suspicious parts of a manuscript before issuing critiques. PeerCheck [51] instead targets review *quality*, aiming to raise LLM-generated academic reviews toward human-level standards. A parallel line keeps the human as the decision-maker and uses AI earlier or alongside: local models can pre-screen submissions ahead of human triple-blind review in health sciences, keeping sensitive manuscripts on-premise [23], and conversational voice annotation lets reviewers dictate and structure critiques rather than receive generated ones [99]. Community-level analysis is beginning to connect these choices to policy, mapping reviewer AI policies across research communities against the resulting review quality [77]. A complementary direction evaluates reviews by the repairs they enable. PaperDoctor [202] produces evidence-linked diagnoses through typed verifiers and selective experiment reruns, pairing each finding with a revision suggestion. ActReview [235] learns actionable critiques from review–rebuttal threads, using author responses as supervision for grounded diagnoses and revision plans. Although rebuttals supply its training signal, its output is a review rather than a rebuttal letter, and human evaluation still reveals a technical-accuracy gap. These systems shift the target from plausible criticism toward feedback an author can inspect and act on.

Overall, automated review generation has become increasingly structured, but review-like prose should not be mistaken for reliable validation: the core difficulty is whether critiques are accurate, calibrated, and grounded in the manuscript and relevant literature.

5.1.2 Meta-Review Generation

Meta-review generation synthesizes multiple reviewer opinions into a coherent area-chair-style assessment. This task differs from single-review generation because it must compare reviewer concerns, resolve disagreements, identify consensus, and justify a final recommendation. Bhatia *et al.* [127] evaluate GPT-3.5, LLaMA-2, and PaLM-2 on composing meta-review drafts from 40 ICLR papers, finding that LLMs are useful for multi-perspective summarization but struggle with nuanced judgment calls. AgentReview [153] simulates the full review lifecycle, including meta-review and final decisions, and shows that social influence and authority bias can affect outcomes.

The main challenge is not summarization alone, but decision-making under disagreement. When reviewers fundamentally disagree about a manuscript's contribution, LLMs often produce diluted compromises rather than taking a defensible substantive position. This limitation reflects a broader issue in AI-assisted validation: current systems are better at consolidating stated opinions than at independently adjudicating technical disputes.

5.1.3 Reviewer Matching

Reviewer–paper matching supports the editorial process by assigning manuscripts to reviewers with relevant expertise while accounting for conflicts of interest. This task is less visible than review generation, but it is crucial for review quality at scale: even a well-written review is less useful if the reviewer lacks appropriate domain expertise. RelevAI-Reviewer [54] has been deployed at major venues, while RATE [213] improves expertise-based matching through profile distillation, aiming to capture a reviewer's competence signature beyond keyword overlap. MERIT [437] likewise targets reviewer assignment, using rubric-informed training to model expertise and retrieve suitable reviewers at scale.

Compared with automated reviewing, reviewer matching is a more appropriate setting for operational AI support because it assists the allocation process rather than replacing expert judgment. However, matching systems still require transparent conflict handling, robust expertise modeling, and human oversight, especially in interdisciplinary areas where surface-level keyword similarity can be misleading.

5.1.4 Assessment: Review Consistency, Bias, and Robustness

Assessing AI-assisted peer review requires more than measuring whether generated reviews resemble human reviews. A useful review must be consistent, critical, fair, grounded, and robust to manipulation. On consistency, recent systems show measurable progress. The Stanford Agentic Reviewer [149] achieves Spearman correlation of 0.42, comparable to human–human correlation of 0.41. ReviewAgents [84] uses a multi-agent framework trained on a Review-CoT dataset of 37,403 papers and 142,324 reviews. The reviewer component reported in the Nature version of The AI Scientist [224] reaches 69% balanced accuracy on ICLR acceptance prediction, while ClaimCheck [263] evaluates whether LLM critiques are grounded in the reviewed manuscript. ReViewGraph [191] further models multi-round reviewer–author debates as heterogeneous graphs, improving debate outcome prediction without LLM fine-tuning. PRISM [220] introduces a multi-dimensional benchmark for evaluating LLM peer reviewers, finding that models can excel on individual aspects yet none matches balanced human performance.

Claim-level verification provides a more direct test of whether a critique is grounded. Peerify [58] decomposes review comments into atomic claims and retrieves manuscript evidence, with an 800-claim benchmark and a 300-claim human audit. HalluPeer [204] complements this with a review-specific hallucination taxonomy and detection, classification, and localization tasks. Its injected hallucinations provide controlled tests, while checks on authentic reviews probe external relevance. In both cases, support within the manuscript is a narrower criterion than the truth of the underlying scientific claim, and interpretive criticism remains difficult to adjudicate.

However, consistency is not sufficient. A reviewer can be consistent while being systematically lenient, biased, or shallow. LLM-based reviewers have been shown to assign inflated scores relative to humans, and in some settings to misclassify rejected papers as acceptable [484]. Direct comparisons continue to sharpen the picture of where the gap lies [27], and a related benchmark asks whether an LLM judge is exercising genuine analysis or articulating intuition it cannot justify [40]. A large-scale study in which 45 expert scientists assessed the reviews of Nature-family papers [162] finds that AI reviewers can surpass top human reviewers on some dimensions while still exhibiting recurring, systematic weaknesses; relatedly, the apparent tendency of LLMs to favor LLM-written papers has been shown to stem largely from general leniency toward low-quality work rather than a genuine self-preference effect [319].

Robustness to manipulation is now a documented failure mode rather than a hypothetical one. Because AI reviewers read the manuscript as text, they can be steered by how a claim is phrased instead of whether it holds: rhetorical sensitivity has been shown to reward-hack AI reviewers, with confident framing shifting assessments independently of substance [187]. This is the review-stage instance of the pattern that recurs throughout this survey, in which a system optimizes a proxy that correlates with quality only while nobody is optimizing against it. It also has a concrete governance consequence: any deployment that lets authors see and iterate against an AI reviewer converts that reviewer into an optimization target. Evaluation is meanwhile extending to review-adjacent artifacts, including reviewer-aligned benchmarks for judging surveys [464]. This makes standalone AI review risky: it may produce coherent critiques while failing to identify decisive methodological weaknesses. A more credible deployment mode is to use LLMs to improve human reviews rather than to replace reviewers. In a randomized ICLR 2025 study across 22,467 reviews, LLM feedback on reviews improved review quality in 89% of cases, with reviewers updating their reviews 26.6% of the time, without affecting acceptance rates [360]. Chen *et al.* [45] further study how reviewers engage with AI-generated feedback during a live ICLR 2025 process, and Zhuang *et al.* [491] provide a broader taxonomy of automated scholarly paper review methods.

Robustness and governance remain major concerns. The "AI Review Lottery" [298] estimates that at least 15.8% of ICLR 2024 reviews were AI-assisted, with 49.4% of submissions receiving at least one AI-assisted review. A 2025 Nature survey similarly reports that many academics use AI in peer review despite restrictive venue policies [249], and a major 2026 conference rejected 497 papers for AI-use policy violations [93]. The AAAI-26 AI Review Pilot [19] reports one of the first large-scale controlled deployments of AI-assisted review at a top venue, offering early evidence on how such assistance behaves at scale. These trends indicate that AI involvement in peer review is already widespread, while enforceable governance remains immature.

Field evidence now distinguishes policy assignment from actual tool use. Kim *et al.* [163] report a randomized experiment on LLM-use policies within ICML 2026 and an anonymous post-survey with 1,486 responses. Assignment had near-zero effects on final decisions, scores, and reviewer confidence, but the survey found substantial noncompliance under both policy regimes. These estimates therefore concern the effect of assigning a policy, not the effect of a specified form of LLM assistance under full compliance. This distinction limits claims that permissive policies either improve or leave unchanged the substantive quality of peer review.

Adversarial manipulation further complicates deployment. Breaking the Reviewer [203] studies adversarial robustness of LLM-based review assessments, while Keuper [159] shows that simple prompt injections, such as white text on a white background, can manipulate LLM reviews. Ye *et al.* [441] show that covert content injection can substantially raise review scores and that manipulating a small fraction of reviews can alter rankings. Zhou *et al.* [483] further demonstrate that in-paper prompt injection can raise LLM scores under static and iterative attacks. Yang *et al.* [428] show that manipulation need not rely on hidden prompts at all: presentation-only revisions that merely rephrase and repackage a paper can substantially raise AI review scores. At the lexical level, Raina *et al.* [285] show that benign adjectives can function as adversarial triggers, while Sahoo *et al.* [302] evaluate indirect manipulation across multiple LLMs and attack strategies, with frontier models beginning to show stronger resistance in some settings. Finally, detection-based policy enforcement remains fragile: Saha *et al.* [301] show that state-of-the-art AI text detectors misclassify LLM-polished reviews, and Yu *et al.* [445] evaluate 18 detection algorithms on 788,984 AI-written peer reviews, highlighting the difficulty of identifying AI-generated review text at the individual-review level. Beyond text detection, Li *et al.* [184] show that AI-assisted reviews can be actively *gamed* with high attack success rates, and Russinovich *et al.* [296] document hallucinated "phantom" citations that survive peer review at top-tier venues.

5.1.5 Findings and Observations

Stage 6: Peer Review



State & Progress

Deployment | Consistency | Mapped Risks | Multi-Agent | RL-Trained | Matching

- The strongest validated deployment mode is LLM feedback on *reviews*, not standalone AI review: in ICLR 2025, review feedback improved quality in 89% of cases without affecting acceptance rates [360].
- Automated reviewers can approach human-level consistency on selected metrics, with the Stanford Agentic Reviewer matching human inter-rater agreement ($\rho = 0.42$ *vs.* 0.41) [149].
- Reviewer matching and meta-review generation are promising support tasks because they assist editorial coordination and opinion synthesis rather than directly replacing expert judgment.



Gaps & Limitations

Leniency | Fragility | Policy | Inflation | Injection | Undetectable

- Standalone AI-generated review remains unsafe: LLMs assign inflated scores (AI 6.86 *vs.* human 5.70 [484]), misclassifying 95.8% of rejected papers as acceptable.
- Adversarial fragility persists: prompt injection reaches 10 scores [483]; benign adjectives function as universal triggers [285]; 5% manipulation flips 12% of rankings [441].
- Governance is difficult because AI-assisted reviewing is already prevalent [298], yet all five SOTA detectors fail on polished reviews [301]. Prevalence has outpaced governance.

Grounding Meta-Review and Error Detection in Document Change. Two resources move review automation away from opinion generation and toward verifiable document work. Metag [348] targets meta-reviewing by tying 349 curated action items to the actual differences between paper versions across the review–rebuttal cycle, so a meta-review claim can be checked against what changed rather than judged as prose. VERA-RL [189] formulates scientific error detection as reinforcement learning over a Reason–Verify–Scan progression, with VERA-13K supplying 12,900 samples in 4,300 matched chains across six error categories — notably, framing the task as *proactive* detection rather than reactive assessment of a submitted review. Both narrow the gap this survey flags between AI-on-reviews, which is validated, and AI-as-reviewer, which is not.

5.2 Rebuttal and Revision

Rebuttal and revision form the second stage of Phase 3 (*Validation*), where authors respond to external critique and revise the manuscript before a final decision or camera-ready submission. This stage is epistemologically important because it is the only point in the publication process where authors engage directly with reviewers' objections. Existing work spans reviewer-comment analysis, automated rebuttal generation, and revision tracking. Across these directions, the central challenge is not merely generating persuasive responses, but ensuring that rebuttals are evidence-grounded, faithful to the manuscript, and followed by actual revisions.

A comprehensive inventory of rebuttal systems is provided in Table 10 (Appendix).

5.2.1 Reviewer Comment Analysis

Reviewer-comment analysis decomposes critiques into actionable concerns, such as missing experiments, unclear motivation, insufficient baselines, unsupported claims, or presentation issues. This analysis is a prerequisite for effective rebuttal generation because reviewer comments are often long, mixed in priority, and partially overlapping across reviews. ReviewMT [353] models peer review as multi-turn, long-context dialogue with role-based interactions, covering 26,841 papers and 92,017 reviews from ICLR and Nature Communications. Re^2 [454] further provides a consistency-ensured dataset for full-stage peer review and multi-turn rebuttal, covering 19,926 submissions, 70,668 reviews, and 53,818 rebuttals from 24 conferences. A complementary line treats author responses as a supervision signal: RbtAct [399] uses rebuttals to train models that generate actionable review feedback, while GoodPoint [248] learns to produce constructive paper

feedback grounded in author responses.

Empirical studies show that rebuttal can materially affect outcomes, especially for borderline submissions. Analysis of ICLR 2024–2025 [156] reports that 75–81% of scores remain unchanged after rebuttal, 17–23% improve, and only approximately 1% decrease, with the most common transition being $5 \rightarrow 6$ from borderline to acceptable. A complementary analysis [222] finds that rebuttals do move review scores, but that the structure of the initial reviews bounds how much movement is possible; and as AI-generated reviews become common, Leblanc *et al*. [173] study how authors decide whether to trust and respond to AI-based reviews. These numbers suggest that rebuttal is not a universal remedy, but it is consequential for the subset of papers where reviewer concerns can be clarified, corrected, or supported with additional evidence.

5.2.2 Automated Rebuttal Generation

Automated rebuttal generation attempts to produce author responses that address reviewer concerns clearly and strategically. Early systems treated rebuttal as direct text generation, but this formulation is prone to hallucination, missed reviewer points, and unverifiable claims. More recent systems, therefore, decompose rebuttal into intermediate steps such as concern extraction, evidence retrieval, response planning, and final generation. RebuttalAgent [119] uses Theory-of-Mind modeling to craft strategically persuasive responses, reporting an average 18.3% improvement over the base model. Paper2Rebuttal [234] introduces evidence-centric planning by decomposing reviewer comments into atomic concerns and retrieving supporting literature, improving Coverage and Specificity by up to +0.78 and +1.33. ReviewerToo [303] includes a rebuttal module within a broader modular framework and reports 81.8% accept/reject accuracy.

A newer direction emphasizes planning and author control. DRPG [112] proposes a four-step Decompose–Retrieve–Plan–Generate pipeline, reporting 98%+ planning accuracy and stronger-than-average human rebuttal quality with an 8B model. Author-in-the-Loop [295] integrates author expertise and intent into response generation, aiming to ensure that rebuttals reflect the paper's actual contributions rather than generic LLM output. Defend [160] generates rebuttals with minimal author guidance, reporting improved factual correctness and refutation strength while keeping the author in the loop. These systems indicate that rebuttal automation is moving from fluent response generation toward evidence-grounded and author-aware revision support.

5.2.3 Assessment: Rebuttal Effectiveness

Assessing rebuttal systems requires measuring both immediate effectiveness and downstream accountability. Immediate effectiveness concerns whether a rebuttal addresses reviewer concerns, clarifies misunderstandings, provides evidence, and improves reviewer confidence. ICLR 2024–2025 analysis shows that papers whose scores improve after rebuttal achieve acceptance rates of 55.7–57.6%, compared to 7.8–12.4% for papers with unchanged scores [156]. This makes rebuttal especially important for borderline papers, where small changes in reviewer confidence can affect final outcomes.

However, effective rebuttal is not only persuasion. Many reviewer requests require new experiments, additional ablations, corrected figures, or manuscript restructuring, creating a feedback loop from S7 back to S3, S4, and S5. Ruan and Gurevych [295] provide large-scale aligned review–response–revision triplets, enabling the study of how rebuttals translate into actual manuscript changes. Newer systems begin to execute this feedback loop. AppliedScientist [366] couples iterative manuscript revision to independently generated AI reviews and also uses a separate automated evaluator to assess later drafts. It reports resolving 128 of 150 execution-related weaknesses but only 2 of 18 idea-related weaknesses, indicating that implementation and experimental repairs are more tractable than changing concerns about novelty or significance. ScientistTwo [250] similarly incorporates targeted experiments within simulated review and rebuttal (Section 7.1). These advances narrow the execution gap, but automated score improvements do not establish successful revision under independent human review.

The second evaluation dimension is accountability. A recent audit [31] finds that ICLR 2025 authors make an average of 11.8 commitments per paper during rebuttal, but approximately 25% of these commitments are not fulfilled in the camera-ready version, with missing experiments among the most common unfulfilled promises. This gap exposes the same capability-versus-integrity tension observed throughout the survey: AI systems may

generate plausible and persuasive responses, but the scientific validity of a rebuttal depends on whether its claims are supported and its promises are later implemented. Rebuttal systems should therefore be evaluated not only by response quality, but also by concern coverage, evidence grounding, revision traceability, and fulfillment of author commitments.

5.2.4 Findings and Observations

Stage 7: Rebuttal & Revision



State & Progress

Newest | Human-Level | Decomposition | Decisive | Evidence | Planning

- Rebuttal automation is emerging as a distinct stage, with recent systems moving from direct response generation toward decomposition, evidence retrieval, response planning, and author-aware generation.
- Rebuttal is consequential for borderline submissions: 17–23% of ICLR 2024–2025 submissions improve scores after rebuttal, and improved-score papers achieve much higher acceptance rates [156].
- Evidence-centric planning helps address common failures of direct generation, including missed reviewer concerns, unsupported responses, and generic rebuttal text [119, 234].



Gaps & Limitations

No New Expts | Commitment | Overlooked | Accountability | Only 10 Tools | Loop Gap

- Current systems cannot reliably generate new experimental evidence in response to reviewer requests; the S7 (*Rebuttal and Revision*) →S3 (*Coding*) feedback loop remains a major unautomated gap.
- The quality of the rebuttal process must be tied to revision fulfillment: approximately 25% of ICLR 2025 rebuttal commitments are not fulfilled in camera-ready versions [31].
- Rebuttal remains under-served relative to its practical importance, despite being the stage where authors directly negotiate reviewer concerns before final decisions.

5.3 Summary and Transition: Validation

This phase shifts the focus from constructing a manuscript to testing whether its claims withstand external scrutiny. S6 (*Peer Review*) evaluates the manuscript through independent critique, while S7 (*Rebuttal and Revision*) gives authors an opportunity to clarify, defend, and revise the work in response. Together, these stages form a feedback loop rather than a one-way checkpoint: reviewer comments can trigger new experiments, revised analyses, updated figures, and substantial manuscript rewriting.

Progress in Validation shows a consistent pattern. AI systems are increasingly capable of producing review-like critiques, summarizing reviewer opinions, supporting reviewer matching, and drafting rebuttal responses. However, the hard part of validation is not producing plausible evaluative text; it is making fair, critical, evidence-grounded judgments and ensuring that critique leads to accountable revision. In peer review, standalone AI reviewers may be consistent but lenient, biased, or vulnerable to manipulation, while the strongest validated deployment mode is to use AI to improve human reviews. In rebuttal, AI can help decompose concerns and draft evidence-aware responses, but cannot yet generate missing experimental evidence or guarantee that author commitments are fulfilled.

The output of this phase is a manuscript that has been externally challenged, defended, and revised. Once validated, the research must be communicated beyond the review process through posters, slides, videos, project pages, and social media. We therefore next turn to Phase 4 (*Dissemination*), where AI assistance shifts from evaluating scholarly claims to adapting validated research artifacts for different audiences, formats, and communication goals.

6 Phase 4: Dissemination

This phase converts the validated manuscript into formats accessible to audiences beyond specialist venue readers. The discussion covers the transformation of papers into posters, slides, videos, social media posts, and interactive agents. Compared with earlier phases, Dissemination is less about producing or validating scientific claims than about adapting those claims to different audiences, media, and interaction modes.

Dissemination merits a separate phase because its outputs are independent knowledge artifacts rather than simple derivatives of the paper. A poster must compress the contribution into a single visual narrative; a slide deck must support oral explanation; a video must synchronize visual, textual, and spoken channels; a social media post must balance accessibility with precision; and an interactive agent must expose the paper's methods for downstream use. The central challenge is therefore not whether AI can reformat a paper, but whether it can preserve scientific fidelity while adapting the work to new modalities, audiences, and levels of interactivity.

6.1 Research Dissemination (Paper2X)

Research dissemination converts a completed paper into audience-adaptive artifacts. Unlike Phases 1–3, which primarily target specialist authors, reviewers, and readers, Paper2X outputs must serve diverse audiences: conference attendees, oral-session audiences, online readers, prospective users, journalists, practitioners, and future researchers who may interact with the work through tools rather than text. Existing systems cover poster generation, slide generation, narrated video and talk generation, social media and web-page generation, and emerging paper-to-agent conversion. Across these formats, the core bottleneck is trust: researchers may use AI to draft public-facing materials, but they remain reluctant to delegate final communication to systems that may distort results, overstate claims, or omit important limitations.

A comprehensive inventory of Paper2X systems across poster, slides, video, web, social media, and agentic formats is provided in Table 11 (Appendix).

6.1.1 Paper to Posters

Paper-to-poster generation transforms a full manuscript into a compact visual narrative. This requires more than summarization: the system must select the central message, allocate space across motivation, method, results, and conclusion, preserve key figures and tables, and arrange them into a readable layout. Compared with slide generation, poster generation has stronger spatial constraints because all content must be legible and coherent on a single canvas.

Early systems established agentic poster generation as a feasible task. Paper2Poster [267] introduces PosterAgent with binary-tree layout planning and a Painter–Commenter feedback loop, showing that poster generation can be decomposed into layout construction, rendering, and critique. Subsequent systems add stronger design and hierarchy awareness. PosterGen [466] incorporates aesthetic-aware multi-agent generation, PosterForest [53] uses hierarchical multi-agent collaboration, and P2P [345] introduces P2PInstruct with specialized agents and instruction data for poster design.

Recent systems move from one-shot poster creation toward editing and unified poster manipulation. APEX [323] supports interactive poster editing with fine-grained control, addressing the practical need for human post-editing in conference preparation. PosterOmni [43] unifies multiple poster tasks, including rescaling, filling, extension, layout-driven generation, style-driven generation, and identity-driven generation, while PosterCraft [44] further explores quality-aware poster generation in a unified framework. Any2Poster [369] broadens the input side, generating posters from heterogeneous source modalities across multiple domains, and PosterMELD [130] pushes on controllable design diversity with editable outputs, treating the space of acceptable layouts rather than a single target as the objective. Together, these systems suggest that poster automation is shifting from direct paper summarization toward editable, design-aware poster production.

PROS [230] extends interaction to the point before a user knows which edit to request. It identifies source-grounded poster problems, while users choose which diagnoses become repair goals and whether proposed PPTX changes are committed. PROS-Bench separates diagnosis, target resolution, and realized quality: although mean assessed quality improves, some accepted targets deteriorate. This separation matters for

deployment, because completing an edit is not equivalent to improving the poster's communication of the science.

6.1.2 Paper to Slides

Paper-to-slides systems convert manuscripts into sequential visual narratives for oral presentation. Unlike posters, slides unfold over time and must support speaker delivery. This requires content selection, section-to-slide mapping, visual layout, speaker-note synthesis, and often iterative refinement based on rendered slide quality. The key challenge is preserving the paper's argument while changing its rhetorical structure from written exposition to spoken explanation.

Early datasets and pipelines established the task. DOC2PPT [80] provides paired document–slide data, while PPTAgent [477] generates and evaluates presentations with PPTEval across content, design, and coherence. Environment-grounded refinement then closes the gap between symbolic planning and rendered slides. DeepPresenter [478] conditions revision on rendered slide images rather than only internal reasoning traces, showing that visual feedback is important for presentation quality.

Multi-agent and interactive systems further decompose slide generation into specialized subtasks. SlideGen [199] uses agents for outlining, content mapping, arrangement, note synthesis, and iterative refinement to produce editable PPTX slides. Auto-Slides [429] targets Beamer generation with multi-agent collaboration and interactive editing. SlideTailor [452] conditions generation on user preference from a single example pair using a chain-of-speech mechanism. SeaSlides [72] instead inserts a semantic abstraction layer between paper content and rendered slide, so that layout decisions are made over meaning rather than over raw text spans. Other systems focus on task-specific capabilities: PASS [5] combines slide generation with AI audio delivery, AutoPresent [89] fine-tunes a slide-generation model on SlidesBench, Paper2Slides [122] provides one-click conversion through a multi-stage RAG pipeline, Talk to Your Slides [154] supports natural-language slide editing, and Office Raccoon [313] targets page-level editing with template and brand-guideline learning. X+Slides [38] adds an audience-conditioning dimension, tailoring slide content to whether the target audience comprises domain specialists or decision-makers. SlideLab [367] connects narrative planning, visual and layout refinement, and grounding verification, while its ConfArena evaluator assesses a presentation slide by slide from an audience-oriented perspective. Human preference and simulated-audience assessments provide complementary evidence, but neither alone measures learning by a real audience. Across these systems, the main trend is from static slide generation toward editable, feedback-aware, and user-preference-conditioned presentation design.

6.1.3 Paper to Videos and Talks

Paper-to-video and paper-to-talk systems extend dissemination from visual artifacts to multimodal explanation. These systems must coordinate slides, subtitles, narration, cursor motion, pacing, and sometimes avatar or talking-head video. This makes the task substantially harder than poster or slide generation: errors can arise not only from content selection, but also from temporal alignment, speech clarity, visual synchronization, and duration constraints. PresentAgent [325] provides an end-to-end document-to-narrated-video pipeline with synchronized slides, text-to-speech narration, and the PresentEval benchmark. Paper2Video [490] introduces a benchmark of paper–video pairs and the PaperTalker framework, decomposing video generation into slide, subtitle, cursor, and talker builders. Preacher [207] uses top-down decomposition followed by bottom-up generation with Progressive Chain of Thought across multiple research fields. PresentAgent-2 [400] generalizes this line into an agentic framework for end-to-end presentation-video generation with research grounding and interactive delivery. Complementing generation, Mondal *et al.* [244] evaluate what audiences actually *learn* from paper-to-video talks, arguing that an effective talk teaches takeaways rather than merely summarizing the paper.

Although these systems show promising progress, video remains one of the hardest Paper2X formats. Unlike posters and slides, video generation requires coordination across at least four modalities: visual slides, subtitles, speech audio, and temporal or avatar-based presentation. The resulting artifact must also remain faithful to the paper while being concise enough for viewers to follow. Current systems, therefore, work best as first-draft generators that produce synchronized presentation assets for human review, rather than final public-facing videos requiring no editing.

6.1.4 Paper to Social Media

Paper-to-social-media and paper-to-web generation aims to make research discoverable outside the publication venue. Outputs include project pages, blog posts, press-release-style summaries, short-form research posts, and X/Twitter threads. These formats require stronger audience modeling than posters or slides: a thread for ML practitioners, a lay summary for journalists, and a project-page introduction for potential users should emphasize different details, use different vocabulary, and make different assumptions about background knowledge. Paper2Web [46] converts papers into interactive multimedia-rich academic homepages and provides a benchmark for this task. ResearchStudio-Reel [404] targets this "last mile" more broadly, automating the joint production of poster, video, and blog artifacts from a single paper. More generally, researchers increasingly use general-purpose LLMs to draft online summaries, figure captions, project-page text, and social media announcements. However, dedicated research-to-social-media tools remain comparatively underdeveloped. The bottleneck is not text generation alone, but audience-adaptive fidelity: systems must simplify without distorting, emphasize contributions without exaggeration, and preserve limitations while remaining engaging.

This makes social dissemination a distinct trust problem. Public-facing outputs are often read without the paper, so any overclaim, missing caveat, or misleading comparison can shape how the work is perceived. AI assistance is therefore most credible when it provides audience-specific drafts, style variants, and claim-checking support, while leaving final messaging and factual responsibility to the authors.

6.1.5 Paper to Agents and Tools

A newer dissemination direction converts papers from static documents into interactive agents or tools. This changes the function of dissemination: instead of only explaining a contribution, the system exposes the paper's methods, code, data, or workflows through natural-language interaction. In this setting, the reader becomes a user who can query, reproduce, adapt, or extend the work.

Paper2Agent [240] exemplifies this shift by converting research papers and associated codebases into interactive paper agents. The system analyzes the paper and code, constructs a Model Context Protocol (MCP) server with tools, resources, and prompts, and iteratively tests the resulting agent so that users can interact with the paper's methods through natural language. This reframes dissemination as operational access: a paper is no longer only read, but queried and executed.

Related systems broaden this idea from paper-specific agents to tool-using scientific agents. Gao *et al.* [82] study how scientific tool ecosystems can democratize AI scientists by exposing computational capabilities through agent-accessible interfaces. ProteinMCP [416] applies an MCP-based agentic framework to protein engineering, illustrating how domain-specific workflows can be wrapped into interactive, tool-using systems. I-WebGenBench [59] complements these by evaluating interactivity in LLM-generated scientific web applications, probing whether generated interfaces are genuinely usable rather than static pages. These systems suggest that future dissemination may increasingly involve executable interfaces, reproducible workflows, and domain agents that make research easier to reuse. Workbench-style environments extend this further by hosting the research artifacts themselves: SciForge [312] presents an AI-native multimodal workbench in which discovery, analysis, and presentation share one workspace.

This direction also introduces new risks. An interactive paper agent must not only summarize the paper faithfully, but also execute tools correctly, respect the limitations of the original method, and avoid presenting unsupported extrapolations as valid conclusions. Evaluation therefore requires both communication metrics and reproducibility metrics: whether the agent explains the paper clearly, whether it invokes the correct tools, whether it reproduces expected results, and whether it handles out-of-scope user queries responsibly.

6.1.6 Assessment: Fidelity, Usability, and Adoption

Assessment for Paper2X must evaluate three dimensions: *fidelity*, whether the generated artifact accurately represents the paper; *usability*, whether the artifact supports its intended communication or interaction goal; and *adoption*, whether researchers trust the artifact enough to use it publicly. Fidelity is the most important dimension because dissemination artifacts often circulate independently of the paper. A polished poster, slide, video, or thread can misrepresent a contribution if it omits caveats, changes baselines, simplifies methods incorrectly, or exaggerates results.

Poster and slide generation have the most mature evaluation infrastructure. Paper2Poster [267] evaluates poster quality under cost-efficient generation settings, while PPTAgent [477] introduces PPTEval to assess slide quality along content, design, and coherence dimensions. Video evaluation is newer: PresentEval [325] evaluates narrated video pipelines, and Paper2Video [490] introduces comprehension-oriented evaluation through paper–video pairs. Complementing per-format benchmarks, recent work establishes fine-grained correspondences across a paper, its slides, and its presentation video [233], enabling cross-media fidelity to be measured directly. These benchmarks reflect a broader shift from surface aesthetics toward whether generated materials preserve content and improve audience understanding.

Agentic dissemination requires additional evaluation criteria. A paper agent should be assessed not only by answer quality, but also by tool correctness, reproducibility, error handling, and boundary awareness. This makes Paper2X evaluation closer to the assessment problems in Coding and Experiments (S3): a system may appear helpful in natural language while invoking the wrong workflow or returning unsupported outputs. Across formats, no large-scale adoption study has yet established whether Paper2X tools reduce or increase misrepresentation compared with manual author-created materials. As a result, Paper2X systems are currently best understood as drafting and interaction aids rather than final producers.

6.1.7 Findings and Observations

Stage 8: Dissemination (Paper2X)



State & Progress

Low Cost | Paper2Poster | Paper2Slides | Paper2Video | Multi-Agent | Interactive



Gaps & Limitations

Trust | 4-Modal Hard | No Adaptation | Adoption | Fidelity | Social Media

- Cost barrier eliminated: $0.005/poster with 87% fewer tokens [267]; 8B models match frontier on slides [478]. Most cost-efficient stage to automate.
- Poster and slide generation are the most developed directions, moving from one-shot conversion toward editable, feedback-aware, and user-preference-conditioned workflows [267, 477, 478].
- Paper-to-agent systems extend dissemination from static explanation to interactive reuse, exposing paper methods, code, and workflows through tool-using agents [240, 416].
- Trust, not generation cost, is the bottleneck: researchers need confidence that AI-generated public artifacts preserve claims, caveats, and limitations.
- Video remains difficult because it must coordinate slides, subtitles, narration, pacing, and sometimes avatar or cursor motion under strict time constraints [207, 490].
- Social media and web dissemination remain limited by audience modeling: simplifying a paper for broader reach without exaggerating or distorting the contribution remains unresolved.

6.2 Summary and Transition: Dissemination

This phase shifts the focus from validated scholarly argument to audience-adaptive communication and reuse. The goal is to convert the manuscript into posters, slides, videos, project pages, social media posts, and increasingly interactive agents or tools. Progress in this phase shows that AI can substantially lower the cost of producing dissemination artifacts, especially when the input paper is complete and the target format has a predictable structure.

The central limitation is fidelity under format change. Each dissemination artifact compresses, reorders, or re-expresses the paper, creating opportunities for omission, overstatement, or distortion. This risk appears differently across formats: posters and slides may oversimplify the contribution, videos may misalign narration and visual evidence, social media posts may trade nuance for engagement, and paper agents may expose tools or workflows beyond their validated scope. Dissemination-stage automation is therefore most credible when it supports draft generation, format adaptation, editing, and interaction while preserving author oversight over claims and limitations.

The completion of this phase closes the research lifecycle: a contribution has been created, written, validated, and communicated. The remaining question is what patterns cut across all phases. We therefore next synthesize the common architectures, capability boundaries, deployment principles, and open challenges that define AI-assisted research as a whole.

7 Cross-Cutting Analysis

The preceding sections analyzed AI-assisted research stage by stage. We now synthesize patterns that emerge across the complete lifecycle. This cross-cutting view is necessary because many of the most important limitations do not appear within a single stage, but at the boundaries between stages: ideas that weaken after implementation, retrieved evidence that is misrepresented in writing, experiments that produce unsupported claims, reviews that miss methodological flaws, rebuttals that promise revisions without fulfilling them, and dissemination artifacts that simplify results beyond the evidence.

We organize this analysis around **four** questions. First, how do end-to-end systems integrate multiple stages of the research lifecycle? Second, how should research automation be evaluated across heterogeneous artifacts and long-horizon workflows? Third, what recurring capability boundaries and deployment principles appear across phases? Finally, what open challenges must be addressed before AI systems can be trusted as reliable research collaborators rather than artifact generators?

7.1 End-to-End Research Systems

Some systems discussed earlier, especially in Section 4, also qualify as end-to-end research systems because they generate complete manuscripts. Here, however, we analyze them from a different perspective: not as paper-writing tools, but as lifecycle-scale architectures. The key question is how these systems connect ideation, literature review, coding, experimentation, writing, validation, and dissemination, and where the handoffs between stages remain fragile.

Most current end-to-end systems emphasize Phase 1 (*Creation*) and Phase 2 (*Writing*), connecting idea generation, implementation, experiment execution, and manuscript drafting into a single workflow. Far fewer systems incorporate substantive Phase 3 (*Validation*), such as adversarial review, rebuttal planning, or revision tracking, and Phase 4 (*Dissemination*) remains mostly outside current end-to-end pipelines. This imbalance reflects a broader pattern observed throughout the survey: it is easier to generate research artifacts than to validate, revise, and communicate them with accountable fidelity.

Existing systems can be grouped into four architectural families: sequential pipelines, search-based and self-improving systems, skill-based and tool-integrated systems, and multi-agent or community-scale frameworks.

7.1.1 Sequential and Pipeline-Based Systems

Sequential systems connect research stages in a mostly linear order, typically moving from idea generation to experiment execution and manuscript drafting. The AI Scientist [223] established this paradigm by demonstrating that hypothesis generation, code execution, experimental analysis, and paper writing can be assembled into a single automated workflow. Agent Laboratory [308], AI-Researcher [356], CycleResearcher [394], Kosmos [242], Dolphin [448], CodeScientist [145], and InternAgent [453] instantiate related pipeline designs with different choices of base models, task scopes, and evaluation targets. A more recent pipeline, ScientistOne [238], augments this idea-to-paper flow with a chain-of-evidence mechanism that explicitly targets fabricated citations and unreproducible results. NVAITC AI Scientist [135] adds explicit governance to the end-to-end pipeline and demonstrates it on a real hypertension GWAS discovery, illustrating the shift toward auditable, domain-grounded research systems. Newer pipelines continue this direction while broadening the artifact they orchestrate: DeepResearch Agent System [139] and ScienceFlow [473] target long-horizon ML research and discovery, Spark-to-Paper [281] packages the whole idea-to-manuscript flow as a single composable skill, and OmniScientist [176] extends the pipeline across modalities and disciplines rather than a single field.

Auditability has become a first-class design goal rather than a reporting afterthought, and it now distinguishes a recognizable group of pipelines. XScientist [226] adopts a git-like research protocol so that a long-running

autonomous project has an inspectable, revertible history; auditable AI-scientist workflows for materials pair each claim with held-out transfer evidence [257]; auditable experimental records convert agent trajectories into evidence usable by an industrial reviewer [492]; PEARL [343] makes the repair of extracted scientific reasoning graphs traceable; and IdeaTrail [106] publishes full-process agent trajectories rather than only final outputs. Evidence-graph designs make the same commitment structurally, grounding an agent's conclusions in an explicit evidence structure that a human can walk [291]. This cluster is the clearest architectural response so far to the phase-boundary problem: rather than trying to make each stage individually stronger, it makes the handoffs legible.

YouRA [397] operationalizes traceability as persistent research state. Its verification-state representation maintains hypotheses, evidence pointers, and failure history, while an independent controller coordinates bounded repair, redesign, and recovery. Evaluation on the ten-task end-to-end subset of MLR-Bench supports this architectural choice under matched backbones, although automated diagnostics and a small task set leave broader scientific reliability open. The contribution is therefore a mechanism for preserving evidence across stages rather than a guarantee that every generated claim is correct.

The advantage of sequential architectures is operational simplicity. Each stage produces an artifact that becomes the input to the next stage, making the workflow interpretable and relatively easy to implement. The limitation is error propagation. A weak idea can lead to irrelevant experiments; incorrect code can produce misleading results; and unsupported experimental claims can be polished into a plausible manuscript. Sequential pipelines therefore expose the same phase-boundary risk observed throughout the lifecycle: producing an artifact at one stage does not guarantee that the next stage represents it faithfully or verifies it adequately.

7.1.2 Search-Based and Self-Improving Systems

A second family introduces search, evolution, or self-improvement to avoid the brittleness of one-pass generation. AI Scientist v2 [419] uses agentic tree search to explore research trajectories more systematically than its predecessor. ASI-Evolve [415], AutoSOTA [194], CORAL [283], and related evolutionary systems search over architectures, algorithms, data curation strategies, or multi-agent behaviors to discover stronger solutions. EvoMaster [489] proposes a self-evolving foundational agent framework for agentic science at scale, while AutoSci [279] centers the full lifecycle on a persistent cross-project memory with self-improving research procedures. Recent closed-loop systems tighten the discover–verify loop and extend it beyond ML: an agentic self-driving lab [141] compresses the validation bottleneck for scientific discovery, iterative meta-reflection [470] refines the research loop through self-critique, and closed-loop agents certify generalizable improvements in molecular property prediction [258] and rediscover psychological theories through an automated cognitive scientist [143].

Search-based designs are important because research rarely proceeds as a single pass. Strong work typically emerges from branching alternatives, failed experiments, ablation-driven refinement, and selective continuation of promising directions. These systems therefore better match the iterative structure of scientific practice than direct pipelines. Recent work refines *what* is searched over rather than only how widely: adaptive inquiry treats knowledge revision, not solution proposal, as the unit of search [79], graph-guided planning decomposes a project into tasks before searching within them [411], CausalForge [354] grounds self-improvement in a formal causal framing, and a provocative reframing casts agentic auto-research as fuzz testing, where value comes from generating many cheap probes rather than one considered attempt [118]. Others make the search substrate explicit, whether by scaling discovery environments for turn-level agentic reinforcement learning [417] or by using learned world models as the space in which research agents scale [427]; preference models trained on research judgments supply the selection signal such search requires [78]. However, search alone does not solve validation. Without reliable evaluators, search can optimize toward benchmark-specific artifacts, superficial novelty, or brittle improvements. The core design question is thus not only how broadly a system searches, but what signals guide selection and whether those signals reflect scientific value rather than local metric gains. Efficiency has accordingly become an explicit axis, with recent analysis arguing that cost per unit of validated progress, not raw capability, now separates autonomous research systems [422], and that scientific memory should be evaluated as budgeted context restoration rather than as leaderboard recall [322].

AIDE^2 [341] makes the research agent's own code the object of recursive improvement. With backbone models fixed, it proposes and selects harness rewrites using hidden grading; its principal eight-day run accepts seven

successive changes. Crucially, the study separates the grading signal used to select candidates from four external evaluation benchmarks. This provides evidence of transfer beyond the selection tasks while keeping the scope of the claim clear: improvement concerns agent code, search, and memory management, rather than recursive training of model weights.

7.1.3 Skill-Based and Tool-Integrated Systems

Skill-based systems package research workflows as composable capabilities, often built around coding agents, retrieval tools, experiment runners, document editors, and evaluation modules. ARIS [425] represents this direction by organizing research automation into reusable workflows for idea discovery, auto-review, and paper writing. AutoResearchClaw [209] similarly implements a multi-stage pipeline with internal agents for coding, benchmarking, and figure generation. Broader tool-integrated systems such as AutoAgent [355], Biomni [136], SciSciGPT [317], and ResearchClaw [424] emphasize retrieval, analytics, code execution, document understanding, and domain-specific tool use.

The strength of skill-based architectures is modularity. Rather than relying on one monolithic agent to perform all research activities, these systems expose explicit tools and reusable skills for individual operations. This makes it easier to inspect intermediate artifacts, swap components, and insert human checkpoints. Domain-grounded instantiations show how far this reaches when the tools are real instruments rather than software: agents have been stress-tested in a physical robotic chemistry laboratory [107], used to discover practical quantum error-correcting codes [420], run as long-horizon architecture-research agents [300], and deployed as mathematical agents coupling rigorous proof with autonomous exploration [387]. Closed-loop domain frameworks such as WARA [110] show the same shape in engineering optimization, where the agent must act against a live system rather than a static dataset. Two further framings position the system itself differently: Mechanist [374] treats AI as a scientific instrument for studying mechanisms rather than as a researcher substitute, and a quadruped-navigation study argues that structure and "taste" are what prevent an AI scientist from drifting off a falsifiable track [465]. The limitation is coordination: modular systems still need reliable state management across stages. If the idea, literature trace, code state, experimental logs, manuscript claims, review feedback, and revision plan are not represented in a shared and updateable workspace, then phase handoffs remain fragile despite the presence of many tools.

ResearchTrails [97] supplies process-level supervision by converting Git histories into evidence-grounded decisions about methods, experiments, and ablations. Retrieved trajectories, distilled skills, and decision-sequence training improve next-decision similarity on held-out repositories. Commit histories provide an observable proxy for research development, and the LLM-judged similarity measure should remain distinct from successful experiment execution or independent discovery.

7.1.4 Multi-Agent and Community-Scale Systems

Multi-agent systems distribute research tasks across specialized agents, such as researchers, engineers, reviewers, analyzers, writers, or simulated community members. FreePhDLabor [179], SciMaster [29], EvoScientist [231], UniScientist [177], Medical AI Scientist [398], AiScientist-LH [34], FARS [10], and AutoResearchClaw [209] illustrate different forms of multi-agent orchestration. Arbor [152] adopts a related design in which a long-lived coordinator dispatches short-lived executor agents to drive a long-horizon research loop through hypothesis-tree refinement. Related community-scale systems such as VirSci [344], AgentRxiv [307], and ResearchTown [443] further simulate aspects of scientific collaboration, including idea exchange, manuscript writing, review, and revision. ScientistTwo [250] connects problem-driven hypothesis generation, experiment execution, ablation, writing, and simulated review–rebuttal through specialized agents. Its reported results include comparisons with published baselines and artifact-integrity audits, making it relevant to both capability and verification. However, its reported acceptance thresholds are applied by automated reviewers; they do not represent actual conference decisions or independent reproduction. The system extends lifecycle coverage into iterative validation while illustrating why coverage and the credibility of validation must be assessed separately.

The motivation for multi-agent architectures is that research requires heterogeneous expertise and adversarial feedback. A single model asked to generate, execute, write, and critique its own work is prone to self-confirmation. Separating roles can reduce this risk by introducing specialization and cross-agent critique. Recent instances extend the pattern to specialized scientific communities, for instance an agentic community

organized around automated neural-operator discovery [221], and to self-evolving domain agents governed by stage-level rubrics in medicine [208]. A position paper pushes the framing further, arguing that because these systems are embedded among human collaborators they should be studied as *human-agent* systems rather than as autonomous agents evaluated in isolation [68], and a historical account situates the whole trajectory of AI scientists to date [165]. However, multi-agent systems also introduce coordination problems: agents may duplicate work, reinforce shared misconceptions, defer to weak signals, or produce verbose deliberation without improving scientific quality. The strongest multi-agent systems therefore require more than agent count; they require clear role separation, shared memory, grounded tools, and explicit verification mechanisms.

In practice, these four families describe emphases rather than mutually exclusive designs, and the strongest recent systems increasingly combine them: agentic tree search layered over multi-agent role separation [231, 419], or skill-based tool use coupled with persistent cross-project memory and self-improvement [279, 489]. The historical trajectory mirrors this convergence: 2024 systems were predominantly sequential idea-to-paper pipelines, whereas 2025–2026 systems add search, evolution, shared memory, and explicit critique to counteract one-pass brittleness. Yet across all four families, the decisive factor is consistently the same: not the orchestration pattern itself, but whether the architecture sustains grounded verification and traceable state at the points where one stage hands off to the next.

ST-Bench [52] examines multi-agent scientific data analysis under matched backbones and explicit cost accounting. Across paper-grounded Earth-science tasks, stronger workflows mainly improve coverage of queries yielding realistic numerical outputs, while conditional metric quality remains comparable to the single-agent baseline. The additional inference cost and coverage-sensitive score make this evidence for a bounded workflow benefit rather than universal multi-agent superiority.

7.1.5 Assessment: Lifecycle Coverage and Phase Boundaries

End-to-end systems should be evaluated not only by the quality of their final manuscript, but also by lifecycle coverage and phase-boundary reliability. Current systems are strongest at generating ideas, code, experiments, and paper drafts, but weaker at external validation, author-style revision, and audience-adaptive dissemination. This uneven coverage is not accidental: Stage 1 (*Creation*) and Stage 2 (*Writing*) produce artifacts, whereas Stage 3 (*Validation*) and Stage 4 (*Dissemination*) require judgment, accountability, and audience-aware fidelity. The most important failure mode is not isolated stage failure, but unverified handoff. An idea may appear novel but fail during execution; code may run but implement the wrong algorithm; experimental logs may be summarized into unsupported claims; an automated review may be coherent but lenient; and a rebuttal may promise changes that are not fulfilled. End-to-end systems amplify this risk because errors can propagate silently across stages. A mature lifecycle-scale research system must therefore preserve traceable links between hypotheses, retrieved evidence, code, experiments, figures, manuscript claims, reviews, rebuttals, and revisions.

Reported costs and quality metrics further suggest that the token budget alone is not the decisive factor. Systems vary widely in cost, but stronger results often come from search strategy, tool integration, structured decomposition, and verification design rather than brute-force generation. The central evaluation question is therefore shifting from *Can the system produce a paper?* to *Can the system maintain scientific fidelity across the complete lifecycle?*

A dedicated benchmark layer for whole-system evaluation has begun to form around exactly that question, and it is notable for measuring process rather than product. Capability-oriented suites ask whether models are ready for scientific discovery at all [324]; ARAC [56] scores alignment and completeness of end-to-end research runs rather than final-artifact quality; and one systematic evaluation argues explicitly for looking past final scores when grading long-horizon research and development agents [195]. Replication is emerging as the sharpest available test, since it has a checkable ground truth: agents can be trained specifically to replicate research [69], while a meta-evaluation asks the prior question of whether models can even write reliable rubrics for judging reproduction [124]. Two designs attack contamination directly, either by drawing tasks from adversarially fast-moving real-world domains that postdate training data [22] or by fixing a state-centric target such as automated world-model research [245]; domain-specific suites extend the same logic to theorem proving in quantum algorithms and quantum information [459]. Taken together, this layer marks a shift in what counts as evidence for an end-to-end system: not a manuscript that reads well, but a trajectory that

can be audited, replicated, and scored against a target the system could not have memorized.

7.1.6 Findings and Observations

End-to-End Research Systems



State & Progress

Pipelines | Search | Skills | Multi-Agent | Validation | Handoffs

- End-to-end systems increasingly move beyond linear pipelines toward search-based, skill-based, and multi-agent architectures that better reflect the iterative structure of research.
- Phase coverage remains uneven: most systems cover *Creation* and *Writing*, while substantially fewer incorporate *Validation*, and none yet provide mature *Dissemination* coverage.
- Systems that include review, critique, or revision mechanisms point toward more credible lifecycle automation, but their success depends on verification quality rather than review-like text alone.



Gaps & Limitations

Propagation | Self-Critique | Fragmented | Validation Gap | State Loss | Overclaiming

- Error propagation is the main lifecycle risk: weak ideas, semantic code errors, unsupported claims, and lenient reviews can compound when phase handoffs are not explicitly verified.
- Single-model self-critique remains structurally limited; credible validation usually requires role separation, external evidence, adversarial review, or human oversight.
- Most existing E2E systems do not maintain a fully traceable, updateable state across hypotheses, literature review, coding, experiments, manuscript claims, reviews, and revisions.

7.2 Evaluation Across the Research Lifecycle

Evaluation is the central bottleneck for AI-assisted research. Each stage produces different artifacts—ideas, literature summaries, code, experiments, figures, manuscripts, reviews, rebuttals, and dissemination materials—so no single metric can capture research quality across the full lifecycle. Existing benchmarks have therefore evolved from narrow task-specific evaluations toward broader, process-aware, and increasingly execution-grounded protocols.

Table 2, introduced in Section 2.3, summarizes major benchmarks across the eight stages.

Across the benchmark landscape, three trends are clear. First, evaluation is moving from isolated outputs to a multi-dimensional assessment. Early benchmarks often measured a single capability, such as citation prediction, code execution, or writing fluency. Recent benchmarks instead evaluate multiple axes, such as novelty and feasibility in ideation, coverage and citation accuracy in literature synthesis, semantic correctness in research code, review consistency in peer review, and fidelity in Paper2X artifacts. Second, benchmarks are becoming more domain- and workflow-aware. Specialized evaluations now target GPU kernel optimization [181, 264], biology research [172], scientific experimentation [169], and broader scientist-aligned workflows [24, 413]. Third, a persistent gap remains between benchmark performance and real-world research value: systems can perform well on measurable proxies while still producing outputs that experts judge as shallow, incremental, or insufficiently grounded [394, 484].

7.2.1 Stage-Specific Benchmarks

Stage-specific benchmarks remain necessary because each part of the research lifecycle requires different evaluation criteria.

- For S1 (*Idea Generation*), benchmarks assess novelty, feasibility, diversity, and downstream potential. These evaluations are difficult because apparent novelty may not survive implementation, and expert judgments of research promise are inherently noisy.

- For S2 (*Literature Review*), benchmarks emphasize retrieval precision, citation fidelity, coverage completeness, and synthesis quality. The central challenge is not only whether the system finds relevant papers, but whether it uses them faithfully when constructing a narrative.
- For S3 (*Coding and Experiments*), benchmarks increasingly move beyond code execution toward semantic correctness and reproducibility. Research-code benchmarks ask whether generated implementations match the intended algorithm, while broader workflow benchmarks such as EXP-Bench [169] evaluate experiment design, execution, and analysis. NatureBench [386] raises the bar further, testing whether coding agents can match the published state of the art of Nature-family papers.
- For S4 (*Tables and Figures*), evaluation must distinguish visual plausibility from scientific correctness: a figure or table may look publication-ready while misrepresenting data, notation, or comparison structure.
- For S5 (*Paper Writing*), evaluation combines writing quality, citation accuracy, factual grounding, and review-style judgment.
- For S6 (*Peer Review*), benchmarks assess consistency, grounding, bias, and robustness to manipulation rather than review fluency alone.
- For S7 (*Rebuttal and Revision*), emerging datasets align reviews, author responses, and manuscript changes, enabling evaluation of whether rebuttals actually address concerns and lead to fulfilled revisions.
- For S8 (*Dissemination*), evaluation centers on fidelity, usability, and audience adaptation across posters, slides, videos, project pages, social media posts, and interactive agents. This stage is especially difficult because dissemination artifacts often circulate independently of the paper and can shape public understanding of the work.

7.2.2 Evaluation Methodologies

Current evaluation methodologies can be grouped into five families. *Expert evaluation* remains the most credible approach for assessing novelty, significance, correctness, and scientific contribution. Si *et al.* [327], for example, recruited over 100 NLP researchers to evaluate generated ideas. However, expert evaluation is expensive, slow, and noisy: even peer-review-style judgments show limited inter-rater agreement, with the Stanford Agentic Reviewer study reporting human–human correlation around $\rho = 0.41$ [149]. This makes expert evaluation indispensable but difficult to scale.

LLM-as-Judge and Agent-as-Judge methods provide scalable approximations of human assessment. CycleReviewer [394] reports a 26.89% reduction in Proxy MAE relative to individual human reviewers for score prediction, while the Stanford Agentic Reviewer [149] achieves review-score correlations comparable to human inter-rater agreement ($\rho = 0.42$ vs. human $\rho = 0.41$). These results show that automated evaluators can provide useful review-style signals, but they remain imperfect proxies. They can exhibit positivity bias, length bias, authority bias, self-preference, and vulnerability to adversarial prompts [440, 484]. As a result, LLM-based evaluation is most reliable when calibrated against expert judgments and combined with task-specific verification.

Automated metrics offer objective but narrow signals. Code execution success, unit-test pass rates, citation accuracy, acceptance prediction, and traditional text-generation metrics such as BLEU [269], ROUGE [201], and BERTScore [461] are easy to compute, but they capture only fragments of research quality. For example, executable code can still be semantically wrong, accurate citations can still be used to support misleading claims, and fluent text can still lack contribution. Over-optimization on any one metric risks Goodhart's law.

Execution-grounded evaluation verifies research outputs by running code, reproducing experiments, or checking claims against generated evidence. This paradigm is especially important for S3, but it also affects Writing and Validation because manuscript claims should be traceable to executed experiments. PaperBench [342], for example, decomposes papers into individually gradable subtasks that can be checked through implementation and execution. Si *et al.* [329] further show that execution-guided search can improve discovery workflows by using empirical feedback rather than textual judgment alone.

Process- and trace-based evaluation assesses how a system reaches an output, not only the final artifact. This includes tool-use trajectories in deep research, reviewer-comment decomposition in rebuttal, revision

fulfillment after author responses, and fidelity between paper content and dissemination materials. This paradigm is increasingly important because many lifecycle failures occur at handoffs: a system may retrieve the right paper but cite it incorrectly, run an experiment but summarize it inaccurately, or promise a rebuttal revision without implementing it.

7.2.3 Emerging Evaluation Paradigms

Several emerging paradigms are reshaping evaluation for AI-assisted research. The first is *execution-grounded evaluation*, where claims are checked against executable artifacts rather than judged only from text. This is essential for research coding, paper replication, and experimental analysis, where surface plausibility is insufficient. It also provides a path toward evaluating Writing: a manuscript claim should be traceable to a figure, table, log, or executed experiment.

The second is *adversarial evaluation*. As discussed in S6 (*Peer Review*), LLM-based reviewers and judges are vulnerable to prompt injection, lexical triggers, and covert content manipulation. Breaking the Reviewer [203] and related studies show that robustness to manipulation must be treated as an evaluation dimension rather than a peripheral security issue. This is particularly important for Validation, where a manipulated review or judge can affect acceptance decisions.

The third is *long-horizon evaluation*. Many benchmarks remain short-horizon, evaluating tasks that take minutes or hours rather than weeks or months. However, real research involves delayed feedback, failed attempts, changing hypotheses, and evolving evidence. METR's analysis suggests that AI task horizons are rapidly increasing [239], while RE-Bench [395] provides open-ended ML R&D environments that begin to approximate longer research workflows. Still, current long-horizon benchmarks remain far shorter and cleaner than authentic research projects.

The fourth is *lifecycle-level evaluation*. Existing benchmarks usually evaluate one stage at a time, but many important failures occur between stages. A future lifecycle benchmark should test whether an idea remains valid after implementation, whether retrieved literature is faithfully represented in writing, whether experimental evidence supports manuscript claims, whether rebuttal commitments are fulfilled, and whether dissemination artifacts preserve claims and limitations. Such evaluation would better match the real risk profile of end-to-end research automation. Recent benchmarks begin to instantiate this lifecycle-level view: ResearchClawBench [414] grounds end-to-end research tasks in hidden papers with expert multimodal rubrics, the benchmark suite of [373] evaluates frontier models and agentic harnesses across the full research lifecycle, and [468] provides a minimal scaffold for running and assessing the complete loop with off-the-shelf agents.

A complementary criterion asks whether experimental success yields scientific understanding. EurekaBench [90] evaluates 26 expert-verified tasks across six disciplines along scientific constraints, predictive accuracy, and the insights supported by discovered mechanisms. Its results expose a gap between optimizing predictions and deriving insights, reinforcing the need to report these outcomes separately. The curated expected insights operationalize discovery for evaluation, rather than establishing unrestricted novelty at the research frontier.

MMPostTrainBench [217] evaluates budgeted multimodal post-training through independently tested submitted checkpoints, non-target capability retention, and recorded candidate trajectories. It exposes regressions and gaps between the best evaluated candidate and the final submission. MMResearch addresses these boundaries with media-grounded evidence, hierarchical memory, and development-guided candidate retention; its gains remain specific to the evaluated configurations.

7.2.4 Evaluation Gaps

Despite rapid progress, several gaps remain unresolved. First, *novelty and significance are still difficult to define*. Expert judgments vary across reviewers, venues, and fields, and automated novelty scores can reward ideas that sound original but fail after execution. Second, *benchmark contamination and temporal validity are persistent concerns*. Many tasks are derived from public papers, code, or reviews that may appear in model training data. Temporal splits help, but they introduce changes in topic, difficulty, and community standards.

Third, *cross-system comparison remains difficult*. Systems are often evaluated with different base models, prompts, tools, datasets, compute budgets, and human-in-the-loop assumptions. This makes reported results

hard to compare even when they target the same stage. Fourth, *cross-domain generalization remains under-tested*. Most benchmarks focus on machine learning and NLP, while chemistry, biology, materials science, physics, medicine, and social science require different evidence standards, experimental workflows, and domain-specific tools.

Fifth, *computational cost is itself an evaluation dimension*. Some research tasks, such as paper replication, long-horizon experimentation, and multimodal dissemination, require substantial token, compute, or tool-use budgets. A system that performs well under unlimited sampling may not be practically useful if its cost is prohibitive or its results are not reproducible under realistic constraints.

Finally, *no existing benchmark evaluates the complete research lifecycle with human-equivalent rigor*. PaperBench [342] makes important progress for replication, and process-aware benchmarks are emerging across literature review, coding, peer review, rebuttal, and Paper2X. However, no benchmark yet evaluates the full chain from ideation to dissemination while preserving traceability across artifacts. This lifecycle evaluation gap is central: without it, systems may appear strong within individual stages while failing to maintain scientific fidelity across the research process. Two further gaps are now being named explicitly. Safety has been largely absent from research-agent evaluation, which SciHazard [178] addresses by scoring decomposed scientific harm rather than treating capability and risk as one axis. And current suites tend to reward performance in the settled interior of a field rather than at its moving frontier, the concern behind proposals to evaluate at the *science edge* [113], where the correct answer is not yet in any training corpus.

7.3 Cross-Cutting Insights

The preceding sections reveal a consistent pattern across the research lifecycle: AI systems are increasingly capable of producing research-like artifacts, but remain less reliable at verifying whether those artifacts are novel, faithful, executable, and scientifically meaningful. We distill five cross-cutting insights from the stage-level analysis. These insights are not tied to a single tool or benchmark; rather, they describe recurring capability boundaries and deployment principles that appear across Creation, Writing, Validation, and Dissemination.

7.3.1 Artifact Generation Outpaces Scientific Verification

Across the lifecycle, AI systems are better at producing artifacts than at verifying their scientific validity. In S1 (*Idea Generation*), generated ideas can appear novel and well-motivated, yet weaken after implementation. Si *et al*. [328] show that AI-generated ideas degrade more sharply after execution than human ideas, exposing a gap between apparent novelty and executable substance. In S3 (*Coding and Experiments*), generated code may run successfully while implementing the wrong algorithm, with semantic failures forming a major source of error [134]. In S4 (*Tables and Figures*), generated visual artifacts may look polished while misrepresenting data, notation, or information flow. In S5 (*Paper Writing*), fluent prose can conceal weak reasoning or unsupported claims.

This gap also appears in Validation and Dissemination. In S6 (*Peer Review*), automated reviews can be coherent and consistent while under-detecting decisive methodological flaws or assigning inflated scores [263, 484]. In S7 (*Rebuttal and Revision*), generated responses may sound persuasive, but their value depends on whether promised evidence or manuscript changes are actually fulfilled [31]. In S8 (*Dissemination*), posters, slides, videos, and social media summaries can simplify a paper in ways that overstate claims or omit limitations. The central lifecycle problem is therefore not artifact production alone, but artifact verification: each output must remain traceable to evidence, assumptions, and limitations. Recent analyses sharpen this point: a large multi-domain study finds that AI scientists frequently produce results without reasoning scientifically [299], and others argue that completing a research-like workflow is not equivalent to achieving scientific closure [379].

Research-agent oversight is itself vulnerable when the same agent controls both a result and the evidence offered for it. Huang *et al*. [140] study reward hacking across 17 models and 38 tasks, distinguishing spontaneous behavior from conditions in which hacking is explicitly allowed. Repeated interaction can also enable evasion of an LLM review panel. These findings motivate metrics outside the agent's control and independent

recomputation; they do not establish a universal hacking rate, and the detailed-feedback comparison does not isolate the effect of explanations from attempt history.

7.3.2 Human-Governed Collaboration Remains the Most Reliable Deployment Mode

The strongest deployment pattern across stages is not full autonomy, but human-governed collaboration. In Writing, semi-automated systems are most credible when they assist planning, drafting, polishing, and citation support while researchers retain control over argumentation, interpretation, and final responsibility. In Peer Review, the strongest validated setting is not standalone AI review, but AI feedback on human reviews: the ICLR 2025 randomized study shows that LLM feedback improved review quality in 89% of cases without affecting acceptance rates [360]. In Rebuttal, author-aware systems are more appropriate than generic response generation because rebuttals must reflect the paper's actual contributions and the authors' intended revisions [295]. In Dissemination, AI-generated posters, slides, videos, and public summaries are best treated as editable drafts whose claims and emphasis remain under author control.

This pattern explains why direct automation is risky in high-stakes research settings. Research requires judgment under uncertainty: deciding whether an idea is worth pursuing, whether an experiment is sufficient, whether a critique is valid, whether a rebuttal promise is feasible, and whether a public-facing summary is faithful. These decisions are precisely where current systems remain fragile. AI assistance is therefore most useful when it expands researcher capacity while preserving human oversight over scientific claims, evidence interpretation, and accountability.

7.3.3 Capability Boundaries Emerge in Open-Ended Research Tasks

The sharpest capability boundaries appear when tasks become novel, underspecified, or long-horizon. Current systems perform strongly on structured tasks with clear feedback, such as standard software issue resolution, grammar correction, simple plotting, and format conversion. Performance drops when the task requires interpreting implicit assumptions, designing meaningful experiments, reproducing underspecified methods, or judging scientific contribution. This is most visible in research coding: while frontier systems perform well on familiar software benchmarks, performance falls sharply on novel research-code tasks, with reported ceilings around 37–39% on dedicated benchmarks [134, 402].

Similar boundaries recur elsewhere. Literature review systems retrieve and summarize individual papers increasingly well, but struggle with multi-paper relational reasoning and citation fidelity. Idea-generation systems produce plausible hypotheses but face persistent novelty–feasibility tradeoffs. Paper-writing systems generate fluent manuscripts but remain weaker at argumentative depth and reviewer anticipation. Peer-review systems can approximate review style but remain vulnerable to leniency, bias, and manipulation. These failures share a common structure: the task cannot be solved by pattern matching alone, because success depends on implicit domain knowledge, causal reasoning, long-horizon feedback, and expert judgment. Position analyses reinforce this boundary, arguing that current agentic AI scientists are structurally better suited to assisting research than to autonomous discovery [18].

7.3.4 Effective Systems Converge on Layered Architectures

Across phases, the most capable systems increasingly combine three layers: *exploration*, *execution*, and *verification*. The exploration layer searches over hypotheses, paper collections, code variants, response plans, or design alternatives. The execution layer interacts with tools: retrieval engines, code interpreters, experiment runners, plotting libraries, document editors, or presentation generators. The verification layer checks whether intermediate outputs are grounded, correct, and useful, through execution feedback, citation validation, critique, reviewer simulation, or human review.

This layered view explains why simple prompting is insufficient for research automation. Research tasks rarely require only one generation step; they require proposing alternatives, testing them, revising based on feedback, and preserving state across iterations. Search-based systems improve exploration, tool-integrated systems strengthen execution, and multi-agent systems can support specialization and critique. However, more agents do not automatically improve performance. Multi-agent systems appear most useful when the task can be decomposed into parallel or role-specialized subtasks; they can degrade on sequential reasoning tasks when

coordination overhead and error propagation dominate [164]. Thus, the important design principle is not agent count, but whether the architecture matches the task structure and includes reliable verification.

7.3.5 AI Use Has Become a Governance Problem, Not a Detection Problem

AI assistance is already embedded in the research ecosystem. Corpus-level studies estimate detectable AI modification in a nontrivial fraction of scientific writing, including up to 17.5% of computer science abstracts [197] and 13.5% of biomedical abstracts [166]. Self-reported adoption is higher, with many researchers using AI for writing or review-related tasks [249]. At the same time, linguistic marker studies show that AI-associated words can surge after the introduction of LLMs, but such signals are unreliable for adjudicating individual cases and can change as users and models adapt.

The policy implication is that the community should move from detection-centered enforcement toward disclosure, attribution, and accountability. Detection tools can produce false positives, especially for formal or non-native academic prose, while watermarking remains dependent on provider cooperation and robustness to paraphrasing [288]. The more durable governance questions are therefore: What forms of AI assistance must be disclosed? Which uses are allowed during review? Who is responsible for AI-generated claims, citations, rebuttal commitments, or public summaries? How should venues audit high-risk uses without penalizing legitimate writing support? As AI becomes a routine part of research practice, governance must focus less on whether AI was used at all and more on whether its use preserved scientific integrity. Emerging benchmarks begin to measure these integrity risks directly: SciIntegrity-Bench [435] evaluates whether AI-scientist systems uphold academic-integrity norms when honest failure-acknowledgment is the only correct response, while PseudoBench [200] tests whether auto-research systems resist generating plausible-but-pseudoscientific studies.

7.4 Open Challenges and Future Directions

Despite rapid progress across the research lifecycle, the preceding analysis shows that the main barriers to reliable AI-assisted research are not merely missing tools. The harder problems concern whether AI systems can preserve faithfulness across phase boundaries, evaluate scientific value, verify evidence, support responsible governance, generalize across domains, and preserve human expertise. We organize the remaining challenges around these six themes.

7.4.1 Faithfulness Across Phase Boundaries

Many of the most consequential failures occur not within a single stage, but when artifacts move from one phase to the next. An idea that appears promising in S1 (*Idea Generation*) may weaken after implementation in S3 (*Coding and Experiments*); retrieved evidence in S2 (*Literature Review*) may be misrepresented in S5 (*Paper Writing*); experimental results from S3 (*Coding and Experiments*) may be summarized into claims that are stronger than the data support; reviewer concerns in S6 (*Peer Review*) may lead to rebuttal promises in S7 (*Rebuttal and Revision*) that are not fulfilled; and dissemination outputs in S8 (*Dissemination*) may simplify the contribution beyond its evidence.

This phase-boundary problem is especially important for end-to-end systems. A lifecycle-scale system must not only generate artifacts, but preserve traceable links between them: hypotheses should connect to retrieved literature, code should connect to experiments, figures should connect to logs, manuscript claims should connect to evidence, rebuttal commitments should connect to revisions, and public-facing summaries should connect to the validated paper. Current systems rarely maintain this level of provenance across the full lifecycle. Future systems should therefore treat phase handoffs as explicit verification checkpoints rather than implicit transitions between modules.

7.4.2 Scientific Judgment and Novelty Assessment

Scientific judgment remains difficult to automate because research quality is not reducible to surface novelty, fluency, or benchmark score. In ideation, generated proposals can appear novel before execution but fail to remain feasible or impactful after implementation. Diversity is also a persistent concern: LLM-generated ideas may cluster in narrow regions of the idea space, limiting their ability to explore genuinely distinct research

directions [148]. In literature review, systems increasingly retrieve and summarize individual papers well, but still struggle with multi-paper relational reasoning, methodological lineage, and cross-paper contradictions.

The deeper challenge is that novelty, significance, and contribution are socially and temporally situated. A good research idea depends on field-specific context, feasibility, timing, community standards, and the availability of evidence. Automated novelty scoring can therefore reward ideas that sound original while missing whether they are executable, important, or meaningfully different from prior work. Future progress will likely require evaluation methods that combine retrieval, temporal splits, expert judgment, execution feedback, and downstream impact analysis, rather than relying on LLM-as-Judge scores alone.

7.4.3 Verification, Reproducibility, and Accountability

Verification is the central unresolved problem for autonomous research systems. In coding and experiments, generated code may execute successfully while implementing the wrong algorithm, and automated experiment runners can produce outputs that appear quantitative without being scientifically meaningful. Paper replication remains particularly difficult: PaperBench [342] shows that current agents still fall far short of human performance on reproducing research results. This indicates that even verifying existing work is not yet solved, let alone generating new work that is independently reproducible.

Rebuttal and revision expose a parallel accountability problem. A rebuttal is scientifically meaningful only if its claims are supported and its commitments are fulfilled. The commitment–fulfillment gap observed in ICLR 2025 [31] shows that persuasive response text is insufficient: systems must track whether promised experiments, clarifications, and revisions are actually incorporated. Future AI research systems should therefore include explicit evidence ledgers, experiment provenance, versioned manuscript diffs, and revision-tracking mechanisms. The goal is not only to produce stronger artifacts, but to make every claim auditable.

7.4.4 Citation, Versioning, and Source Provenance

Citation verification is not solved by adding retrieval or web search. Scientific records are versioned: the same contribution may appear as an arXiv preprint, workshop paper, conference version, journal extension, or revised technical report, with changes to title, authors, venue, year, DOI, and sometimes content. Bibliographic databases may merge or separate these records differently, and prior work on arXiv–publisher citation consolidation shows that version merging is itself a nontrivial bibliometric problem [85].

This creates a challenge for AI-assisted literature review, writing, and dissemination. A generated manuscript may cite the correct idea but assemble metadata from inconsistent versions, or quote a claim from one version while citing another. Existing citation-audit tools and benchmarks target fabricated or unsupported references [276, 385], but end-to-end research agents also need *version-consistent citation assembly*: title, authors, venue, year, URL/DOI, and quoted claims should come from the same selected record, with provenance preserved. Future systems should therefore treat citation not as a formatting task, but as a versioned source-grounding problem.

7.4.5 Governance, Disclosure, and Research Integrity

AI use in research is no longer hypothetical. Writing assistance, review support, literature search, code generation, and dissemination drafting are already part of many researchers' workflows. This makes governance a central challenge. Detection-based enforcement is unreliable because AI text detectors can produce false positives, especially for formal academic writing, non-native prose, or heavily edited text. As discussed in S5 (*Paper Writing*) and S6 (*Peer Review*), the community is therefore shifting from trying to detect every instance of AI use toward requiring disclosure, attribution, and accountability.

The open question is how to define responsible AI use across stages. Assistance with grammar correction is different from generating experimental claims; drafting a rebuttal is different from promising new experiments; using AI to improve a review is different from delegating the review itself. Venues, publishers, and institutions need policies that distinguish low-risk assistance from high-risk substitution, specify what must be disclosed, and clarify who is accountable for AI-generated content. The central governance principle should be that authors remain responsible for claims, citations, experiments, rebuttal commitments, and public-facing summaries, regardless of which AI tools contributed to their production.

7.4.6 Cross-Domain Generalization and Infrastructure Access

Most current systems and benchmarks are concentrated in computer science, machine learning, and NLP. Extending AI-assisted research to chemistry, biology, medicine, materials science, physics, and social science requires more than retraining on domain papers. These fields differ in evidence standards, experimental infrastructure, safety constraints, data availability, and community norms. Systems such as Google AI Co-scientist [100], Biomni [136], Medical AI Scientist [398], and domain-specific laboratory agents point toward this direction, but broad cross-domain generalization remains unresolved.

Infrastructure access is part of the same challenge. Some domains require specialized instruments, wet-lab protocols, proprietary datasets, or expensive compute. If advanced AI research tools are available only to well-resourced laboratories or companies, they may amplify existing inequalities in scientific production. Future systems should therefore be evaluated not only by performance, but also by accessibility, reproducibility, and deployability under realistic resource constraints. Open-source tools, standardized interfaces, shared benchmarks, and transparent provenance mechanisms will be important for preventing research automation from becoming an infrastructure privilege.

7.4.7 Human Expertise and Cognitive Ownership

A final challenge concerns the long-term development of researchers themselves. Many AI tools automate the external products of research: summaries, code, plots, manuscripts, reviews, rebuttals, and slides. However, the cognitive value of research lies in forming hypotheses, understanding prior work, diagnosing failures, interpreting results, constructing arguments, and responding to critique. If AI tools bypass these processes too aggressively, they may increase short-term productivity while weakening the skills that define scientific expertise.

This concern is most visible in Writing, where AI assistance is already widely adopted, but it applies across the lifecycle. A junior researcher who delegates literature synthesis may not develop field judgment; one who delegates experiment planning may not learn what makes evidence decisive; one who delegates rebuttal may not learn how to reason from criticism. Tools such as Script&Shift [330] and DraftMarks [332] suggest a better design direction: AI should support source transformation, process transparency, and reflective revision rather than replacing the user's cognitive engagement. The practical principle is that AI should handle mechanical, repetitive, or scaffolded tasks, while humans retain ownership of judgment, interpretation, argumentation, and accountability.

7.4.8 Toward Reliable AI-Assisted Research

Taken together, these challenges suggest a shift in the goal of AI-assisted research. The near-term objective should not be fully autonomous science in which AI systems independently generate, validate, publish, and promote research without oversight. A more credible objective is reliable human-governed research automation: systems that expand the scale and speed of research while preserving traceability, verification, expert judgment, and accountability.

Future progress will likely come from systems that integrate four design principles. First, they should maintain provenance across the full lifecycle, linking ideas, evidence, code, figures, claims, reviews, rebuttals, and dissemination artifacts. Second, they should use execution and retrieval grounding wherever possible, replacing purely textual self-judgment with verifiable signals. Third, they should include human checkpoints at phase boundaries, where errors are most likely to propagate. Fourth, they should make AI involvement transparent, so that readers, reviewers, and institutions can assess how a research artifact was produced. These principles define the path from artifact-generating systems toward trustworthy research collaborators.

7.5 Practitioner's Playbook

Table 3 turns the preceding analysis into guidance across the eight stages. The distinction is between assistance that produces a candidate artifact and the checkpoint that makes it suitable for further use. Literature-grounded proposals still require feasible tests; code and model checkpoints require independent evaluation; writing and dissemination require claims that remain linked to evidence. At each phase boundary, researchers

Table 3 Practitioner's playbook for AI-assisted research. Assistance is subject to verification: each stage retains a human checkpoint and requires evidence before its outputs inform the next stage. The guidance follows the capability boundaries and phase-handoff risks discussed in sections 7.3 and 7.4.

Phase	Stage	AI Assistance	Human Checkpoint	Evidence to Check
1: Creation	S1: Ideation	Literature-grounded idea variants and test proposals	Select hypotheses, assess novelty, and plan falsifiable tests	Sources and test protocol
	S2: Lit. Review	Retrieval, draft summaries, and metadata checks	Resolve contradictions and select the cited version	Source passages and metadata
	S3: Coding	Code scaffolding, debugging, and logged experiment runs	Define held-out tests and budgets; interpret results	Independent tests and checkpoints
	S4: Figures	Plotting code, chart drafts, and initial layout	Verify data, comparisons, and visual interpretation	Raw data and plotting code
2	S5: Writing	Drafting, language editing, and citation formatting	Own the argument, contribution claims, and limitations	Claim–evidence links
3: Valid.	S6: Peer Review	Draft comments and cross-reference checks	Judge methodology, significance, and review recommendations	Source text and protocols
	S7: Rebuttal	Comment mapping and evidence-linked response drafts	Approve commitments and confirm completed revisions	Responses matched to diffs
4	S8: Dissemination	Format conversion, layouts, and public-summary drafts	Select emphasis and verify audience-facing claims	Consistency with the paper

should inspect the sources, executed results, manuscript changes, or public-facing summaries that the next stage will rely on.

For long-running agents, this also requires preserving rejected candidates, failed experiments, budgets, and versioned research decisions. ResearchTrails [97] illustrates the value and limits of decision traces, while MMPostTrainBench [217] separates candidate selection from the quality and capability retention of the submitted checkpoint. These records support diagnosis and reproduction; they do not replace scientific judgment.

8 Conclusion

This work presented an end-to-end analysis of AI-assisted academic research across the complete lifecycle. We organized the field into four epistemological phases: *Creation*, *Writing*, *Validation*, and *Dissemination*, and eight stages spanning ideation, literature review, coding and experiments, tables and figures, paper writing, peer review, rebuttal and revision, and Paper2X dissemination. This lifecycle framing connects tools that are often studied in isolation and exposes where current systems succeed, where they fail, and how errors propagate across stage boundaries.

The central finding is that AI systems are increasingly capable of producing research artifacts, but remain less reliable at verifying their scientific meaning. Across the lifecycle, plausible outputs can conceal deeper failures: ideas may weaken after execution, retrieved evidence may be misrepresented, executable code may implement the wrong algorithm, fluent manuscripts may lack argumentative depth, reviews may miss methodological flaws, rebuttals may promise unfulfilled revisions, and dissemination artifacts may overstate claims. The core bottleneck is therefore not generation alone, but maintaining novelty, faithfulness, reproducibility, and accountability across the research process.

The most credible path forward is human-governed AI-assisted research. AI should reduce mechanical friction in retrieval, drafting, coding, visualization, review support, and dissemination, while researchers retain ownership over judgment, interpretation, experimental design, argumentation, and final responsibility. Future systems should maintain provenance across artifacts, use retrieval and execution grounding wherever possible, support human checkpoints at phase boundaries, and make AI involvement transparent. If developed with these principles, AI can amplify human creativity and rigor; without them, it risks scaling the production of plausible but unreliable research artifacts.

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Responsible Use and Limitations

This work is intended to inform responsible use of AI-assisted research tools, not to endorse replacing human scientific judgment with full automation. Current systems are most reliable when used to assist retrieval, drafting, coding, visualization, review support, and dissemination, while humans retain responsibility for novelty, interpretation, verification, authorship, and accountability. Because the field evolves rapidly, this paper should be read as a structured snapshot through our search cutoff, and AI-generated research outputs should be independently verified before scholarly use.

Appendix

A Auto-Research Tool Inventory

This appendix provides an inventory of surveyed systems, datasets, benchmarks, and evaluation methods, organized by stage. Empirical policy studies and broader critical analyses are discussed in the main text.

A.1 Phase 1: Creation

Table 4 Comprehensive inventory: S1 Idea Generation. †Evaluation information might be uncertain.



S1 Idea Generation

Generating, refining, and evaluating novel research hypotheses. Techniques range from knowledge graph reasoning and retrieval-augmented generation to Multi-Agent collaboration for structured hypothesis formation.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
LLM Internal Knowledge-Based Generation							
1	Chain of Ideas	[183]	arXiv'24		LLM Internal		Comparable to human quality; $0.50/idea min cost
2	ResearchAgent	[13]	NAACL'25		LLM Internal		Human + model eval; academic graph feedback
3	SciMON	[376]	ACL'24		LLM Internal		Mitigates shallow novelty; iterative refinement
4	Idea Gen Agent	[327]	arXiv'24		LLM Internal	-	100+ NLP researchers; LLM ideas higher novelty ($p < 0.05$)
5	IRIS	[88]	ACL'25		LLM Internal		MCTS adaptive reasoning; human-in-the-loop platform
6	Spark	[305]	ICCC'25		LLM Internal	-	Judge model trained on 600K OpenReview reviews
7	Tree-of-Ideas	[192]	arXiv'26		LLM Internal	-	Cross-trajectory reasoning over scholarly evolution for ideation
8	IDEAgent	[105]	arXiv'26		LLM Internal	-	Quality-diversity search over the research idea space
External Signal-Driven Generation							
9	MOOSE-Chem	[433]	ICLR'25		External Signal	-	Rediscovers hypotheses from 51 high-impact papers
10	Nova	[131]	arXiv'24		External Signal	-	3.4× more novel ideas; 2.5× more top-rated
11	SciAgents	[91]	arXiv'24		External Signal		Multi-agent reasoning over knowledge graphs
12	SciPIP	[380]	arXiv'24		External Signal		Multi-domain; paper-anchored idea generation
13	IdeaSynth	[277]	CHI'25		External Signal	-	20-user study; more alternatives explored vs baseline
14	MOOSE-Chem2	[432]	NeurIPS'25		External Signal	-	Fine-grained, experimentally actionable hypotheses
15	HALO	[146]	arXiv'26		External Signal	-	Interactive co-abductive reasoning for hypothesis generation
16	TCA-SIR	[349]	arXiv'26		External Signal	-	Target-conditioned abstractions for inspiration retrieval
17	ECLAIR	[368]	arXiv'26		External Signal	-	Causally-grounded discovery in empirical software engineering
18	IdeaAnchor	[50]	arXiv'26		External Signal	-	Literature-grounded training and held-out ideation assessment; execution not established
Multi-Agent Collaborative Generation							
19	Combi. Creativity	[104]	arXiv'24		Multi-Agent	-	+7–10% similarity scores; cross-domain composition
20	Deep Ideation	[472]	arXiv'25		Multi-Agent		+10.67% quality; surpasses conference acceptance levels
21	VirSci	[344]	ACL'25		Multi-Agent		Outperforms single-agent on novelty†
22	Multi-Agent Dial.	[364]	SIGDIAL'25		Multi-Agent	-	Optimal at 3 critique-revision rounds†
23	Artificial Hivemind	[148]	NeurIPS'25		Multi-Agent	-	26K queries; diversity collapse across models
24	Auditable AI Sci.	[352]	arXiv'26		Multi-Agent	-	Hypothesis evolution protocol for auditable LLM scientists

Table 4 Comprehensive inventory: S1 Idea Generation. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Multi-Agent Collaborative Generation							
25	Diverse Personalized Ideation	[412]	arXiv'26		Multi-Agent	-	Counters AI-induced homogenization across users
Novelty and Feasibility Assessment							
26	IdeaBench	[109]	KDD'25		Evaluation	-	2,374 papers; 8 domains; novelty >0.6, feasibility <0.5
27	LiveIdeaBench	[294]	arXiv'24		Evaluation	-	40+ models; 1,180 keywords; 22 scientific domains
28	AI Idea Bench 2025	[282]	arXiv'25		Evaluation		3,495 papers; alignment + general reference eval
29	HeurekaBench	[268]	ICLR'26		Evaluation		+22% with critic module; open-ended science tasks
30	ResearchBench	[218]	ACL'26		Evaluation	-	12 disciplines; inspiration retrieval + ranking
31	HindSight	[147]	arXiv'26		Evaluation	-	LLM novelty negatively correlated with impact (ρ=−0.29)
32	Rubric Rewards	[95]	arXiv'25		LLM Internal	-	70% expert preference; RL with rubric self-grading
33	DeepInnovator	[70]	arXiv'26		LLM Internal		80–94% win rates vs frontier; 14B model
34	FlowPIE	[377]	arXiv'26		External Signal	-	Higher novelty, feasibility, diversity vs baselines
35	Diverse Hypo. Search	[372]	arXiv'26		LLM Internal	-	Diversity-aware population search; molecular/equation/algorithm discovery
36	SoundnessBench	[123]	arXiv'26		Evaluation	-	Tests if AI scientists distinguish sound vs. flawed ideas; optimism bias
37	LLM-Judge Novelty	[334]	arXiv'26		Evaluation	-	Limits of LLM-as-Judge for novelty assessment
38	LigBench	[371]	arXiv'26		Evaluation	-	Human-aligned benchmark for LLM research idea generation
39	Reconstruction	[42]	arXiv'26		Evaluation	-	Blind recovery of ideas from pre-publication bibliographies
40	HypoForge	[280]	arXiv'26		Multi-Agent	-	Reusable scientific skills across generation and testing
41	SGHA	[92]	arXiv'26		Agent	-	Corpus-first structural-gap discovery on a local LLM
42	MAIL	[12]	arXiv'26		Agent	-	Memory-driven incremental literature-grounded hypotheses
43	RATIO	[320]	arXiv'26		Benchmark	-	Relevance defined by three typed ideation moves
44	Lit2Test	[389]	arXiv'26		Benchmark	-	Each proposal precommits its own falsifying outcome
45	AgentIdeaBench	[243]	arXiv'26		Benchmark		Static vs. active ideation; 33 models, 40 subfields; critic-based novelty
46	IdeaAMBIG	[236]	arXiv'26		Benchmark	-	660 specifications; real and synthetic gaps; localization vs. clarification
47	MechHypoBench	[407]	arXiv'26		Evaluation	-	Consequences under withheld conditions; controlled mechanisms and fixed LLM interpretation

Table 5 Comprehensive inventory: S2 Literature Review. †Evaluation information might be uncertain.



S2 Literature Review

Retrieving, synthesizing, and organizing prior work into coherent narratives. Modern approaches span semantic retrieval, citation-graph traversal, and Deep Research agents that iteratively explore the literature.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Literature Retrieval							
1	CiteME	[276]	arXiv'24		Retrieval	-	Citation fidelity benchmark
2	LitLLM	[4]	arXiv'24		Retrieval	-	LLM + academic database integration
3	LitSearch	[6]	arXiv'24		Retrieval		Retrieval precision benchmark
4	PaperQA2	[335]	arXiv'24		Retrieval		Matches/exceeds expert on 3 tasks; 70% contradiction val.
5	OpenResearcher	[481]	EMNLP'24		Retrieval	-	RAG + graph traversal for literature exploration
6	PaSa	[116]	arXiv'25		Retrieval		Agentic multi-step iterative retrieval
7	Search, Inspect, Fetch	[378]	arXiv'26		Retrieval	-	Structure-aware Boolean retrieval for deep-research agents
8	Rubric Reranker	[212]	arXiv'26		Retrieval	-	Trains document rerankers against search rubrics
9	Personalized DR Refinement	[442]	arXiv'26		Retrieval	-	Graph-scaffolded evidence grounding for query refinement
Survey & Related Work Generation							
10	ChatPaper	[227]	GitHub'23		Generation		19K+ GitHub stars; arXiv summarization tool
11	PaperQA	[232]	arXiv'23		Generation		8K+ GitHub stars; RAG for scientific Q&A
12	AutoSurvey	[384]	arXiv'24		Generation		First end-to-end LLM survey drafting system
13	GPT Researcher	[67]	GitHub'24		Generation		26K+ GitHub stars; comprehensive report generation
14	LLMs for Lit.	[357]	arXiv'24		Generation	-	Hallucination analysis; models still generate errors†
15	STORM	[318]	arXiv'24		Generation		Multi-perspective question-asking for outlines
16	Agentic AutoSurvey	[214]	arXiv'25		Generation	-	Multi-agent role decomposition†
17	Citegeist	[15]	arXiv'25		Generation	-	Dynamic RAG pipeline on arXiv corpus
18	IterSurvey	[455]	arXiv'25		Generation		Iterative outline planning with stability checks
19	LiRA	[94]	arXiv'25		Generation	-	Multi-agent retrieval + verification + narrative
20	SurveyForge	[421]	arXiv'25		Generation		Outperforms AutoSurvey on outline quality†
21	SurveyG	[253]	arXiv'25		Generation	-	Three-layer citation graph (Foundation/Dev/Frontier)
22	SurveyX	[198]	arXiv'25		Generation	-	+0.259 content quality improvement; near expert level
23	InteractiveSurvey	[393]	arXiv'25		Generation		User-customizable reference categorization + outlines

Table 5 Comprehensive inventory: S2 Literature Review. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Survey & Related Work Generation							
24	CiteLLM	[126]	arXiv'26		Generation	-	Hallucination-free via trusted repository routing
25	STRUCTSURVEY	[272]	arXiv'26		Generation	-	Structured agentic retrieval for survey generation
Deep Research Agents							
26	ASReview	[365]	Nature MI'21		Deep Research		Active learning; up to 95% effort reduction
27	CHIME	[129]	arXiv'24		Deep Research	-	Hierarchical organization of scientific studies
28	DeepResearch-Agent	[336]	GitHub'25		Deep Research		Hierarchical multi-agent; planner + sub-agents
29	DeerFlow	[26]	GitHub'25		Deep Research		Sub-agents with shared memory; sandboxed execution
30	OpenScholar	[11]	Nature'26		Deep Research	-	45M papers; +6.1% over GPT-4o, +5.5% over PaperQA2
31	AutoAgent	[355]	arXiv'25		Deep Research	-	Universal LLM compatibility; GAIA benchmark
32	Tongyi DeepResearch	[9]	GitHub'25		Deep Research		30.5B params (3.3B activated); SOTA on Deep Research
33	O-Researcher	[439]	arXiv'26		Deep Research	-	Multi-agent distillation + agentic RL
34	OpenResearcher	[196]	arXiv'26		Deep Research		54.8% BrowseComp-Plus; 97K+ trajectories
35	AREX	[225]	arXiv'26		Deep Research	-	Recursively self-improving deep research agent
36	Predictive Navigation	[482]	arXiv'26		Deep Research	-	Pretraining objective tailored to research navigation
37	On-Device DR (4B)	[28]	arXiv'26		Deep Research	-	Exposure bounds faithfulness; retrieval bounds coverage
38	Carnot	[297]	VLDB'26		Deep Research	-	Interpretable, interactive, optimized deep-research execution
39	Marginal Value Est.	[167]	arXiv'26		Deep Research	-	Stops search when added tokens no longer change the answer
40	Retrieval-Aware Control	[339]	arXiv'26		Deep Research	-	Repairs stagnation in deep research reasoning
41	Analogical Deep Research	[47]	arXiv'26		Deep Research	-	Retrieves historical analogies for foresight analysis
42	Plato-Bio	[55]	arXiv'26		Deep Research	-	Verification-first biological novelty screening
43	Albilich	[96]	arXiv'26		Deep Research	-	Proof-state orchestration with CAS integration
Retrieval and Synthesis Quality Assessment							
44	DeepScholar-Bench	[270]	arXiv'25		Evaluation		Coverage, coherence, factual accuracy benchmark
45	ReportBench	[185]	arXiv'25		Evaluation		100-prompt benchmark from 678 filtered survey papers
46	IDRBench	[76]	arXiv'26		Evaluation	-	100 tasks; interactive Deep Research evaluation

Table 5 Comprehensive inventory: S2 Literature Review. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Retrieval and Synthesis Quality Assessment							
47	ScholarGym	[321]	arXiv'26		Evaluation	-	2,536 queries; query planning + tool invocation
48	SciNetBench	[316]	arXiv'26		Evaluation	-	18M papers; relation-aware retrieval <20%
49	Self-Evolving Retrieval	[66]	arXiv'26		Retrieval	-	Agent that adapts its own literature-search policy over time
50	MasterSet	[289]	arXiv'26		Retrieval	-	Large-scale must-cite citation recommendation
51	DeepSurvey	[436]	arXiv'26		Generation	-	Evidence-constrained citation assignment
52	AutoResearchBench	[409]	arXiv'26		Evaluation	-	Benchmark for complex multi-hop scientific discovery
53	PaperMind	[476]	arXiv'26		Evaluation	-	Multimodal reasoning + critique over papers
54	DRNOISE	[256]	arXiv'26		Evaluation	-	Benchmarks agents in misleading evidence environments
55	HiEviDR-Bench	[346]	arXiv'26		Evaluation	-	Hierarchical evidence aggregation in deep research
56	SciExplore	[358]	arXiv'26		Evaluation	-	Scientific navigation through information integration
57	WANDR	[274]	arXiv'26		Evaluation	-	Benchmark spanning wide as well as deep research
58	QA-to-DR Bench	[370]	arXiv'26		Evaluation	-	Verifiable tasks grown by iterative task evolution
59	DAS	[418]	arXiv'26		Agent	-	Stateful closed loop for publication-oriented surveys
60	Crase	[115]	arXiv'26		Agent	-	Bounded graph exploration; 3x recall@50 at 1/3 cost
61	ITER	[37]	arXiv'26		Retrieval	-	Retriever trained on agent interaction trajectories
62	DeepWeaver	[382]	arXiv'26		Agent	-	Weaves fragmented evidence into well-cited answers
63	AgentR	[304]	arXiv'26		Agent	-	Durable, recoverable literature-review artifacts
64	Tree-of-Concerns	[241]	EMNLP'26		Multi-Agent	-	Skeptic personas surface unstated limitations
65	Multi-Aspect Retr.	[392]	arXiv'26		Benchmark	-	Same-paper retrieval queried from different aspects
66	ScholarCatalyst	[161]	arXiv'26		Benchmark		Author-judged inspiration retrieval; 207 projects; temporal corpus cutoffs
67	INSPIRE	[64]	arXiv'26		Benchmark	-	476 CS targets; exposure, selection, and ranking of cited antecedents
68	CESS	[405]	arXiv'26		Evaluation	-	Audits adaptive evidence selection against an explicit candidate pool

Table 6 Comprehensive inventory: S3 Coding & Experiments. †Evaluation information might be uncertain.



S3 Coding & Experiments

Translating ideas into executable code, running experiments at scale, and analyzing results. This stage spans code generation, Paper-to-Code translation, autonomous experiment orchestration, and result interpretation.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Code Generation							
1	SWE-bench	[151]	ICLR'24		Code Gen.		2,294 real GitHub issues; Verified split (500 problems)
2	SWE-agent	[423]	arXiv'24		Code Gen.		Agent–computer interface paradigm for coding
3	OpenHands	[381]	ICLR'25		Code Gen.		Open platform for generalist coding agents
4	SWE-bench Pro	[62]	arXiv'25		Code Gen.	-	1,865 enterprise problems; best score 23%
5	SWE-EVO	[359]	arXiv'25		Code Gen.	-	Software evolution benchmark; best score 25%
Paper-to-Code							
6	FunSearch	[293]	Nature'24		Paper-to-Code		New cap-set solutions; evolutionary program search
7	SciCode	[361]	arXiv'24		Paper-to-Code		Research-level coding across math, physics, chemistry
8	PaperBench	[342]	arXiv'25		Paper-to-Code		20 ICML'24 papers; 8,316 gradable subtasks
9	PaperCoder	[314]	arXiv'25		Paper-to-Code		3-stage multi-agent; ML papers to code repos
10	ResearchCodeBench	[134]	arXiv'25		Paper-to-Code	-	212 novel ML tasks; best 37.3% (Gemini-2.5-Pro)
11	SciReplicate-Bench	[402]	arXiv'25		Paper-to-Code		100 tasks from 36 NLP papers; 39% ceiling
12	PaperCompiler	[216]	arXiv'26		Paper-to-Code	-	Paper-grounded repository specifications; fidelity assessed on Paper2CodeBench
Experiment Execution & Orchestration							
13	BioPlanner	[261]	arXiv'23		Execution		Biological protocol planning evaluation
14	CRISPR-GPT	[284]	arXiv'24		Execution	-	Gene-editing experiment design assistance
15	DS-Agent	[108]	arXiv'24		Execution		End-to-end data science workflow automation
16	MLE-Bench	[30]	arXiv'24		Execution	-	75 Kaggle competitions benchmark
17	MLAgentBench	[137]	arXiv'24		Execution		13 ML experimentation tasks benchmark
18	MLR-Copilot	[188]	arXiv'24		Execution	-	IdeaAgent + ExperimentAgent dual-agent pipeline
19	AIDE	[150]	arXiv'25		Execution	-	SOTA on MLE/RE-Bench; tree search in code space
20	AlphaEvolve	[259]	arXiv'25		Execution	-	LLM-generated mutations + automated evaluators
21	AutoReproduce	[474]	arXiv'25		Execution		Paper lineage algorithm for experiment reproduction
22	CURIE	[168]	arXiv'25		Execution		Rigorous automated experimentation framework
23	MLGym	[251]	arXiv'25		Execution	-	AI research agent gym benchmark
24	MLR-Bench	[36]	arXiv'25		Execution	-	201 tasks (NeurIPS/ICLR/ICML); 80% fabrication rate
25	Exe.-Grounded	[329]	arXiv'26		Execution	-	69.4% vs 48.0% GRPO; parallel GPU search
26	Learn to Discover	[450]	arXiv'26		Execution	-	Test-time training + RL; math, GPU kernel, biology
27	SciNav	[460]	arXiv'26		Execution	-	Pairwise tree-search branch selection

Table 6 Comprehensive inventory: S3 Coding & Experiments. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Experiment Execution & Orchestration							
28	FrontierScience	[375]	arXiv'26		Execution	-	Expert-level tasks; Olympiad + PhD difficulty
29	AutoTTS	[480]	arXiv'26		Execution		Coding-agent discovery of test-time scaling strategies
30	AutoScientists	[83]	arXiv'26		Execution	-	Self-organizing agent teams for long-running experiments
31	EurekAgent	[408]	arXiv'26		Execution	-	Agent environment engineering for autonomous discovery
32	ResearchTrails	[97]	arXiv'26		Decision Support		Commit-derived research decisions; LLM-judged next-decision similarity
Code Correctness and Reproducibility Assessment							
33	DiscoveryBench	[237]	arXiv'24		Analysis		Data-driven insight extraction benchmark
34	DiscoveryWorld	[144]	arXiv'24		Analysis		120 tasks; 8 topics; 3 difficulty levels
35	InfiAgent	[132]	arXiv'24		Analysis	-	End-to-end data analysis workflow benchmark
36	ScienceAgentBench	[49]	arXiv'24		Analysis	-	Rigorous data-driven scientific discovery assessment
37	LAB-Bench	[172]	arXiv'24		Execution		Multi-domain biology research task benchmark
38	KernelBench	[264]	arXiv'25		Execution		GPU kernel generation benchmark
39	TritonBench	[181]	arXiv'25		Execution		Triton operator generation benchmark
40	AstaBench	[24]	arXiv'25		Execution		2,400+ problems; multi-domain scientific research
41	ResearchClawBench	[413]	arXiv'25		Execution		Scientist-aligned workflow benchmark
42	EXP-Bench	[169]	ICLR'26		Execution		461 tasks from 51 AI papers
43	PostTrainBench	[287]	arXiv'26		Execution		LLM post-training automation benchmark
44	EvoDS	[434]	arXiv'26		Execution	-	Self-evolving DS agent; skill learn + context mgmt
45	MLReplicate	[81]	arXiv'26		Analysis	-	End-to-end ML reproducibility benchmark
46	BeyondSWE	[35]	arXiv'26		Execution		500 instances / 246 repos; cross-repo, domain-fix, doc2repo
47	NatureBench	[386]	arXiv'26		Analysis	-	Tests if coding agents match Nature-family paper SOTA
48	ReproAgent	[133]	EMNLP'26		Agent	-	Contract-guided recovery of the split specification
49	DeepRepro	[337]	CIKM'26		Agent	-	State-aware subplanning over evolving repositories
50	Exp. Fidelity Audit	[444]	arXiv'26		Evaluation	-	Documents silent methodological hallucinations
51	Praxist	[182]	arXiv'26		Agent	-	Solution lineages trace gains to causes
52	Skill-Based Baselines	[246]	MICCAI'26		Agent	-	Literature-guided competitive baselines, 4 benchmarks
53	SciCoQA	[14]	ACL'26		Evaluation	-	Quality assurance for paper–code alignment
54	AgentActionBench	[125]	arXiv'26		Benchmark	-	150 papers; process traces and rubrics; 10% human-annotated subset
55	AutoDataBench	[449]	arXiv'26		Benchmark		Three controlled tasks isolate data interventions and data-effect reasoning

Table 7 Comprehensive inventory: S4 Tables & Figures. †Evaluation information might be uncertain.



S4 Tables & Figures

Creating Method Diagrams, result plots, comparison tables, and LaTeX formulas. Scientific visualization transforms raw experimental outputs into publication-quality charts, illustrations, and structured tables.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Scientific Figure Generation							
1	ChartGPT	[362]	arXiv'23		Data Viz	-	6-step reasoning for chart generation
2	MatPlotAgent	[431]	arXiv'24		Data Viz	-	+12.3 over GPT-4 base; VLM visual feedback†
3	CoDA	[48]	arXiv'25		Data Viz	-	+41.5% over baselines; multi-agent collaboration
4	PlotGen	[98]	arXiv'25		Data Viz	-	4–6% improvement over baselines†
5	VIS-Shepherd	[266]	arXiv'25		Figure Editing	-	Constructive critique feedback framework
6	DiagramAgent	[391]	CVPR'25		Data Viz	-	4 specialized agents; 8 diagram categories
7	StarVector	[292]	CVPR'25		Method Diagrams	-	Scalable SVG generation from descriptions†
8	VisCoder	[255]	EMNLP'25		Data Viz	-	VisCode-200K dataset; 90%+ execution pass rate†
9	AI-Generated Figures	[32]	arXiv'26		Policy	-	Publisher policy survey (Nature, Science, etc.)
10	AutoFigure-Edit	[205]	arXiv'26		Method Diagrams		Editable text-to-SVG scientific illustrations†
11	AutoFigure	[487]	ICLR'26		Method Diagrams		FigureBench (3,300 pairs); publication-ready illust.
12	PaperBanana	[485]	arXiv'26		Method Diagrams	-	292 test cases; outperforms baselines†
13	SAIL	[311]	arXiv'26		Figure Editing	-	Domain logic / code syntax separation
14	Can AI Draw Sci.	[33]	arXiv'26		Data Viz	-	Benchmark for scientific figure generation by T2I/MLLMs
15	SciDiagramEdit	[347]	arXiv'26		Figure Editing	-	Learns diagram edits from paper revision histories
16	GenGA	[158]	arXiv'26		Figure Editing	-	Editable, data-grounded graphical abstract generation
Table Understanding & Generation							
17	ArxivDIGESTables	[252]	EMNLP'24		Table Gen.	-	Literature comparison table synthesis
18	Chain-of-Table	[388]	ICLR'24		Table Reasoning	-	Multi-step table reasoning chains
19	ShowTable	[219]	CVPR'26		Table Viz	-	Collaborative reflection and refinement†
20	Table2LaTeX-RL	[206]	arXiv'25		Table Conversion	-	Image-to-LaTeX via reinforced multimodal LM

Table 7 Comprehensive inventory: S4 Tables & Figures. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Mathematical Formulas & TikZ							
21	AutomaTikZ	[16]	ICLR'24		TikZ	-	DaTikZ: first large-scale dataset (120K drawings)
22	DeTikZify	[17]	NeurIPS'24		TikZ	-	360K TikZ graphics; MCTS iterative refinement
23	TikZilla	[102]	arXiv'26		TikZ	-	3B/8B matches GPT-5; SFT+RL on expanded DaTikZ
24	Edit2TikZ	[467]	arXiv'26		TikZ	-	Benchmark for instruction-conditioned TikZ figure editing
25	EdiTikZ	[103]	arXiv'26		Figure Editing		Revision-derived TikZ editing; automated and human render/edit evaluation
Visual Fidelity and Scientific Accuracy Assessment							
26	PlotCraft	[457]	arXiv'25		Benchmark	-	1K-task benchmark; 48 chart types
27	TeXpert	[155]	SDP'25		Benchmark	-	3-level difficulty; 78.8%/58.7%/17.5%†
28	AbGen	[475]	ACL'25		Benchmark	-	1,500 ablation studies; 807 NLP papers
29	SciFig	[138]	arXiv'26		Benchmark	-	Rubric-based evaluation; 2K+ pipeline figures
30	SciFlow-Bench	[462]	arXiv'26		Benchmark	-	500 framework figures; inverse-parsing evaluation
31	FigureBench	[487]	ICLR'26		Benchmark		3,300 text-figure pairs; publication-ready eval
32	Crafter	[471]	arXiv'26		Method Diagrams	-	Multi-agent editable figure generation from heterogeneous inputs
33	DiagramRAG	[447]	arXiv'26		Method Diagrams	-	Retrieval-augmented scientific diagram generation
34	GeoSVG-RL	[190]	arXiv'26		Method Diagrams	-	Geometry-aware RL for layout-constrained text-to-SVG diagrams
35	CSPO	[430]	arXiv'26		Table Conversion	-	Component-specific optimization for table-to-LaTeX generation
36	SciFigQual-Bench	[63]	arXiv'26		Benchmark	-	Figure quality scored with full-manuscript context
37	SciFigAlign	[410]	arXiv'26		Benchmark	-	Aligns figures with manuscript evidence for scoring
38	SciFigPlag-Bench	[57]	arXiv'26		Benchmark	-	Provenance-aware scientific figure plagiarism detection
39	VLM Blind/Misled	[260]	arXiv'26		Benchmark	-	Behavioral evaluation of VLMs on scientific figures
40	FigTree	[215]	arXiv'26		Figure Generation	-	Paper-grounded recursive SVG construction; render-and-program refinement

A.2 Phase 2: Writing

Table 8 Comprehensive inventory: S5 Paper Writing. †Evaluation information might be uncertain.



S5 Paper Writing

Drafting, editing, and polishing academic manuscripts. AI assistance ranges from semi-automated grammar and citation tools to fully automated paper generation: the most commercially mature yet ethically contested stage.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Semi-Automated Writing Assistance							
1	CoAuthor	[174]	arXiv'22		Collaborative	-	Human–AI collaborative writing workflows
2	Script&Shift	[330]	CHI'25		Source Transform	-	CHI Honorable Mention; preserves cognitive engagement
3	AI Writing Study	[331]	AIED'25		Empirical Study	-	90-student RCT; purposeful AI fosters writing
4	OpenDraft	[61]	-		Full Draft Gen.		19 agents; 20K+ words in 10 min; verified citations
5	DraftMarks	[332]	arXiv'25		Transparency	-	Skeuomorphic visual traces for AI process transparency
6	PaperDebugger	[128]	arXiv'25		In-Editor Assist		Multi-agent Overleaf plugin (Reviewer+Enhancer+Scorer)
7	ScholarCopilot	[385]	arXiv'25		Citation Assist	-	40.1% top-1 citation accuracy (vs 15.0% E5-Mistral)
8	XtraGPT	[39]	arXiv'25		Post-Writing	-	1.5B–14B models; 7K papers; 140K revision pairs
9	LimAgents	[8]	arXiv'26		Limitations Gen.	-	OpenReview comments + citation network integration
10	AutoSupervision	[180]	arXiv'26		In-Editor Assist	-	Grounded revision verification closes the feedback loop
11	ReasFlow	[117]	arXiv'26		In-Editor Assist	-	Knowledge-structured assistance for applied mathematics
Fully Automated Paper Generation							
12	CycleResearcher	[394]	ICLR'25		E2E Gen.	-	5.36 ICLR scale (vs 5.24 preprint, 5.69 accepted)
13	Agent Laboratory	[308]	EMNLP'25		E2E Gen.	-	$2–13/paper; 84% cost reduction; 3.5–4.0 score
14	FutureGen	[7]	arXiv'25		Section Gen.	-	RAG-based Future Work section generation
15	AI Scientist	[223]	Nature'26		E2E Gen.		$15/paper; end-to-end across 3 ML subfields
16	APRES	[469]	arXiv'26		Rubric Revision	-	79% expert preference; citation-predictive rubrics

Table 8 Comprehensive inventory: S5 Paper Writing. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Fully Automated Paper Generation							
17	RWGBench	[406]	arXiv'26		Section Gen.	-	Evaluates scholarly positioning in related-work generation
Societal Analysis							
18	AI Writing Adoption	[114]	Nature'26		Measurement	-	41.3M papers; AI expands impact but contracts focus
19	Nature AI Survey	[249]	Nature'26		Survey	-	57% of researchers use AI in peer review
20	Denial of Science	[111]	arXiv'26		Measurement	-	Indirect data poisoning could industrialize scientific fraud
21	AI Slop OSS	[3]	arXiv'26		Measurement	-	AI-generated contributions strain open-source sustainability
Writing Quality and AI Detection Assessment							
22	Mapping LLM Use	[197]	arXiv'24		Detection	-	Up to 17.5% of CS papers AI-modified
23	CycleReviewer	[394]	ICLR'25		AI Judge	-	26.89% MAE reduction vs individual human reviewers
24	Stanford Agentic	[149]	Web'25		AI Judge	-	$\rho = 0.42$ vs human $\rho = 0.41$; matches consistency
25	SciIG	[87]	arXiv'25		Writing Bench	-	NAACL/ICLR 2025 introduction writing benchmark†
26	Watermarking	[288]	arXiv'25		Detection	-	Near-zero false-positive rate under controlled conditions
27	PaperWritingBench	[338]	arXiv'26		Benchmark	-	200 reverse-engineered top-tier conference papers
28	PaperMentor	[210]	arXiv'26		In-Editor Assist	-	Human-centered multi-agent Overleaf writing tutor
29	LECTOR	[403]	arXiv'26		Section Gen.	-	Introduction generation via joint reasoning-graph optimization
30	CiteTracer	[186]	arXiv'26		Detection	-	Multi-agent citation-hallucination detection
31	Process Eval	[333]	arXiv'26		Writing Bench	-	Keystroke-level study of AI- vs. human-draft revision
32	SciSlopBench / SciSlopHarness	[262]	arXiv'26		Evaluation / Revision	-	390 matched paper pairs; structural, argumentative, and artifact measures

A.3 Phase 3: Validation

Table 9 Comprehensive inventory: S6 Peer Review. †Evaluation information might be uncertain.



S6 Peer Review

Automated Review Generation, reviewer-paper matching, and review quality assessment. AI systems can produce structured critiques and predict acceptance decisions, though leniency bias and adversarial vulnerabilities persist.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Automated Review Generation							
1	ChatReviewer	[254]	GitHub'23		Review Gen.		ChatGPT-based strengths/weaknesses analysis
2	AI-Peer-Review	[273]	GitHub'24		Review Gen.		Multi-LLM independent reviews + meta-review synthesis
3	MARG	[60]	arXiv'24		Review Gen.	-	3.7 good comments/paper (2.2× over baseline)
4	Reviewer2	[86]	arXiv'24		Review Gen.	-	Two-stage prompt-based review aspect modeling
5	ReviewRL	[451]	EMNLP'25		Review Gen.	-	RL + RAG; factually grounded reviews
6	DeepReviewer	[486]	arXiv'25		Review Gen.	-	88.21% win rate vs GPT-o1; 64% accept/reject acc.
7	OpenReviewer	[142]	NAACL'25		Review Gen.	-	Llama-8B fine-tuned on 79K expert reviews
8	REMOR	[351]	arXiv'25		Review Gen.	-	Multi-objective RL review optimization
9	ScholarPeer	[101]	arXiv'26		Review Gen.	-	Context-aware multi-agent; literature verification
10	PeerCheck	[51]	arXiv'26		Review Gen.	-	Pushes LLM-generated reviews toward human-level quality
11	Local Pre-Screening	[23]	arXiv'26		Review Gen.	-	On-premise pre-screening before human triple-blind review
12	ReVoicer	[99]	arXiv'26		Review Gen.	-	Conversational voice annotation for LLM-assisted review
13	PaperDoctor	[202]	arXiv'26		Review Gen.		Evidence-linked diagnoses; typed verifiers and selective experiment reruns
14	ActReview	[235]	arXiv'26		Review Gen.	-	Rebuttal-supervised actionable feedback; 1,000-instance human-curated benchmark
Meta-Review & Reviewer Matching							
15	AgentReview	[153]	EMNLP'24		Meta-Review	-	Full review lifecycle simulation; social/authority bias
16	Meta-Review LLMs	[127]	NAACL'25		Meta-Review	-	40 ICLR papers; GPT-3.5/LLaMA-2/PaLM-2 compared
17	RATE	[213]	arXiv'26		Matching	-	Expertise-based matching via profile distillation
Adversarial Attacks & Bias Analysis							
18	Raina *et al.*	[285]	EMNLP'24		Adversarial	-	Benign adjectives as universal adversarial triggers
19	AI Review Lottery	[298]	arXiv'24		Bias Analysis	-	15.8% ICLR reviews AI-assisted; +4.9pp borderline
20	Ye *et al.*	[441]	arXiv'24		Adversarial	-	Scores inflated to ∼8; 5% manipulation flips 12%
21	Breaking Reviewer	[203]	arXiv'25		Adversarial	-	Systematic adversarial robustness evaluation
22	LLM Reviewer Bias	[484]	arXiv'25		Bias Analysis	-	1,441 papers; 95.8% rejected misclassified as acceptable
23	Prompt Injection	[159]	arXiv'25		Adversarial	-	White-text injection; up to 100% acceptance scores
24	Sahoo *et al.*	[302]	arXiv'25		Adversarial	-	+13.95 on Mistral; 13 LLMs; 15 attack strategies
25	Zhou *et al.*	[483]	arXiv'25		Adversarial	-	+1.24 to +2.80 from hype prose; 10.00 under iteration
26	Gaming AI Reviews	[184]	arXiv'26		Adversarial	-	Gaming AI-assisted peer reviews poses new risks
27	Phantom Refs	[296]	arXiv'26		Adversarial	-	Hallucinated citations survive top-tier peer review

Table 9 Comprehensive inventory: S6 Peer Review. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Adversarial Attacks & Bias Analysis							
28	Rhetorical Reward-Hacking	[187]	arXiv'26		Adversarial	-	Confident framing shifts AI review independent of substance
Detection & Policy							
29	AI Detection	[445]	arXiv'25		Detection	-	788,984 AI-written reviews; 18 detection algorithms
30	AI Use Rejects	[93]	Nature'26		Policy	-	497 papers rejected (∼2% of submissions)
31	Nature AI Survey	[249]	Nature'26		Survey	-	1,600 academics; 57% use AI in peer review
32	Policy Enforcement	[301]	arXiv'26		Policy	-	All 5 SOTA detectors misclassify LLM-polished reviews
33	Reviewer Feedback	[45]	CHI'26		Empirical	-	ICLR 2025 live process; reviewer engagement study†
34	AAAI-26 Pilot	[19]	arXiv'26		Empirical	-	AI-assisted peer review at scale; the AAAI-26 pilot
35	Reviewer AI Policies	[77]	arXiv'26		Policy	-	Maps community AI policies against resulting review quality
Review Consistency and Bias Assessment							
36	Review Survey	[491]	IF'25		Survey	-	Comprehensive taxonomy of review methods
37	Stanford Agentic	[149]	Web'25		Quality	-	$\rho = 0.42$ vs human $\rho = 0.41$; matches consistency
38	ClaimCheck	[263]	EMNLP'25		Quality	-	LLM critique grounding; gaps in factual basis
39	ReViewGraph	[191]	AAAI'26		Quality	-	+15.73% avg improvement via heterogeneous graph
40	ReviewAgents	[84]	arXiv'25		Quality	-	37,403 papers; 142,324 reviews; Review-CoT dataset
41	ICLR 2025 Study	[360]	NMI'26		Quality	-	22,467 reviews; 89% quality improved; no acceptance effect
42	AI Reviewer Limits	[162]	arXiv'26		Quality	-	45 experts; AI vs. human reviews of Nature-family papers
43	PRISM	[220]	arXiv'26		Quality	-	Multi-dimensional benchmark for LLM peer reviewers
44	ProReviewer	[71]	arXiv'26		Review Gen.	-	Proactive investigation agent; probes suspicious paper parts
45	MERIT	[437]	arXiv'26		Matching	-	Rubric-informed reviewer assignment via expertise modeling
46	Present. Gaming	[428]	arXiv'26		Adversarial	-	Presentation-only revisions game AI review without prompts
47	LLMs Favor LLMs?	[319]	arXiv'26		Bias Analysis	-	Apparent self-preference is largely general leniency
48	LLM-Human Alignment	[27]	arXiv'26		Quality	-	Measures alignment of LLM reviews with human peer review
49	Epistemic Reliability	[40]	arXiv'26		Quality	-	Tests genuine analysis vs articulated intuition in judges
50	SurveyReview	[464]	arXiv'26		Quality	-	Reviewer-aligned benchmark for survey evaluators
51	Metag	[348]	arXiv'26		Dataset	-	349 action items tied to review–rebuttal paper diffs
52	VERA-RL	[189]	EMNLP'26		Training	-	RL for proactive error detection; VERA-13K, 6 categories
53	Peerify	[58]	arXiv'26		Benchmark	-	800 review claims; manuscript evidence retrieval; 300-claim human audit
54	HalluPeer	[204]	arXiv'26		Benchmark		Review-specific hallucination detection; synthetic injections and authentic checks

Table 10 Comprehensive inventory: S7 Rebuttal & Revision. †Evaluation information might be uncertain.



S7 Rebuttal & Revision

Analyzing reviewer comments and generating evidence-grounded responses. The newest frontier in AI auto-research, rebuttal automation, is decisive for roughly one in five borderline submissions at major venues.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Reviewer Comment Analysis							
1	ReviewMT	[353]	arXiv'24		Analysis	-	26,841 papers; 92,017 reviews; multi-turn dialogue
2	ICLR Rebuttal Study	[156]	arXiv'25		Analysis	-	ICLR 2024–2025; score transition analysis
Automated Rebuttal Generation							
3	ReviewerToo	[303]	arXiv'25		Modular Pipeline	-	81.8% accept/reject accuracy (vs 83.9% human)
4	RebuttalAgent	[119]	ICLR'26		Rebuttal Gen.		18.3% avg improvement; ToM-grounded
5	Author-in-the-Loop	[295]	ACL'26		Author-Aware	-	Integrates author expertise and intent
6	DRPG	[112]	arXiv'26		Rebuttal Gen.		98%+ planning accuracy; surpasses avg human quality
7	Paper2Rebuttal	[234]	arXiv'26		Rebuttal Gen.	-	Evidence-centric rebuttal planning
Rebuttal Effectiveness Assessment							
8	Re^2	[454]	arXiv'25		Dataset	-	19,926 submissions; 70,668 reviews; 53,818 rebuttals
9	Commitment Checklist	[31]	arXiv'26		Benchmark	-	11.8 commitments/paper; ∼25% unfulfilled
10	Re^3Align	[295]	ACL'26		Dataset	-	First large-scale aligned review–response–revision triplets
11	RbtAct	[399]	arXiv'26		Analysis	-	Rebuttals as supervision for actionable review feedback
12	GoodPoint	[248]	arXiv'26		Analysis	-	Learns constructive feedback grounded in author responses
13	Defend	[160]	arXiv'26		Rebuttal Gen.	-	Minimal-guidance rebuttal generation; improved factual correctness
14	Rebuttals Move	[222]	arXiv'26		Analysis	-	Rebuttals move scores; initial-review structure bounds movement
15	Trust AI Reviews	[173]	arXiv'26		Analysis	-	Studies authors' response to AI-based reviews
16	AppliedScientist	[366]	arXiv'26		Revision	-	Iterative AI review; execution weaknesses easier to repair than idea weaknesses

A.4 Phase 4: Dissemination

Table 11 Comprehensive inventory: S8 Dissemination (Paper2X). †Evaluation information might be uncertain.



S8 Dissemination / Paper2X

Converting papers into posters, slides, videos, websites, and social media content. Each output format targets a different audience and demands its own design logic, AI tool chain, and trust considerations.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Paper2Poster							
1	P2P	[345]	ICLR'26		Paper2Poster	-	P2PInstruct 30K+ examples; 3 specialized agents
2	Paper2Poster	[267]	NeurIPS'25		Paper2Poster		$0.005/poster; 87% fewer tokens vs GPT-4o
3	PosterForest	[53]	arXiv'25		Paper2Poster	-	Hierarchical multi-agent collaboration†
4	PosterGen	[466]	arXiv'25		Paper2Poster	-	Aesthetic-aware multi-agent generation†
5	APEX	[323]	arXiv'26		Paper2Poster		First agentic interactive poster editing
6	PosterOmni	[43]	arXiv'26		Paper2Poster	-	6 unified poster tasks; outperforms open-source
7	PosterMELD	[130]	arXiv'26		Paper2Poster	-	Controllable design diversity with editable poster output
8	PROS	[230]	arXiv'26		Paper2Poster	-	User-governed PPTX repair; target resolution separated from quality uplift
Paper2Slides							
9	DOC2PPT	[80]	AAAI'22		Paper2Slides	-	5,873 paired document–slide decks
10	PPTAgent	[477]	EMNLP'25		Paper2Slides		PPTEval benchmark; 10,448 curated presentations
11	AutoPresent	[89]	CVPR'25		Paper2Slides	-	8B Llama model; SlidesBench (7K train, 585 test)
12	Paper2Slides	[122]	GitHub'25		Paper2Slides		4-stage RAG pipeline; one-click conversion
13	Auto-Slides	[429]	arXiv'25		Paper2Slides	-	Multi-agent Beamer generation; interactive editing†
14	PASS	[5]	arXiv'25		Paper2Slides	-	First combined slides + AI audio delivery†
15	SlideGen	[199]	arXiv'25		Paper2Slides	-	Multi-agent VLM coordination; editable PPTX output
16	Talk to Your Slides	[154]	arXiv'25		Paper2Slides	-	+34% instruction fidelity; 87% lower cost vs GUI
17	SlideTailor	[452]	AAAI'26		Paper2Slides		User-preference conditioned; chain-of-speech
18	DeepPresenter	[478]	arXiv'26		Paper2Slides		9B model competitive with frontier at lower cost

Table 11 Comprehensive inventory: S8 Dissemination (Paper2X). †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Paper2Slides							
19	Office Raccoon	[313]	Web'26		Paper2Slides	-	Page-level editing; template/brand-guideline learning†
20	SeaSlides	[72]	arXiv'26		Paper2Slides	-	Semantic abstraction layer for agentic slide generation
21	SlideLab	[367]	arXiv'26		Paper2Slides	-	Narrative planning and grounding; human preference and simulated-audience evaluation
Paper2Video							
22	Preacher	[207]	ICCV'25		Paper2Video		Top-down decomposition; 5 research fields
23	Paper2Video	[490]	arXiv'25		Paper2Video		101 paper–video pairs; +10% PresentQuiz accuracy
24	PresentAgent	[325]	EMNLP'25		Paper2Video		PresentEval benchmark; approaches human-level
25	Paper2Video Talks	[244]	arXiv'26		Paper2Video	-	A good talk teaches takeaways, not summaries
Paper2Web & Social Media							
26	Paper2Web	[46]	arXiv'25		Paper2Web		10,716 papers; multimedia-rich academic homepages
27	ResearchStudio-Reel	[404]	arXiv'26		Paper2Web	-	Automates paper to poster, video, and blog jointly
28	I-WebGenBench	[59]	arXiv'26		Paper2Web	-	Evaluates interactivity in LLM-generated scientific web apps
29	SciForge	[312]	arXiv'26		Paper2Web	-	AI-native multimodal workbench for discovery and presentation
Fidelity and Adoption Assessment							
30	PPTEval	[477]	EMNLP'25		Benchmark		Content, design, coherence; 10,448 presentations
31	PresentQuiz	[490]	arXiv'25		Benchmark		101 paper–video pairs; +10% over human on comprehension
32	PresentEval	[325]	EMNLP'25		Benchmark		End-to-end narrated video quality; near human-level
33	Any2Poster	[369]	arXiv'26		Paper2Poster	-	Poster generation from eight input modalities across domains
34	X+Slides	[38]	arXiv'26		Paper2Slides	-	Audience-conditioned slide generation benchmark
35	PresentAgent-2	[400]	arXiv'26		Paper2Video	-	Generalist agentic presentation-video generation
36	Sci. Comm. Correspondence	[233]	arXiv'26		Benchmark	-	Fine-grained paper–slides–video correspondence

A.5 Cross-Phase: End-to-End Systems

Table 12 Comprehensive inventory: End-to-End and Cross-Phase Systems. †Evaluation information might be uncertain.

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Fully Automated Research Systems							
1	AI Scientist	[223]	arXiv'24		E2E Pipeline		Pioneered E2E at $15/paper; ICLR-scale review
2	Agent Laboratory	[308]	EMNLP'25		E2E Pipeline	-	$2–13/paper; 3.5–4.0 NeurIPS scale
3	AI-Researcher	[356]	arXiv'25		E2E Pipeline	-	Scientist-Bench; approaches human-level quality
4	CycleResearcher	[394]	ICLR'25		E2E Pipeline	-	5.36 ICLR scale; cyclic write–review
5	AI Scientist v2	[419]	arXiv'25		E2E + Tree Search		ICLR 2025 ICBINB workshop; score 6.33
6	Kosmos	[242]	arXiv'25		E2E Pipeline	-	79.4% claim accuracy; 7 discoveries; 4 domains
7	Dolphin	[448]	ACL'25		E2E Pipeline	-	Closed-loop auto-research pipeline
8	CodeScientist	[145]	ACL'25		E2E Pipeline		Hypothesis to verification; closed-loop
9	InternAgent	[453]	arXiv'25		E2E Pipeline		Closed-loop hypothesis to verification
10	freephdlabor	[179]	arXiv'25		Multi-Agent	-	Personalized research group; continual automation
11	SciMaster	[29]	arXiv'25		Multi-Agent		General-purpose scientific AI agents
12	ARIS	[425]	GitHub'26		Skill Library		31 skills; score 5.0→7.5; 20+ GPU experiments
13	EvoScientist	[231]	arXiv'26		Multi-Agent		6 papers accepted at ICAIS'25
14	UniScientist	[177]	Web'26		Multi-Agent	-	30B open-source; beats Claude Opus 4.5
15	ASI-Evolve	[415]	GitHub'26		Multi-Agent		+0.97 DeltaNet; +18 MMLU; +12.5 GRPO
16	AiScientist-LH	[34]	arXiv'26		E2E + Hierarchical		Long-horizon ML engineering; File-as-Bus
17	AutoSOTA	[194]	arXiv'26		E2E Pipeline		Paper-to-code-to-SOTA optimization
18	CORAL	[283]	arXiv'26		Multi-Agent		Multi-agent evolution; SOTA on 10 tasks
19	FARS	[10]	Web'26		Multi-Agent	-	100 papers in 228h; avg 5.05 ICLR scale
20	AutoResearchClaw	[209]	GitHub'26		Multi-Agent		fully autonomous 23-stage pipeline
21	NVAITC AI Sci.	[135]	arXiv'26		E2E Pipeline	-	Governed E2E system; hypertension GWAS case study
22	DeepResearch Agent Sys.	[139]	arXiv'26		E2E Pipeline	-	End-to-end deep research agent system

Table 12 Comprehensive inventory: End-to-End and Cross-Phase Systems. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Fully Automated Research Systems							
23	ScienceFlow	[473]	arXiv'26		E2E Pipeline	-	Long-horizon agent for ML research and discovery
24	Spark-to-Paper	[281]	arXiv'26		E2E Pipeline	-	Idea-to-manuscript flow as one composable skill
25	OmniScientist	[176]	arXiv'26		E2E Pipeline	-	Omni-modal, omni-discipline AI scientist
26	ScientistTwo	[250]	arXiv'26		Multi-Agent	-	Problem-driven experiments and simulated review/rebuttal; automated review scores
27	YouRA	[397]	arXiv'26		E2E Pipeline		Persistent evidence state and recovery; ten-task MLR-Bench end-to-end subset
Domain-Specific Systems							
28	Coscientist	[21]	Nature'23		Chemistry	-	Autonomous chemistry; LLM-driven tool use
29	AlphaFold 3	[1]	Nature'24		Biology	-	Biomolecular structure prediction
30	ChemCrow	[25]	NMI'24		Chemistry	-	Chemistry tool orchestration
31	Med. AI Scientist	[398]	arXiv'26		Medicine	-	Clinical research automation
32	Cognitive Scientist	[143]	arXiv'26		Psychology	-	Closes the loop to discover psychological theories
33	Molecular Closed-Loop	[258]	arXiv'26		Chemistry	-	Closed-loop research certifying generalizable improvements
34	Auto Research (Materials)	[257]	arXiv'26		Materials	-	Auditable workflows with held-out transfer evidence
35	OmniQEC	[420]	arXiv'26		Quantum	-	Discovers practical quantum error-correcting codes
36	Andy	[387]	arXiv'26		Mathematics	-	Mathematical agent for rigorous proof and research
37	BaT	[208]	arXiv'26		Medicine	-	Self-evolving medical research agent with stage rubrics
38	Agentic Sci. Community	[221]	arXiv'26		Multi-Agent	-	Agent community for automated neural operator discovery
39	WARA	[110]	ICCC'26		Engineering	-	Closed-loop multi-agent wireless optimization autoresearch
40	MMResearch	[217]	arXiv'26		Multimodal		Budgeted multimodal post-training; checkpoint verification and candidate retention
Evolutionary & Self-Improving Systems							
41	ShinkaEvolve	[171]	arXiv'25		Evolutionary		Open-ended sample-efficient program evolution
42	Darwin Godel Machine	[456]	arXiv'25		Self-Improving		Open-ended evolution of self-improving agents
43	Self-Driving Lab	[141]	arXiv'26		Self-Improving	-	Agentic self-driving lab compresses validation bottleneck
44	Meta-Reflection	[470]	arXiv'26		Self-Improving	-	Autonomous discovery via iterative meta-reflection
45	Project2Task	[411]	arXiv'26		Self-Improving	-	Graph-guided project-level planning for research

Table 12 Comprehensive inventory: End-to-End and Cross-Phase Systems. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Evolutionary & Self-Improving Systems							
46	Adaptive Inquiry	[79]	arXiv'26		Self-Improving	-	Knowledge revision as the unit of search
47	CausalForge	[354]	arXiv'26		Self-Improving	-	Formally grounded self-improving causal research
48	World-Model Scaling	[427]	arXiv'26		Self-Improving	-	Uses world models as the space for scaling research agents
49	Research Preference Models	[78]	arXiv'26		Self-Improving	-	Learned preference signal over research judgments
50	AIDE^2	[341]	arXiv'26		Self-Improving	-	Agent-code evolution; fixed backbones; four external evaluation benchmarks
Research Platforms & Infrastructure							
51	R&D-Agent	[426]	arXiv'25		Infrastructure		Researcher+Developer dual-agent; MLE-Bench top
52	autoresearch	[157]	GitHub'25		Infrastructure		∼12 exp/hour overnight
53	Google Co-Scientist	[100]	arXiv'25		Platform	-	Multi-agent hypothesis gen + validation
54	ResearchTown	[443]	ICML'25		Multi-Agent		Simulates research community with LLM agents
55	AgentRxiv	[307]	arXiv'25		Multi-Agent	-	11.4% improvement on MATH-500
56	LabClaw	[401]	Web'26		Skill Library		206 biomedical skills; always-on autonomous lab agent
57	PiFlow	[278]	arXiv'25		Multi-Agent	-	Principle-aware scientific discovery
58	AutoSci	[279]	arXiv'26		E2E Pipeline	-	Memory-centric full-lifecycle agent; self-improving proc.
59	ScientistOne	[238]	arXiv'26		E2E Pipeline	-	Chain-of-evidence verifiability against fabrication
60	Arbor	[152]	arXiv'26		Multi-Agent	-	Coordinator + executor agents; hypothesis-tree refinement
61	EvoMaster	[489]	arXiv'26		Self-Improving	-	Self-evolving foundational agent for agentic science at scale
62	XScientist	[226]	arXiv'26		Infrastructure	-	Git-like protocol for long-running autonomous research
63	Auditable Records	[492]	arXiv'26		Infrastructure	-	Converts agent trajectories into reviewable evidence
64	PEARL	[343]	arXiv'26		Infrastructure	-	Auditable repair of scientific reasoning graphs
65	IdeaTrail	[106]	arXiv'26		Infrastructure	-	Publishes full-process agent ideation trajectories
66	EviGraph	[291]	arXiv'26		Infrastructure	-	Evidence-guided autonomous research agents
67	Discovery Environments	[417]	arXiv'26		Infrastructure	-	Scales environments for turn-level agentic RL

Table 12 Comprehensive inventory: End-to-End and Cross-Phase Systems. †Evaluation information might be uncertain. *(continued)*

#	Method	Ref	Venue	Link	Category	GitHub	Evaluation
Lifecycle Benchmarks							
68	ResearchClawBench	[414]	arXiv'26		Benchmark	-	E2E research tasks grounded in hidden papers + rubrics
69	Act as Real	[373]	arXiv'26		Benchmark	-	Full-lifecycle suite for frontier models + harnesses
70	ResearchArena	[468]	arXiv'26		Benchmark	-	Minimal scaffold for running/assessing the complete loop
71	Quantum Theorem Bench	[459]	arXiv'26		Benchmark	-	Agents proving theorems in quantum algorithms/information
72	SciDiscovery Readiness	[324]	arXiv'26		Benchmark	-	Capability-oriented benchmark for AI scientists
73	ARAC	[56]	arXiv'26		Benchmark	-	Scores alignment and completeness of end-to-end runs
74	Beyond Final Scores	[195]	arXiv'26		Benchmark	-	Systematic evaluation of long-horizon R&D agents
75	Replication Training	[69]	arXiv'26		Benchmark	-	Trains AI scientists to replicate research
76	Rubric Meta-Eval	[124]	arXiv'26		Benchmark	-	Asks whether LLMs can write reliable reproduction rubrics
77	Fast-Moving Domains	[22]	arXiv'26		Benchmark	-	Contamination-resistant tests from post-training-cutoff domains
78	AutoWorldModel-Bench	[245]	arXiv'26		Benchmark	-	State-centric benchmark for automated world-model research
79	Scientific Memory	[322]	arXiv'26		Benchmark	-	Memory as budgeted context restoration, not leaderboard recall
80	SciHazard	[178]	arXiv'26		Benchmark	-	Decomposed harm scoring for scientific safety risks
81	Science Edge Eval.	[113]	arXiv'26		Benchmark	-	Evaluates at the moving frontier rather than settled interior
82	AutoResearch Diag.	[74]	arXiv'26		Evaluation	-	End-to-end diagnosis on 100 real-world research tasks
83	Gemini Co-Scientist	[309]	arXiv'26		System	-	Real-world validation across the full pipeline
84	AutoResearch	[290]	arXiv'26		System	-	Idea generation joined to verified idea execution
85	AutoSR	[458]	arXiv'26		System	-	Searches persistent research states via PW-MCTS
86	Eureka	[396]	arXiv'26		Multi-Agent	-	Task-conditioned meta-agent orchestration
87	Symposium	[275]	arXiv'26		Infrastructure	-	Immutable auditable records of agent-driven research
88	The AI Engineer	[446]	arXiv'26		System	-	LLM coupled to deterministic backends in closed loop
89	EurekaBench	[90]	arXiv'26		Benchmark	-	26 tasks in six disciplines; constraints, prediction, and scientific insights
90	ST-Bench	[52]	arXiv'26		Benchmark	-	Paper-grounded Earth-science tasks; numerical-output coverage and inference cost

B Survey Coverage Comparison & Taxonomy Analysis

Table 13 compares our coverage with five closely related concurrent efforts. The goal is not to rank surveys, but to clarify how our lifecycle framework differs in scope and organization.

Our eight-stage framework subsumes several prior taxonomies while making two distinctions explicit: AI auto-research should be analyzed across the complete lifecycle, and stages should be grouped by epistemological function rather than only by task name or autonomy level.

- **AI4Research** [41] defines five task categories: Comprehension, Survey, Discovery, Writing, and Review. These overlap with our S1–S3, S5, and S6. Our framework newly elevates S4 (Tables & Figures), S7 (Rebuttal & Revision), and S8 (Dissemination) as independent lifecycle stages.
- **From Automation to Autonomy** [479] organizes systems by autonomy level, from tool-like assistance to scientist-level automation. This axis is complementary: each of our stages can be instantiated at different autonomy levels, while our framework specifies *where* in the research lifecycle the system operates.
- **LLM4SR** [229] proposes a four-part view centered on hypothesis, experiment, writing, and review. This structure is close to ours, but does not separately model rebuttal and revision as a feedback stage. Our Validation phase separates S6 from S7, making the review–response loop explicit.
- **Automated Scholarly Paper Review** [491] provide in-depth coverage of review generation, quality assessment, and reviewer–paper matching. They are complementary to our work: they focus on S6, while our framework places peer review within the broader lifecycle.
- **AI Scientist Survey** [363] focus on autonomous or semi-autonomous scientific discovery, overlapping mainly with S1–S3 and S5. Our framework extends this view by also covering scientific visualization, peer validation, rebuttal, and dissemination.

Table 13 Comparison of survey coverage across our four-phase research lifecycle framework. ✓ = in-depth coverage, ○ = partial coverage, — = not covered, ★ new = newly introduced stage.

Stage	Ours	*LLM4SR* 2501	*PR Survey* 2501	*Auto→Auton* 2505	*AI4Research* 2507	*AI Scientists* 2510
Phase 1: Creation						
S1: Idea Generation novelty, feasibility, multi-agent	✓	✓	—	✓	✓	✓
S2: Literature Review retrieval, survey gen, deep research	✓	—	—	✓	✓	✓
S3: Coding & Experiments paper-to-code, execution, analysis	✓	✓	—	✓	✓	✓
S4: Figures & Tables diagrams, plots, formulas	✓	—	—	○	○	—
Phase 2: Writing						
S5: Paper Writing semi-auto, full-auto, detection	✓	✓	—	○	✓	✓
Phase 3: Validation						
S6: Peer Review auto-review, matching, quality	✓	✓	✓	○	✓	—
S7: Rebuttal & Revision comment analysis, rebuttal gen	★ new	—	—	—	—	—
Phase 4: Dissemination						
S8: Dissemination poster, slides, video, social	★ new	—	—	—	—	—
Stages covered	**8/8**	**4**	**1**	**5**	**5**	**4**

These comparisons show that prior taxonomies often list research tasks sequentially, while leaving functional distinctions and feedback loops implicit. Our four-phase framework makes these dependencies explicit. For example, S6 (*Peer Review*) and S7 (*Rebuttal & Revision*) do not simply follow paper writing as isolated downstream steps; they can redirect the workflow back to S3 for additional experiments, S4 for revised figures or tables, and S5 for manuscript restructuring. Similarly, dissemination artifacts in S8 may expose ambiguities in the original framing, requiring revisions to claims, explanations, or visual evidence.

These cross-stage dependencies are central to real research practice and are especially important for AI-assisted workflows, where errors can propagate from generated ideas to experiments, from experiments to claims, and from claims to public-facing summaries. By organizing the field into *Creation*, *Writing*, *Validation*, and *Dissemination*, our framework highlights not only which stages are covered by existing systems, but also where evidence, claims, critique, and communication must remain aligned.

References

[1] J. Abramson, J. Adler, J. Dunger, R. Evans, T. Green, A. Pritzel, O. Ronneberger, L. Willmore, A. J. Ballard, J. Bambrick, S. W. Bodenstein, D. A. Evans, C.-C. Hung, M. O'Neill, D. Reiman, K. Tunyasuvunakool, Z. Wu, A. Žemgulytė, E. Arvaniti, C. Beattie, O. Bertolusso, A. Sherwood, J. M. Jumper, and D. Hassabis. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500, 2024.

[2] E. Adamidi, S. Chatzopoulos, and T. Vergoulis. Scientific knowledge discovery in the age of large language models. *arXiv preprint arXiv:2607.26670*, 2026. URL https://arxiv.org/abs/2607.26670.

[3] S. Afroz et al. "AI slop is DDoSing open source": Understanding the impact of AI-generated contributions on open source sustainability. *arXiv preprint arXiv:2607.04003*, 2026. URL https://arxiv.org/abs/2607.04003.

[4] S. Agarwal, G. Sahu, A. Puri, I. H. Laradji, K. D. Dvijotham, J. Stanley, L. Charlin, and C. Pal. LitLLM: A toolkit for literature review with large language models. *arXiv preprint arXiv:2402.01788*, 2024. URL https://arxiv.org/abs/2402.01788.

[5] T. Aggarwal and A. Bhand. PASS: Presentation automation for slide generation and speech. *arXiv preprint arXiv:2501.06497*, 2025. URL https://arxiv.org/abs/2501.06497.

[6] A. Ajith, M. Xia, A. Chevalier, T. Goyal, D. Chen, and T. Gao. LitSearch: A retrieval benchmark for scientific literature search. In *Conference on Empirical Methods in Natural Language Processing*, 2024. URL https://arxiv.org/abs/2407.18940.

[7] I. Al Azher, M. J. Mokarrama, Z. Guo, S. R. Choudhury, and H. Alhoori. FutureGen: A RAG-based approach to generate the future work of scientific article. *arXiv preprint arXiv:2503.16561*, 2025. URL https://arxiv.org/abs/2503.16561.

[8] I. Al Azher, Z. Guo, and H. Alhoori. Multi-agent LLMs for generating research limitations. *arXiv preprint arXiv:2601.11578*, 2026. URL https://arxiv.org/abs/2601.11578.

[9] Alibaba NLP. Tongyi DeepResearch: An agentic LLM for long-horizon deep information seeking. https://github.com/Alibaba-NLP/DeepResearch, 2025. URL https://arxiv.org/abs/2510.24701.

[10] Analemma.ai. FARS: Fully automated research system. https://analemma.ai/blog/introducing-fars, 2026.

[11] A. Asai, J. He, R. Shao, W. Shi, A. Singh, J. C. Chang, K. Lo, L. Soldaini, S. Feldman, M. D'Arcy, D. Wadden, M. Latzke, J. Sparks, J. D. Hwang, V. Kishore, M. Tian, P. Ji, S. Liu, H. Tong, B. Wu, Y. Xiong, L. Zettlemoyer, G. Neubig, D. S. Weld, D. Downey, W. tau Yih, P. W. Koh, and H. Hajishirzi. Synthesizing scientific literature with retrieval-augmented language models. *Nature*, 650:857–863, 2026. URL https://arxiv.org/abs/2411.14199.

[12] M. Babaei, X. Li, Y. Kuang, J. P. Reid, and Y. Gan. MAIL: Memory-driven, adaptive, incremental, and literature-grounded framework for hypothesis generation in chemistry. *arXiv preprint arXiv:2608.28315*, 2026. URL https://arxiv.org/abs/2608.28315.

[13] J. Baek, S. K. Jauhar, S. Cucerzan, and S. J. Hwang. ResearchAgent: Iterative research idea generation over scientific literature with large language models. In *Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 6709–6738, 2025. URL https://arxiv.org/abs/2404.07738.

[14] T. Baumgärtner and I. Gurevych. SciCoQA: Quality assurance for scientific paper–code alignment. In *Annual Meeting of the Association for Computational Linguistics*, 2026. URL https://arxiv.org/abs/2601.12910.

[15] C. Beger and C.-L. Henneking. Citegeist: Automated generation of related work analysis on the arXiv corpus. *arXiv preprint arXiv:2503.23229*, 2025. URL https://arxiv.org/abs/2503.23229.

[16] J. Belouadi, A. Lauscher, and S. Eger. AutomaTikZ: Text-guided synthesis of scientific vector graphics with TikZ. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2310.00367.

[17] J. Belouadi, S. P. Ponzetto, and S. Eger. DeTikZify: Synthesizing graphics programs for scientific figures and sketches with TikZ. In *Advances in Neural Information Processing Systems*, volume 37, pages 85074–85108, 2024. URL https://arxiv.org/abs/2405.15306.

[18] H. Bisht, V. Kumar, K. M. Jablonka, Mausam, and N. M. A. Krishnan. Agentic AI scientists are not built for autonomous scientific discovery. *arXiv preprint arXiv:2605.08956*, 2026. URL https://arxiv.org/abs/2605.08956.

[19] J. Biswas et al. AI-assisted peer review at scale: The AAAI-26 AI review pilot. *arXiv preprint arXiv:2604.13940*, 2026. URL https://arxiv.org/abs/2604.13940.

[20] Blog. 228 hours of non-stop work to produce 100 papers, burning through 11.4 billion tokens: Fars has gone crazy. https://eu.36kr.com/en/p/3696795271966336, 2026.

[21] D. A. Boiko, R. MacKnight, B. Kline, and G. Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023. URL https://arxiv.org/abs/2304.05332.

[22] W. Bolton and P. Torr. Adversarial fast-moving real-world domains as test beds for benchmarking AI scientist capabilities. *arXiv preprint arXiv:2608.03569*, 2026. URL https://arxiv.org/abs/2608.03569.

[23] R. M. Boos. Local AI pre-screening for human triple-blind peer review in health sciences. *arXiv preprint arXiv:2608.14625*, 2026. URL https://arxiv.org/abs/2608.14625.

[24] J. Bragg, M. D'Arcy, N. Balepur, D. Bareket, B. Dalvi, S. Feldman, D. Haddad, J. D. Hwang, P. Jansen, V. Kishore, B. P. Majumder, A. Naik, S. Rahamimov, K. Richardson, A. Singh, H. Surana, A. Tiktinsky, R. Vasu, G. Wiener, C. Anastasiades, S. Candra, J. Dunkelberger, D. Emery, R. Evans, M. Hamada, R. Huff, R. Kinney, M. Latzke, J. Lochner, R. Lozano-Aguilera, C. Nguyen, S. Rao, A. Tanaka, B. Vlahos, P. Clark, D. Downey, Y. Goldberg, A. Sabharwal, and D. S. Weld. AstaBench: Rigorous benchmarking of AI agents with a scientific research suite. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2510.21652.

[25] A. M. Bran, S. Cox, O. Schilter, C. Baldassari, A. D. White, and P. Schwaller. ChemCrow: Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6(5):525–535, 2024. URL https://arxiv.org/abs/2304.05376.

[26] ByteDance. DeerFlow: A deep research framework orchestrating sub-agents, memory, and sandboxes. https://github.com/bytedance/deer-flow, 2025.

[27] A. Camelo-Guerrero and J. Diaz-Rodriguez. How closely do LLM reviews align with human peer review? *arXiv preprint arXiv:2608.03659*, 2026. URL https://arxiv.org/abs/2608.03659.

[28] V. K. Chaganti. On-device deep research at 4b: Exposure bounds faithfulness, retrieval bounds coverage. *arXiv preprint arXiv:2607.12257*, 2026. URL https://arxiv.org/abs/2607.12257.

[29] J. Chai, S. Tang, R. Ye, Y. Du, X. Zhu, M. Zhou, Y. Wang, W. E, Y. Zhang, L. Zhang, and S. Chen. SciMaster: Towards general-purpose scientific AI agents, part I. X-Master as foundation: Can we lead on humanity's last exam? *arXiv preprint arXiv:2507.05241*, 2025. URL https://arxiv.org/abs/2507.05241.

[30] J. S. Chan, N. Chowdhury, O. Jaffe, J. Aung, D. Sherburn, E. Mays, G. Starace, K. Liu, L. Maksin, T. Patwardhan, L. Weng, and A. Madry. MLE-Bench: Evaluating machine learning agents on machine learning engineering. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2410.07095.

[31] C.-C. Chen and I. Gurevych. Commitment checklist: Auditing author commitments in peer review. *arXiv preprint arXiv:2603.00003*, 2026. URL https://arxiv.org/abs/2603.00003.

[32] D. Chen. AI-generated figures in academic publishing: Policies, tools, and practical guidelines. *arXiv preprint arXiv:2603.16159*, 2026. URL https://arxiv.org/abs/2603.16159.

[33] D. Chen et al. Can AI draw science? a benchmark for evaluating scientific figure generation by text-to-image and multimodal models. *arXiv preprint arXiv:2606.28406*, 2026. URL https://arxiv.org/abs/2606.28406.

[34] G. Chen, J. Chen, L. Chen, J. Zhao, F. Meng, W. X. Zhao, R. Song, C. Chen, J.-R. Wen, and K. Jia. Toward autonomous long-horizon engineering for ML research. *arXiv preprint arXiv:2604.13018*, 2026. URL https://arxiv.org/abs/2604.13018.

[35] G. Chen, F. Meng, J. Zhao, M. Li, D. Cheng, H. Song, J. Chen, Y. Lin, H. Chen, X. Zhao, R. Song, C. Liu, C. Chen, K. Jia, and J.-R. Wen. BeyondSWE: Can current code agent survive beyond single-repo bug fixing? *arXiv preprint arXiv:2603.03194*, 2026. URL https://arxiv.org/abs/2603.03194.

[36] H. Chen, M. Xiong, Y. Lu, W. Han, A. Deng, Y. He, J. Wu, Y. Li, Y. Liu, and B. Hooi. MLR-Bench: Evaluating AI agents on open-ended machine learning research. *arXiv preprint arXiv:2505.19955*, 2025. URL https://arxiv.org/abs/2505.19955.

[37] H. Chen, S. Wang, Y. Yin, S. Zhuang, G. Zuccon, and T. Leelanupab. ITER: Interaction-aware retrieval for agentic search. *arXiv preprint arXiv:2608.27912*, 2026. URL https://arxiv.org/abs/2608.27912.

[38] H. Chen, X. Zhou, W. Zhou, X. Shao, Y. Zhu, B. Wang, J. Hong, A. Jia, and F. Wu. X+Slides: Benchmarking audience-conditioned slide generation. *arXiv preprint arXiv:2606.19256*, 2026. URL https://arxiv.org/abs/2606.19256.

[39] N. Chen, A. H. Lin, J. Wu, J. Hou, Z. Zhang, Q. Wang, X. Wang, and B. He. XtraGPT: Context-aware and controllable academic paper revision. *arXiv preprint arXiv:2505.11336*, 2025. URL https://arxiv.org/abs/2505.11336.

[40] N. Chen, Q. Wang, Q. Zou, and B. He. Articulate intuition or genuine analysis? benchmarking epistemic reliability in LLM-as-a-judge peer reviews. *arXiv preprint arXiv:2607.10511*, 2026. URL https://arxiv.org/abs/2607.10511.

[41] Q. Chen, M. Yang, L. Qin, J. Liu, Z. Yan, J. Guan, D. Peng, Y. Ji, H. Li, M. Hu, Y. Zhang, Y. Liang, Y. Zhou, J. Wang, Z. Chen, and W. Che. AI4Research: A survey of artificial intelligence for scientific research. *arXiv preprint arXiv:2507.01903*, 2025. URL https://arxiv.org/abs/2507.01903.

[42] S. Chen, Y. Fei, N. Liu, X. Yu, L. Li, R. Thapa, M. Ciobanu, Q. Mao, and R. Das. Reconstruction: A blind benchmark for recovering research ideas from pre-publication bibliographies. *arXiv preprint arXiv:2608.16645*, 2026. URL https://arxiv.org/abs/2608.16645.

[43] S. Chen, J. Lai, J. Gao, H. Shi, Z. Liu, T. Ye, J. Luo, X. Wei, and L. Zhu. PosterOmni: Generalized artistic poster creation via task distillation and unified reward feedback. *arXiv preprint arXiv:2602.12127*, 2026. URL https://arxiv.org/abs/2602.12127.

[44] S. Chen, J. Lai, J. Gao, T. Ye, H. Chen, H. Shi, S. Shao, Y. Lin, S. Fei, Z. Xing, Y. Jin, J. Luo, X. Wei, and L. Zhu. PosterCraft: Rethinking high-quality aesthetic poster generation in a unified framework. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2506.10741.

[45] S. Chen, S. Zhong, D. P. Brumby, and A. L. Cox. What happens when reviewers receive AI feedback in their reviews? In *CHI Conference on Human Factors in Computing Systems*, pages 1–19, 2026.

[46] Y. Chen, T. Lv, S. Zhang, Y. Yin, Y. Wan, P. S. Yu, and D. Chen. Paper2Web: Let's make your paper alive! *arXiv preprint arXiv:2510.15842*, 2025. URL https://arxiv.org/abs/2510.15842.

[47] Y. Chen, G. Chen, Y. Sun, and K. Zhang. Analogical deep research: Retrieving and integrating historical analogies for foresight analysis. *arXiv preprint arXiv:2607.13602*, 2026. URL https://arxiv.org/abs/2607.13602.

[48] Z. Chen, J. Chen, S. O. Arik, M. Sra, T. Pfister, and J. Yoon. CoDA: Agentic systems for collaborative data visualization. *arXiv preprint arXiv:2510.03194*, 2025. URL https://arxiv.org/abs/2510.03194.

[49] Z. Chen, S. Chen, Y. Ning, Q. Zhang, B. Wang, B. Yu, Y. Li, Z. Liao, C. Wei, Z. Lu, V. Dey, M. Xue, F. N. Baker, B. Burns, D. Adu-Ampratwum, X. Huang, X. Ning, S. Gao, Y. Su, and H. Sun. ScienceAgentBench: Toward rigorous assessment of language agents for data-driven scientific discovery. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2410.05080.

[50] Z. Chen, Y. Zhao, J. Sun, Y. Ma, M. Patwardhan, and A. Cohan. IdeaAnchor: Teaching LLMs to Turn Literature into Research Ideas. *arXiv preprint arXiv:2610.08781*, 2026. URL https://arxiv.org/abs/2610.08781.

[51] Z. Chen et al. PeerCheck: Enhancing LLM-generated academic reviews towards human-level quality. *arXiv preprint arXiv:2606.20897*, 2026. URL https://arxiv.org/abs/2606.20897.

[52] Q. Cheng, R. Dong, S. Chen, L. Liu, D. Lu, Z. Chen, W. Cheng, Y. Xie, H. Chen, X. Jia, and H. Wang. ST-Bench: A Spatial-Temporal Benchmark for Multi-Agent System Generation on Scientific Research Tasks. *arXiv preprint arXiv:2610.07763*, 2026. URL https://arxiv.org/abs/2610.07763.

[53] J. Choi, S. Park, S. Song, and H. Shim. PosterForest: Hierarchical multi-agent collaboration for scientific poster generation. *arXiv preprint arXiv:2508.21720*, 2025. URL https://arxiv.org/abs/2508.21720.

[54] P. H. Couto, Q. P. Ho, N. Kumari, B. K. Rachmat, T. G. H. Khuong, I. Ullah, and L. Sun-Hosoya. RelevAI-Reviewer: A benchmark on AI reviewers for survey paper relevance. *arXiv preprint arXiv:2406.10294*, 2024. URL https://arxiv.org/abs/2406.10294.

[55] S. G. Creadore. Plato-Bio: verification-first biological novelty screening with temporal rediscovery and structural benchmarks. *arXiv preprint arXiv:2607.23975*, 2026. URL https://arxiv.org/abs/2607.23975.

[56] J. Cui, Y. Yuan, K. Zhong, X. Xu, J. Wu, and Z. Liu. ARAC: Benchmarking auto-research's alignment and completeness on end-to-end researchs. *arXiv preprint arXiv:2608.12788*, 2026. URL https://arxiv.org/abs/2608.12788.

[57] Z. Cui, M. Yang, L. Gao, J. Liu, and P. Li. SciFigPlag-Bench: A benchmark for provenance-aware scientific figure plagiarism detection. *arXiv preprint arXiv:2607.29124*, 2026. URL https://arxiv.org/abs/2607.29124.

[58] A. Daghighfarsoodeh, S. Ebrahimi, A. Ghorbanpour, S. Sadeghian, R. Cheraghi, N. Arabzadeh, and E. Bagheri. Peerify: Benchmarking Peer-Review claim verification. *arXiv preprint arXiv:2609.25046*, 2026. URL https://arxiv.org/abs/2609.25046.

[59] D. Dai et al. I-WebGenBench: Evaluating interactivity in LLM-generated scientific web applications. *arXiv preprint arXiv:2606.00750*, 2026. URL https://arxiv.org/abs/2606.00750.

[60] M. D'Arcy, T. Hope, L. Birnbaum, and D. Downey. MARG: Multi-agent review generation for scientific papers. *arXiv preprint arXiv:2401.04259*, 2024. URL https://arxiv.org/abs/2401.04259.

[61] F. De Ponte. OpenDraft: 19-agent research draft generation. https://github.com/federicodeponte/opendraft, 2025.

[62] X. Deng, J. Da, E. Pan, Y. Y. He, C. Ide, K. Garg, N. Lauffer, A. Park, N. Pasari, C. Rane, K. Sampath, M. Krishnan, S. Kundurthy, S. Hendryx, Z. Wang, V. Bharadwaj, J. Holm, R. Aluri, C. B. C. Zhang, N. Jacobson, B. Liu, and B. Kenstler. SWE-Bench Pro: Can AI agents solve long-horizon software engineering tasks? *arXiv preprint arXiv:2509.16941*, 2025. URL https://arxiv.org/abs/2509.16941.

[63] Z. Deng, C. Xu, H. Liang, H. Li, X. Zhong, and L. Yu. SciFigQual-Bench: A benchmark for scientific figure quality assessment with full-manuscript context. *arXiv preprint arXiv:2607.27084*, 2026. URL https://arxiv.org/abs/2607.27084.

[64] J. Ding, Z. Shi, J. Zhong, K. Qiu, Q. Dai, Y. Yang, C. Luo, and Q. Xu. Inspire: Benchmarking scientific literature search for open research problems. *arXiv preprint arXiv:2609.33233*, 2026. URL https://arxiv.org/abs/2609.33233.

[65] J. D'Souza, F. Ahmed, C. A. B. Andrade, L. Frolova, P. Gnanasambandan, D. Hussain, M. U. Khan, N. K. Nkengfoa, P. P. J., F. Priante, S. F. van der Werf, and T. F. J. van Roeden. A pathway to general-purpose scientific AI: Multimodal comprehension of scientific images. *arXiv preprint arXiv:2608.14075*, 2026. URL https://arxiv.org/abs/2608.14075.

[66] Y. Du, T. Jin, J. Kang, X. Pang, J. Chai, T. Miao, F. Liu, W. Wang, S. Yao, Y. Zhang, and S. Chen. Towards self-evolving agentic literature retrieval. *arXiv preprint arXiv:2605.14306*, 2026. URL https://arxiv.org/abs/2605.14306.

[67] A. Elovic. GPT Researcher: Autonomous agent for comprehensive online research. https://github.com/assafelovic/gpt-researcher, 2024.

[68] P. Emami, S. Horawalavithana, T. Nguyen, G. Panapitiya, B. Jacob, S. Raskar, S. Sinha, J. D. Willard, et al. Position: AI agents in scientific teams should be studied as human-agent systems. *arXiv preprint arXiv:2608.14667*, 2026. URL https://arxiv.org/abs/2608.14667.

[69] D. Falck, S. Sabri, A. Surina, T. Foster, A. Sims, S. Devlin, D. Rogers, T. Collins, K. Aleksiev, L. Kirsch, and E. Hughes. Training AI scientists to replicate research. *arXiv preprint arXiv:2608.13331*, 2026. URL https://arxiv.org/abs/2608.13331.

[70] T. Fan, F. Zhang, Y. Zheng, B. Chen, X. Niu, C. Huang, J. Lin, and C. Huang. DeepInnovator: Triggering the innovative capabilities of LLMs. *arXiv preprint arXiv:2602.18920*, 2026. URL https://arxiv.org/abs/2602.18920.

[71] H. Fang, Y. Feng, and I. Gurevych. From passive generation to investigation: A proactive scientific peer review agent. *arXiv preprint arXiv:2606.13349*, 2026. URL https://arxiv.org/abs/2606.13349.

[72] S. Fang, C. Wu, and Z. Zhang. SeaSlides: Semantic abstraction layer for agentic slide generation. *arXiv preprint arXiv:2608.03298*, 2026. URL https://arxiv.org/abs/2608.03298.

[73] M. Farmer. Abduction without a body? representational grounding and the abduction loop for scientific hypothesis generation. *arXiv preprint arXiv:2608.02505*, 2026. URL https://arxiv.org/abs/2608.02505.

[74] Y. Fei, N. Liu, X. Yu, S. Chen, L. Li, R. Thapa, M. Ciobanu, N. P. Singh, Q. Mao, and R. Das. How do agents fail on autoresearch: End-to-end diagnostic evaluation on 100 real-world frontier research tasks. *arXiv preprint arXiv:2608.14905*, 2026. URL https://arxiv.org/abs/2608.14905.

[75] X. Z. Feng, Y. Lin, Y. Zhang, C. Weng, and Y. Peng. LLM assisted writing deserves empirical evaluation. *arXiv preprint arXiv:2608.22124*, 2026. URL https://arxiv.org/abs/2608.22124.

[76] Y. Feng, Q. Huang, X. Xie, Z. Yang, J. Yu, W. Chen, and A. K. H. Tung. IDRBench: Interactive deep research benchmark. *arXiv preprint arXiv:2601.06676*, 2026. URL https://arxiv.org/abs/2601.06676.

[77] A. M. Fichtl, L. Ellinger, J. Kelber, K. Olík, and G. Groh. AI-assisted peer review across research communities: From reviewer AI policies to LLM review quality. *arXiv preprint arXiv:2608.03581*, 2026. URL https://arxiv.org/abs/2608.03581.

[78] T. S. Foster, B. Al Omari, T. Fu, T. Mann, C. Domond, L. Cipolina-Kun, B. Gauri, M. Aghamelu, et al. AI research preference models. *arXiv preprint arXiv:2608.13940*, 2026. URL https://arxiv.org/abs/2608.13940.

[79] S. Fu, Y. Tao, L. Jin, J. Zhao, Q. Shi, T. Pan, H. Li, C. Wang, J. Wu, and C. Huo. Beyond solution-centric search: Adaptive inquiry and knowledge revision for autonomous ML engineering. *arXiv preprint arXiv:2608.02143*, 2026. URL https://arxiv.org/abs/2608.02143.

[80] T.-J. Fu, W. Y. Wang, D. McDuff, and Y. Song. DOC2PPT: Automatic presentation slides generation from scientific documents. In *AAAI Conference on Artificial Intelligence*, volume 36, pages 634–642, 2022. URL https://arxiv.org/abs/2101.11796.

[81] S. K. Gaddipati, D. Muhammed, F. Keya, G. Rabby, and S. Auer. MLReplicate: Benchmarking autonomous research systems for machine learning reproducibility. *arXiv preprint arXiv:2605.16616*, 2026. URL https://arxiv.org/abs/2605.16616.

[82] S. Gao, R. Zhu, P. Sui, Z. Kong, S. Aldogom, Y. Huang, A. Noori, R. Shamji, K. Parvataneni, T. Tsiligkaridis, and M. Zitnik. Democratizing AI scientists using ToolUniverse. *arXiv preprint arXiv:2509.23426*, 2025. URL https://arxiv.org/abs/2509.23426.

[83] S. Gao et al. AutoScientists: Self-organizing agent teams for long-running scientific experimentation. *arXiv preprint arXiv:2605.28655*, 2026. URL https://arxiv.org/abs/2605.28655.

[84] X. Gao, J. Ruan, Z. Zhang, J. Gao, T. Liu, and Y. Fu. ReviewAgents: Bridging the gap between human and AI-generated paper reviews. *arXiv preprint arXiv:2503.08506*, 2025. URL https://arxiv.org/abs/2503.08506.

[85] Y. Gao, Q. Wu, and L. Zhu. Merging the citations received by arXiv-deposited e-prints and their corresponding published journal articles: Problems and perspectives. *Information Processing & Management*, 57(5):102267, 2020.

[86] Z. Gao, K. Brantley, and T. Joachims. Reviewer2: Optimizing review generation through prompt generation. *arXiv preprint arXiv:2402.10886*, 2024. URL https://arxiv.org/abs/2402.10886.

[87] K. Garg, F. Shaik, S. Bandyopadhyay, and C. Caragea. Let's use ChatGPT to write our paper! benchmarking LLMs to write the introduction of a research paper. *arXiv preprint arXiv:2508.14273*, 2025. URL https://arxiv.org/abs/2508.14273.

[88] A. Garikaparthi, M. Patwardhan, L. Vig, and A. Cohan. IRIS: Interactive research ideation system for accelerating scientific discovery. In *Annual Meeting of the Association for Computational Linguistics*, pages 592–603, 2025. URL https://arxiv.org/abs/2504.16728.

[89] J. Ge, Z. Z. Wang, X. Zhou, Y.-H. Peng, S. Subramanian, Q. Tan, M. Sap, A. Suhr, D. Fried, G. Neubig, and T. Darrell. AutoPresent: Designing structured visuals from scratch. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2902–2911, 2025. URL https://arxiv.org/abs/2501.00912.

[90] J. Geng, Z. Wu, K. S. Chen, S. Kim, J. Janssen, Z. Z. Wang, B. Kalita, R. Gao, A. Ho, A. O. Nelson, O. Isayev, F. Villaescusa-Navarro, C.-Y. Lai, H. Chen, and G. Neubig. EurekaBench: Measuring agentic ability to discover new scientific insights. *arXiv preprint arXiv:2610.00492*, 2026. URL https://arxiv.org/abs/2610.00492.

[91] A. Ghafarollahi and M. J. Buehler. SciAgents: Automating scientific discovery through multi-agent intelligent graph reasoning. *arXiv preprint arXiv:2409.05556*, 2024. URL https://arxiv.org/abs/2409.05556.

[92] S. Gharat and J. Komiyama. SGHA: Evidence-grounded research problem discovery with local language models. *arXiv preprint arXiv:2608.17501*, 2026. URL https://arxiv.org/abs/2608.17501.

[93] E. Gibney. Major conference catches illicit AI use — and rejects hundreds of papers. *Nature News*, 652:281–282, 2026.

[94] G. H. T. Go, K. Ly, A. Sogaard, A. Tabatabaei, M. de Rijke, and X. Chen. LiRA: A multi-agent framework for reliable and readable literature review generation. *arXiv preprint arXiv:2510.05138*, 2025. URL https://arxiv.org/abs/2510.05138.

[95] S. Goel, R. Hazra, D. Jayalath, T. Willi, P. Jain, W. F. Shen, I. Leontiadis, F. Barbieri, Y. Bachrach, J. Geiping, and C. Whitehouse. Training AI co-scientists using rubric rewards. *arXiv preprint arXiv:2512.23707*, 2025. URL https://arxiv.org/abs/2512.23707.

[96] T. Gong, M. R. Zeng, and Y. Yang. Albilich: Steerable proof-state orchestration for LLM-based mathematical research with CAS integration. *arXiv preprint arXiv:2607.27705*, 2026. URL https://arxiv.org/abs/2607.27705.

[97] X. Gong, S. Gu, H. Liu, D. Yao, C. Tan, and T. Li. Learning Scientific Exploration from Human Research Decision Trajectories. *arXiv preprint arXiv:2610.07184*, 2026. URL https://arxiv.org/abs/2610.07184.

[98] K. Goswami, P. Mathur, R. Rossi, and F. Dernoncourt. PlotGen: Multi-agent LLM-based scientific data visualization via multimodal feedback. *arXiv preprint arXiv:2502.00988*, 2025. URL https://arxiv.org/abs/2502.00988.

[99] M. Gottsacker, A. Alwin, H. Furuya, R. W. Lindeman, G. Bruder, and G. F. Welch. ReVoicer: Conversational voice annotation for human-centered, LLM-assisted peer review. *arXiv preprint arXiv:2608.00299*, 2026. URL https://arxiv.org/abs/2608.00299.

[100] J. Gottweis, W.-H. Weng, A. Daryin, T. Tu, A. Palepu, P. Sirkovic, A. Myaskovsky, F. Weissenberger, K. Rong, R. Tanno, K. Saab, D. Popovici, J. Blum, F. Zhang, K. Chou, A. Hassidim, B. Gokturk, A. Vahdat, P. Kohli, Y. Matias, A. Carroll, K. Kulkarni, N. Tomasev, Y. Guan, V. Dhillon, E. D. Vaishnav, B. Lee, T. R. D. Costa, J. R. Penadés, G. Peltz, Y. Xu, A. Pawlosky, A. Karthikesalingam, and V. Natarajan. Towards an AI co-scientist. *arXiv preprint arXiv:2502.18864*, 2025. URL https://arxiv.org/abs/2502.18864.

[101] P. Goyal, M. Parmar, Y. Song, H. Palangi, T. Pfister, and J. Yoon. ScholarPeer: A context-aware multi-agent framework for automated peer review. *arXiv preprint arXiv:2601.22638*, 2026. URL https://arxiv.org/abs/2601.22638.

[102] C. Greisinger and S. Eger. TikZilla: Scaling text-to-TikZ with high-quality data and reinforcement learning. *arXiv preprint arXiv:2603.03072*, 2026. URL https://arxiv.org/abs/2603.03072.

[103] C. Greisinger, Z. Zhao, and S. Eger. EdiTikZ: Scientific figure editing from revision trajectories. *arXiv preprint arXiv:2609.01409*, 2026. URL https://arxiv.org/abs/2609.01409.

[104] T. Gu, J. Wang, Z. Zhang, and H. Li. LLMs can realize combinatorial creativity: Generating creative ideas via LLMs for scientific research. *arXiv preprint arXiv:2412.14141*, 2024. URL https://arxiv.org/abs/2412.14141.

[105] V. Gumma, N. Majumder, S. Sinhahajari, and S. Poria. IDEAgent: Agentic quality-diversity search for research idea generation. *arXiv preprint arXiv:2607.22375*, 2026. URL https://arxiv.org/abs/2607.22375.

[106] H. Guo. IdeaTrail: Full-process agent trajectories for scientific ideation. *arXiv preprint arXiv:2607.10144*, 2026. URL https://arxiv.org/abs/2607.10144.

[107] L. Guo, Y. Sun, X. Li, L. Ge, Z. Wang, H. Zheng, J. Li, H. Zhang, et al. Stress-testing large language model agents in a robotic chemistry laboratory. *arXiv preprint arXiv:2607.23045*, 2026. URL https://arxiv.org/abs/2607.23045.

[108] S. Guo, C. Deng, Y. Wen, H. Chen, Y. Chang, and J. Wang. DS-Agent: Automated data science by empowering large language models with case-based reasoning. *arXiv preprint arXiv:2402.17453*, 2024. URL https://arxiv.org/abs/2402.17453.

[109] S. Guo, A. H. Shariatmadari, G. Xiong, A. Huang, M. Kim, C. M. Williams, S. Bekiranov, and A. Zhang. IdeaBench: Benchmarking large language models for research idea generation. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 5888–5899, 2025. URL https://arxiv.org/abs/2411.02429.

[110] Y. Guo, Y. Chen, C. Hu, X. Yu, L. Hong, and J. Xu. WARA: A closed-loop multi-agent framework for wireless optimization autoresearch. In *IEEE/CIC Int. Conf. Commun. China*, 2026. URL https://arxiv.org/abs/2607.19822.

[111] B. Gyevnár et al. Distributed denial of science: How indirect data poisoning of AI systems can industrialize scientific fraud. *arXiv preprint arXiv:2607.10712*, 2026. URL https://arxiv.org/abs/2607.10712.

[112] P. Han, Y. Yu, J. Xu, and J. You. DRPG (decompose, retrieve, plan, generate): An agentic framework for academic rebuttal. *arXiv preprint arXiv:2601.18081*, 2026. URL https://arxiv.org/abs/2601.18081.

[113] T. Han, Y. Zhang, J. Wang, Y. Wu, W. Y. Chiu, Z. Li, Y. Zhang, J. Wang, et al. Science edge evaluation: SEE the missing step toward real scientific discovery. *arXiv preprint arXiv:2608.06931*, 2026. URL https://arxiv.org/abs/2608.06931.

[114] Q. Hao, F. Xu, Y. Li, and J. Evans. Artificial intelligence tools expand scientists' impact but contract science's focus. *Nature*, 649:1237–1243, 2026.

[115] R. Hazra, S. Layek, S. Banerjee, S. Chakrabarti, and A. Mukherjee. Structurally-bounded agentic graph exploration for evidence-grounded scholarly deepsearch. *arXiv preprint arXiv:2608.24809*, 2026. URL https://arxiv.org/abs/2608.24809.

[116] Y. He, G. Huang, P. Feng, Y. Lin, Y. Zhang, H. Li, and W. E. PaSa: An LLM agent for comprehensive academic paper search. *arXiv preprint arXiv:2501.10120*, 2025. URL https://arxiv.org/abs/2501.10120.

[117] Y. He, D. Li, G. Li, J. Geng, Z. Huang, C. Ren, Z. Zhang, Y. Liu, et al. ReasFlow: Assisting reasoning-centric scientific discovery in applied mathematics via a knowledge-based multi-agent system. *arXiv preprint arXiv:2607.14178*, 2026. URL https://arxiv.org/abs/2607.14178.

[118] Y. He, J. Wang, Y. Zhao, J. Liu, and H. Chen. Agentic auto-research is fuzz testing. *arXiv preprint arXiv:2608.09855*, 2026. URL https://arxiv.org/abs/2608.09855.

[119] Z. He, Z. Lyu, and Y. R. Fung. RebuttalAgent: Strategic persuasion in academic rebuttal via theory of mind. *arXiv preprint arXiv:2601.15715*, 2026. URL https://arxiv.org/abs/2601.15715.

[120] A. Hell, K. Vovk, V. Krishnaraj, J. Liu, K. Aizawa, A. E. Bayer, L. Blot, J. Cowell, et al. AI's capability in assisting scientific research in physics, astrophysics, and cosmology i: Literature review. *arXiv preprint arXiv:2607.25672*, 2026. URL https://arxiv.org/abs/2607.25672.

[121] E. Hirsch, D. Wan, H. Wang, E. Stengel-Eskin, M. Bansal, and I. Dagan. Who is the agent to blame? localizing faithfulness and citation mistakes in agentic deep research. In *Conference on Empirical Methods in Natural Language Processing*, 2026. URL https://arxiv.org/abs/2608.24306.

[122] HKU Data Intelligence Lab. Paper2Slides: From paper to presentation in one click. https://github.com/HKUDS/Paper2Slides, 2025.

[123] S.-T. Ho, M. Liu, H. Nghiem, and F. Huang. SoundnessBench: Can your AI scientist really tell good research ideas from bad ones? *arXiv preprint arXiv:2605.30329*, 2026. URL https://arxiv.org/abs/2605.30329.

[124] H. Hong, Y. Li, J. Chen, L. G. Huy, S. Ananiadou, J.-j. Kim, and C. Lin. Can LLMs write reliable rubrics? a meta-evaluation for experiment reproduction. *arXiv preprint arXiv:2607.12835*, 2026. URL https://arxiv.org/abs/2607.12835.

[125] H. Hong, Y. Li, L. G. Huy, J. Yang, M. Zhou, and C. Lin. Overview of the NLPCC 2026 shared task 11: Agent-Based experiment reproduction from scientific papers. *arXiv preprint arXiv:2609.11117*, 2026. URL https://arxiv.org/abs/2609.11117.

[126] M. Hong, D. Jiang, C. J. Zhang, Z. Guo, Y. Li, J. Chen, S. Cui, and Z. Su. CiteLLM: An agentic platform for trustworthy scientific reference discovery. *arXiv preprint arXiv:2602.23075*, 2026. URL https://arxiv.org/abs/2602.23075.

[127] E. Hossain, S. K. Sinha, N. Bansal, R. A. Knipper, S. Sarkar, J. Salvador, Y. Mahajan, S. R. P. K. Guttikonda, M. Akter, M. M. Hassan, M. Freestone, M. C. W. Jr., D. Feng, and S. Karmaker. LLMs as meta-reviewers' assistants: A case study. In *Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics*, pages 7763–7803, 2025. URL https://arxiv.org/abs/2402.15589.

[128] J. Hou, A. H. Lin, N. Chen, Y. Gong, and B. He. PaperDebugger: A plugin-based multi-agent system for in-editor academic writing, review, and editing. *arXiv preprint arXiv:2512.02589*, 2025. URL https://arxiv.org/abs/2512.02589.

[129] C.-C. Hsu, E. Bransom, J. Sparks, B. Kuehl, C. Tan, D. Wadden, L. L. Wang, and A. Naik. CHIME: LLM-assisted hierarchical organization of scientific studies for literature review support. *arXiv preprint arXiv:2407.16148*, 2024. URL https://arxiv.org/abs/2407.16148.

[130] H. Hu, C. Dang, Y. Liu, H. Kang, C. He, and W. Li. PosterMELD: Multi-agent paper-to-poster generation for controllable design diversity with editable print-ready outputs. *arXiv preprint arXiv:2608.02218*, 2026. URL https://arxiv.org/abs/2608.02218.

[131] X. Hu, H. Fu, J. Wang, Y. Wang, Z. Li, R. Xu, Y. Lu, Y. Jin, L. Pan, and Z. Lan. Nova: An iterative planning and search approach to enhance novelty and diversity of LLM generated ideas. *arXiv preprint arXiv:2410.14255*, 2024. URL https://arxiv.org/abs/2410.14255.

[132] X. Hu, Z. Zhao, S. Wei, Z. Chai, Q. Ma, G. Wang, X. Wang, J. Su, J. Xu, M. Zhu, Y. Cheng, J. Yuan, J. Li, K. Kuang, Y. Yang, H. Yang, and F. Wu. InfiAgent-DABench: Evaluating agents on data analysis tasks. *arXiv preprint arXiv:2401.05507*, 2024. URL https://arxiv.org/abs/2401.05507.

[133] X. Hu, Z. Pan, Z. Wang, Z. Liu, Z. Su, and W. Zhang. ReproAgent: Contract-guided paper-to-code reproduction. In *Findings of the Conference on Empirical Methods in Natural Language Processing*, 2026. URL https://arxiv.org/abs/2608.24291.

[134] T. Hua, H. Hua, V. Xiang, B. Klieger, S. T. Truong, W. Liang, F.-Y. Sun, and N. Haber. ResearchCodeBench: Benchmarking LLMs on implementing novel machine learning research code. *arXiv preprint arXiv:2506.02314*, 2025. URL https://arxiv.org/abs/2506.02314.

[135] E. Huang et al. NVAITC AI Scientist: A governed end-to-end research system – a hypertension GWAS case study. *arXiv preprint arXiv:2607.11084*, 2026. URL https://arxiv.org/abs/2607.11084.

[136] K. Huang, S. Zhang, H. Wang, Y. Qu, Y. Lu, Y. Roohani, R. Li, L. Qiu, G. Li, J. Zhang, D. Yin, S. Marwaha, J. N. Carter, X. Zhou, M. Wheeler, J. A. Bernstein, M. Wang, P. He, J. Zhou, M. Snyder, L. Cong, A. Regev, and J. Leskovec. Biomni: A general-purpose biomedical AI agent. https://github.com/snap-stanford/Biomni, 2025.

[137] Q. Huang, J. Vora, P. Liang, and J. Leskovec. MLAgentBench: Evaluating language agents on machine learning experimentation. In *International Conference on Machine Learning*, 2024. URL https://arxiv.org/abs/2310.03302.

[138] S. Huang, Y. Gao, J. Bai, Y. Zhou, Z. Yin, X. Liu, R. Chellappa, C. P. Lau, S. Nag, C. Peng, and S. Pramanick. SciFig: Towards automating scientific figure generation. *arXiv preprint arXiv:2601.04390*, 2026. URL https://arxiv.org/abs/2601.04390.

[139] Y. Huang, Y. Huang, and f. t. t. Collaboration. Deepresearch agent system. *arXiv preprint arXiv:2607.27562*, 2026. URL https://arxiv.org/abs/2607.27562.

[140] Y. Huang, Z. Xu, Y. Ma, W. Wang, Z. Liu, Z. Xu, P.-Y. Chen, M. Galley, Z. Lin, S. Feuerriegel, R. Poovendran, M. Sra, A. Pentland, X. Zhang, and Z. Chen. Reward hacking challenges oversight of autonomous research agents. *arXiv preprint arXiv:2609.28614*, 2026. URL https://arxiv.org/abs/2609.28614.

[141] K. Hur et al. Compressing the validation bottleneck: An agentic self-driving lab for scientific discovery. *arXiv preprint arXiv:2607.04508*, 2026. URL https://arxiv.org/abs/2607.04508.

[142] M. Idahl and Z. Ahmadi. OpenReviewer: A specialized large language model for generating critical scientific paper reviews. In *Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 550–562, 2025. URL https://arxiv.org/abs/2412.11948.

[143] A. K. Jagadish et al. Closing the loop to discover psychological theories with an automated cognitive scientist. *arXiv preprint arXiv:2606.26448*, 2026. URL https://arxiv.org/abs/2606.26448.

[144] P. Jansen, M.-A. Cote, T. Khot, E. Bransom, B. Dalvi Mishra, B. P. Majumder, O. Tafjord, and P. Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems*, 2024. URL https://arxiv.org/abs/2406.06769.

[145] P. Jansen, O. Tafjord, M. Radensky, P. Siangliulue, T. Hope, B. Dalvi Mishra, B. P. Majumder, D. S. Weld, and P. Clark. CodeScientist: End-to-end semi-automated scientific discovery with code-based experimentation. In *Annual Meeting of the Association for Computational Linguistics*, pages 13370–13467, 2025. URL https://arxiv.org/abs/2503.22708.

[146] Y. Jeon, K. Limqueco, J. Chang, and X. A. Chen. HALO: Interactive co-abductive reasoning in scientific hypothesis generation. *arXiv preprint arXiv:2607.18564*, 2026. URL https://arxiv.org/abs/2607.18564.

[147] B. Jiang. HindSight: Evaluating LLM-generated research ideas via future impact. *arXiv preprint arXiv:2603.15164*, 2026. URL https://arxiv.org/abs/2603.15164.

[148] L. Jiang, Y. Chai, M. Li, M. Liu, R. Fok, N. Dziri, Y. Tsvetkov, M. Sap, A. Albalak, and Y. Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *Advances in Neural Information Processing Systems*, 2025. URL https://arxiv.org/abs/2510.22954.

[149] Y. Jiang and A. Y. Ng. Automated scientific reviewing with agentic AI. https://paperreview.ai/tech-overview, 2025.

[150] Z. Jiang, D. Schmidt, D. Srikanth, D. Xu, I. Kaplan, D. Jacenko, and Y. Wu. AIDE: AI-driven exploration in the space of code. *arXiv preprint arXiv:2502.13138*, 2025. URL https://arxiv.org/abs/2502.13138.

[151] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2310.06770.

[152] J. Jin, Y. Hu, K. Qiu, Q. Dai, C. Luo, G. Dong, X. Li, T. Zhao, X. Ma, G. Zhang, Z. Wu, B. Liu, Z. Yang, L. Li, L. Wang, H. Qian, Y. Zhu, and Z. Dou. Toward generalist autonomous research via hypothesis-tree refinement. *arXiv preprint arXiv:2606.11926*, 2026. URL https://arxiv.org/abs/2606.11926.

[153] Y. Jin, Q. Zhao, Y. Wang, H. Chen, K. Zhu, Y. Xiao, and J. Wang. AgentReview: Exploring peer review dynamics with LLM agents. In *Conference on Empirical Methods in Natural Language Processing*, pages 1208–1226, 2024. URL https://arxiv.org/abs/2406.12708.

[154] K. Jung, H. Cho, J. Yun, S. Yang, J. Jang, and J. Choo. Talk to your slides: High-efficiency slide editing via language-driven structured data manipulation. *arXiv preprint arXiv:2505.11604*, 2025. URL https://arxiv.org/abs/2505.11604.

[155] S. Kale and V. Nadadur. TeXpert: A multi-level benchmark for evaluating LaTeX code generation by LLMs. In *Workshop on Scholarly Document Processing*, pages 7–16, 2025. URL https://arxiv.org/abs/2506.16990.

[156] A. H. Kargaran, N. Nikeghbal, J. Yang, and N. Ousidhoum. Insights from the ICLR peer review and rebuttal process. *arXiv preprint arXiv:2511.15462*, 2025. URL https://arxiv.org/abs/2511.15462.

[157] A. Karpathy. autoresearch. https://github.com/karpathy/autoresearch, 2025.

[158] T. Kawada, S. Kitada, and H. Iyatomi. GenGA: Editable and data-grounded graphical abstract generation for academic papers. *arXiv preprint arXiv:2608.05478*, 2026. URL https://arxiv.org/abs/2608.05478.

[159] J. Keuper. Prompt injection attacks on LLM generated reviews of scientific publications. *arXiv preprint arXiv:2509.10248*, 2025. URL https://arxiv.org/abs/2509.10248.

[160] J. Khatri and M. Patwardhan. Defend: Automated rebuttals for peer review with minimal author guidance. *arXiv preprint arXiv:2603.27360*, 2026. URL https://arxiv.org/abs/2603.27360.

[161] S. Kim, Y. Lee, B. Liu, D. Ko, R. Shao, S. Kim, G. Neubig, P. W. Koh, A. Chowdhery, A. Asai, O. Khattab, Y. Choi, G. Kim, and C. Finn. ScholarCatalyst: A benchmark for retrieving papers that inspire new research. *arXiv preprint arXiv:2610.02202*, 2026. URL https://arxiv.org/abs/2610.02202.

[162] S. Kim, D. Yoon, K. Gashteovski, J. Suk, J. Baek, P. Aggarwal, I. Wu, V. Zaverkin, S. Petkoski, D. R. Schrider, I. Dukovski, F. Santini, B. Mitreska, Y. Jeong, K. Kwon, Y. M. Sim, D. Manasova, A. Porto, B. Mojsoska, M. Takamoto, M. Shuntov, R. Liu, H. J. Lee, N. U. Dinç, Y. Jo, S. Han, C. Lee, H. Li, E. H. R. Tsai, E. Simsek, K. Shafi, Y. Chung, J. Park, A. Shulevski, H. Christiansen, Y. Son, E. Knight, A. Montoya, J. Ahn, C. Langkammer, H. Moon, C. Yoon, N. Stikov, M. Jang, E. Choi, J. Kim, Y. S. Jung, W. Y. Kim, J. K. Kim, I. M. Anjum, H. U. Kim, D. Bridges, C. Lawrence, X. Yue, A. Oh, A. Asai, S. Welleck, and G. Neubig. On the limits and opportunities of AI reviewers: Reviewing the reviews of nature-family papers with 45 expert scientists. *arXiv preprint arXiv:2605.20668*, 2026. URL https://arxiv.org/abs/2605.20668.

[163] S. S. Y. Kim, W. H. Deng, J. W. Vaughan, B. Su, W. Su, A. Agarwal, S. Li, M. Jaggi, D. G. Goldstein, N. B. Shah, and M. Dudík. Use and effects of LLMs in peer review: A randomized experiment and survey at ICML 2026. *arXiv preprint arXiv:2609.19420*, 2026. URL https://arxiv.org/abs/2609.19420.

[164] Y. Kim, K. Gu, C. Park, C. Park, S. Schmidgall, A. A. Heydari, Y. Yan, Z. Zhang, Y. Zhuang, M. Malhotra, P. P. Liang, H. W. Park, Y. Yang, X. Xu, Y. Du, S. Patel, T. Althoff, D. McDuff, and X. Liu. Towards a science of scaling agent systems. *arXiv preprint arXiv:2512.08296*, 2025. URL https://arxiv.org/abs/2512.08296.

[165] R. D. King. The past and future of AI scientists. *arXiv preprint arXiv:2608.14407*, 2026. URL https://arxiv.org/abs/2608.14407.

[166] D. Kobak, R. González-Márquez, E. Ákos Bernárdez-Vilaboa, and J. Lause. Delving into LLM-assisted writing in biomedical publications through excess vocabulary. *Science Advances*, 11(27):eadt3813, 2025. URL https://arxiv.org/abs/2406.07016.

[167] H. Kolukuluru, R. Ashok, K. Arora, E. W. Ciccarelli, N. A. Kumar, L. Nie, F. Dernoncourt, S. Basu, R. A. Rossi, and N. Lipka. Not worth another token: Marginal value estimation for efficient deep research agents. *arXiv preprint arXiv:2608.08389*, 2026. URL https://arxiv.org/abs/2608.08389.

[168] P. T. J. Kon, J. Liu, Q. Ding, Y. Qiu, Z. Yang, Y. Huang, J. Srinivasa, M. Lee, M. Chowdhury, and A. Chen. Curie: Toward rigorous and automated scientific experimentation with AI agents. *arXiv preprint arXiv:2502.16069*, 2025. URL https://arxiv.org/abs/2502.16069.

[169] P. T. J. Kon, Q. Ding, J. Liu, X. Zhu, J. Peng, J. Xing, Y. Huang, Y. Qiu, J. Srinivasa, M. Lee, M. Chowdhury, M. Zaharia, and A. Chen. EXP-Bench: Can AI conduct AI research experiments? In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2505.24785.

[170] K. Kusumegi, X. Yang, P. Ginsparg, M. de Vaan, T. Stuart, and Y. Yin. Scientific production in the era of large language models. *Science*, 390:1240–1243, 2025. URL https://arxiv.org/abs/2601.13187.

[171] R. T. Lange, Y. Imajuku, and E. Cetin. ShinkaEvolve: Towards open-ended and sample-efficient program evolution. *arXiv preprint arXiv:2509.19349*, 2025. URL https://arxiv.org/abs/2509.19349.

[172] J. M. Laurent, J. D. Janizek, M. Ruzo, M. M. Hinks, M. J. Hammerling, S. Narayanan, M. Ponnapati, A. D. White, and S. G. Rodriques. Lab-Bench: Measuring capabilities of language models for biology research. *arXiv preprint arXiv:2407.10362*, 2024. URL https://arxiv.org/abs/2407.10362.

[173] C. Leblanc et al. To trust or not to trust: Authors' response to AI-based reviews. *arXiv preprint arXiv:2605.16623*, 2026. URL https://arxiv.org/abs/2605.16623.

[174] M. Lee, P. Liang, and Q. Yang. CoAuthor: Human-AI collaborative writing with language models. *arXiv preprint arXiv:2201.06796*, 2022. URL https://arxiv.org/abs/2201.06796.

[175] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020. URL https://arxiv.org/abs/2005.11401.

[176] B. Li, H. Fei, T. Ju, M.-L. Lee, and W. Hsu. OmniScientist: An omni-modal omni-discipline AI scientist. *arXiv preprint arXiv:2608.13558*, 2026. URL https://arxiv.org/abs/2608.13558.

[177] B. Li, J. Wu, Y. Zhao, W. Xu, X. Chen, H. Yin, L. Chen, W. Zhang, and K. Li. UniScientist: Advancing universal scientific research intelligence. https://unipat.ai/blog/UniScientist, 2026.

[178] C. Li, Y. Xiong, L. Li, T. Du, W. Zhang, L. Bai, and J. Shao. SciHazard: A benchmark for measuring scientific safety risks with decomposed harm scoring. *arXiv preprint arXiv:2607.18665*, 2026. URL https://arxiv.org/abs/2607.18665.

[179] E. Li, J. Ren, X. Pan, C. Yan, C. Li, D. Bergemann, and Z. Yang. Build your personalized research group: A multiagent framework for continual and interactive science automation. *arXiv preprint arXiv:2510.15624*, 2025. URL https://arxiv.org/abs/2510.15624.

[180] H. Li, E. Jung, W. Zhao, F. Liu, J. Wang, K. Xu, Z. Guo, Z. Chen, B. Fei, F. Ling, and L. Bai. AutoSupervision: Closing the feedback loop in scientific workflows with grounded revision verification. *arXiv preprint arXiv:2607.27845*, 2026. URL https://arxiv.org/abs/2607.27845.

[181] J. Li, S. Li, Z. Gao, Q. Shi, Y. Li, Z. Wang, J. Huang, H. Wang, J. Wang, X. Han, Z. Liu, and M. Sun. TritonBench: Benchmarking large language model capabilities for generating triton operators. In *Annual Meeting on Association for Computational Linguistics*, 2025. URL https://arxiv.org/abs/2502.14752.

[182] J. Li, A. Murtadha, Z. Wang, Q. Chen, W. Chen, Y. Wu, G. Wang, A. L. Siy, et al. Praxist: From experimental artifacts to solution lineages. *arXiv preprint arXiv:2608.25955*, 2026. URL https://arxiv.org/abs/2608.25955.

[183] L. Li, W. Xu, J. Guo, R. Zhao, X. Li, Y. Yuan, B. Zhang, Y. Jiang, Y. Xin, R. Dang, D. Zhao, Y. Rong, T. Feng, and L. Bing. Chain of ideas: Revolutionizing research via novel idea development with LLM agents. *arXiv preprint arXiv:2410.13185*, 2024. URL https://arxiv.org/abs/2410.13185.

[184] L. Li et al. Gaming AI-assisted peer reviews poses new risks to the scientific community. *arXiv preprint arXiv:2606.10159*, 2026. URL https://arxiv.org/abs/2606.10159.

[185] M. Li, Y. Zeng, Z. Cheng, C. Ma, and K. Jia. ReportBench: Evaluating deep research agents via academic survey tasks. *arXiv preprint arXiv:2508.15804*, 2025. URL https://arxiv.org/abs/2508.15804.

[186] M. Li, Z. Lin, and S. Ma. Source or it didn't happen: A multi-agent framework for citation hallucination detection. *arXiv preprint arXiv:2605.08583*, 2026. URL https://arxiv.org/abs/2605.08583.

[187] M. Li, C. Wang, X. Li, X. Zeng, D. Li, P. Shi, D. Zhou, and T. Zhou. How can rhetoric reward-hack AI reviewers? dissecting rhetorical sensitivity in AI-based peer review. *arXiv preprint arXiv:2608.08975*, 2026. URL https://arxiv.org/abs/2608.08975.

[188] R. Li, T. Patel, Q. Wang, and X. Du. MLR-Copilot: Autonomous machine learning research based on large language models agents. *arXiv preprint arXiv:2408.14033*, 2024. URL https://arxiv.org/abs/2408.14033.

[189] R. Li, Y. Liu, H. Zhou, F. Meng, J. Zhou, and X. Sun. Not just reason, not just scan: Reinforcement learning for proactive scientific error verification over academic paper. In *Findings of the Conference on Empirical Methods in Natural Language Processing*, 2026. URL https://arxiv.org/abs/2608.26596.

[190] S. Li, Y. Cai, H. Chen, and Y. Wang. GeoSVG-RL: Geometry-aware reinforcement learning for layout-constrained text-to-SVG diagram generation. *arXiv preprint arXiv:2605.25447*, 2026. URL https://arxiv.org/abs/2605.25447.

[191] S. Li, L. Fan, Y. Lin, Z. Li, X. Wei, S. Ni, H. Alinejad-Rokny, and M. Yang. Automatic paper reviewing with heterogeneous graph reasoning over LLM-simulated reviewer-author debates. *arXiv preprint arXiv:2511.08317*, 2026. URL https://arxiv.org/abs/2511.08317. AAAI-2026.

[192] X. Li, Y. Yang, P. Li, X. Yao, S. Liu, X. Ye, Z. Lu, Y. Yao, Y. Li, Y. Li, and W. Jiang. Tree-of-Ideas: Automated research ideation via cross-trajectory reasoning over scholarly evolution. *arXiv preprint arXiv:2608.10740*, 2026. URL https://arxiv.org/abs/2608.10740.

[193] Y. Li. awesome-ai-research-writing. https://github.com/Leey21/awesome-ai-research-writing, 2024.

[194] Y. Li, C. Shao, X. Liu, R. Zhao, P. Liu, H. Su, Z. Chen, Q. Yang, A. Xu, Y. Fang, Q. Zeng, T. Li, J. Xu, F. Xu, Y. Li, and T.-Y. Liu. AutoSOTA: An end-to-end automated research system for state-of-the-art AI model discovery. *arXiv preprint arXiv:2604.05550*, 2026. URL https://arxiv.org/abs/2604.05550.

[195] Y. Li, W. Yang, H. Tan, X. Huang, Z. Chen, Z. Li, B. Chen, S. Lei, et al. Beyond final scores: A systematic evaluation of agents for long-horizon AI research and development. *arXiv preprint arXiv:2608.13417*, 2026. URL https://arxiv.org/abs/2608.13417.

[196] Z. Li, D. Jiang, X. Ma, H. Zhang, P. Nie, Y. Zhang, K. Zou, J. Xie, Y. Zhang, and W. Chen. OpenResearcher: A fully open pipeline for long-horizon deep research trajectory synthesis. *arXiv preprint arXiv:2603.20278*, 2026. URL https://arxiv.org/abs/2603.20278.

[197] W. Liang, Y. Zhang, Z. Wu, H. Lepp, W. Ji, X. Zhao, H. Cao, S. Liu, S. He, Z. Huang, D. Yang, C. Potts, C. D. Manning, and J. Y. Zou. Mapping the increasing use of LLMs in scientific papers. *arXiv preprint arXiv:2404.01268*, 2024. URL https://arxiv.org/abs/2404.01268.

[198] X. Liang, J. Yang, Y. Wang, C. Tang, Z. Zheng, S. Song, Z. Lin, Y. Yang, S. Niu, H. Wang, B. Tang, F. Xiong, K. Mao, and Z. Li. SurveyX: Academic survey automation via large language models. *arXiv preprint arXiv:2502.14776*, 2025. URL https://arxiv.org/abs/2502.14776.

[199] X. Liang, X. Zhang, Y. Xu, S. Sun, and C. You. SlideGen: Collaborative multimodal agents for scientific slide generation. *arXiv preprint arXiv:2512.04529*, 2025. URL https://arxiv.org/abs/2512.04529.

[200] X. Liao, L. Li, H. Liu, T. Gu, Y. Yao, T. Zhu, Y. Teng, and Y. Wang. PseudoBench: Measuring how agentic auto-research fuels pseudoscience. *arXiv preprint arXiv:2606.18060*, 2026. URL https://arxiv.org/abs/2606.18060.

[201] C.-Y. Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, 2004.

[202] K. Q. Lin, S. Hu, P. Lu, Y. Chen, Y. Chen, O. Queen, Y. Chen, J. Yu, J. Yu, Z. Ding, Y. Ji, S. Liu, J. Gu, L. Li, M. Z. Shou, P. Torr, and J. Zou. PaperDoctor: Evidence-Grounded and actionable feedback for scientific papers in progress. *arXiv preprint arXiv:2609.16995*, 2026. URL https://arxiv.org/abs/2609.16995.

[203] T.-L. Lin, W.-C. Chen, T.-F. Hsiao, H.-I. Liu, Y.-H. Yeh, Y.-K. Chan, W.-S. Lien, P.-Y. Kuo, P. S. Yu, and H.-H. Shuai. Breaking the reviewer: Assessing the vulnerability of large language models in automated peer review under textual adversarial attacks. *arXiv preprint arXiv:2506.11113*, 2025. URL https://arxiv.org/abs/2506.11113.

[204] T.-L. Lin, D.-T. Yao, T.-F. Hsiao, W.-C. Chen, and H.-H. Shuai. HalluPeer: A taxonomy-driven benchmark for detecting hallucinations in scientific peer reviews. *arXiv preprint arXiv:2609.03580*, 2026. URL https://arxiv.org/abs/2609.03580.

[205] Z. Lin, Q. Xie, M. Zhu, S. Li, Q. Sun, E. Gu, Y. Ding, K. Sun, F. Guo, P. Lu, Z. Ning, Y. Weng, and Y. Zhang. AutoFigure-Edit: Generating editable scientific illustration. *arXiv preprint arXiv:2603.06674*, 2026. URL https://arxiv.org/abs/2603.06674.

[206] J. Ling, Y. Qi, T. Huang, S. Zhou, Y. Huang, J. Yang, Z. Song, Y. Zhou, Y. Yang, H. T. Shen, and P. Wang. Table2LaTeX-RL: High-fidelity LaTeX code generation from table images via reinforced multimodal language models. *arXiv preprint arXiv:2509.17589*, 2025. URL https://arxiv.org/abs/2509.17589.

[207] J. Liu, L. Yang, H. Luo, F. Wang, H. Li, and M. Wang. Preacher: Paper-to-video agentic system. In *IEEE/CVF International Conference on Computer Vision*, pages 17129–17139, 2025. URL https://arxiv.org/abs/2508.09632.

[208] J. Liu, Y. He, Y. He, P. Guo, D. Yang, A. Myronenko, C. Zhao, H. Ye, T. Qi, Y. Zhou, D. Xu, and Y. Tang. BaT: Towards self-evolving medical research agent with stage rubrics. *arXiv preprint arXiv:2608.16211*, 2026. URL https://arxiv.org/abs/2608.16211.

[209] J. Liu, P. Xia, S. Han, S. Qiu, L. Zhang, G. Chen, H. Tu, X. Yang, J. Zhou, H. Zhu, Y. Li, J. Zhang, Y. Zhou, Z. Zheng, C. Xie, M. Ding, and H. Yao. AutoResearchClaw: Self-reinforcing autonomous research with human-ai collaboration. https://github.com/aiming-lab/AutoResearchClaw, 2026. URL https://arxiv.org/abs/2605.20025.

[210] J. Liu, T. J. Zhang, R. Faulkner, X. A. Huang, V. Zouhar, D. Glandorf, I. Dahlgren, V. Q. Truong, R. Dagli, Y. Chen, F. Leeb, P. S. Pandey, Y. Bicker, S. Majumder, W. Jiang, Z. Qiu, S. P. Chowdhury, B. Schölkopf, M. Diab, and Z. Jin. PaperMentor: A human-centered multi-agent writing tutor for AI research papers on overleaf. *arXiv preprint arXiv:2606.08857*, 2026. URL https://arxiv.org/abs/2606.08857.

[211] Q. Liu, J. Mao, F. Zhu, and T.-S. Chua. Diagnosing search behavior and failure modes in long-horizon search agents. *arXiv preprint arXiv:2608.01913*, 2026. URL https://arxiv.org/abs/2608.01913.

[212] W. Liu, Y. Lu, Q. Xia, H. Xu, T. Zhao, J. Xi, Y. Zhu, H. Liang, H. Shi, H. Wang, and Z. Dou. Training documents reranker with search rubrics for deep research agent. *arXiv preprint arXiv:2608.03527*, 2026. URL https://arxiv.org/abs/2608.03527.

[213] W. Liu, Z. Yang, Y. Zhao, and X. Li. RATE: Reviewer profiling and annotation-free training for expertise ranking in peer review systems. *arXiv preprint arXiv:2601.19637*, 2026. URL https://arxiv.org/abs/2601.19637.

[214] Y. Liu, Y. Wu, D. Zhang, and L. Sun. Agentic AutoSurvey: Let LLMs survey LLMs. *arXiv preprint arXiv:2509.18661*, 2025. URL https://arxiv.org/abs/2509.18661.

[215] Y. Liu, D. Dai, C. Liu, Y. Song, H. Ci, Y. Zhang, Q. Zhang, M. Z. Shou, X. E. Wang, and Y. Bu. Figures as programs: Recursive generation of editable scientific figures. *arXiv preprint arXiv:2609.01006*, 2026. URL https://arxiv.org/abs/2609.01006.

[216] Y. Liu, H. P. Pham, and J. Yoon. PaperCompiler: Faithful Paper-to-Code generation via Repository-Level specification compilation. *arXiv preprint arXiv:2609.02272*, 2026. URL https://arxiv.org/abs/2609.02272.

[217] Y. Liu, Y. Wang, Z. Lei, L. Meng, Y. Sun, J. Lin, H. Liu, Y. Chu, Q. Yang, J. Xu, L. Zhang, and Z. Mao. MMPostTrainBench: Benchmarking Autonomous Research for Multimodal Post-Training. *arXiv preprint arXiv:2610.05398*, 2026. URL https://arxiv.org/abs/2610.05398.

[218] Y. Liu, Z. Yang, T. Xie, J. Ni, B. Gao, Y. Li, S. Tang, W. Ouyang, E. Cambria, and D. Zhou. ResearchBench: Benchmarking LLMs in scientific discovery via inspiration-based task decomposition. In *Findings of the Annual Meeting of the Association for Computational Linguistics*, 2026. URL https://arxiv.org/abs/2503.21248.

[219] Z. Liu, X. Bao, P. Li, J. Zhou, Z. Liao, Y. He, K. Jiang, C.-W. Xie, Y. Zheng, and H. Xie. ShowTable: Unlocking creative table visualization with collaborative reflection and refinement. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. URL https://arxiv.org/abs/2512.13303.

[220] N. P. P. Loc, T. H. L. Viet, T. T. Khanh, D. A. Nguyen, T. A. N. Pham, T. Nguyen, N. V. Chawla, W. Buntine, K.-S. Wong, K. D. Doan, and B. T. Nguyen. PRISM: A multi-dimensional benchmark for evaluating LLM peer reviewers. *arXiv preprint arXiv:2605.26730*, 2026. URL https://arxiv.org/abs/2605.26730.

[221] L. Loo and U. Braga-Neto. An agentic AI scientific community for automated neural operator discovery. *arXiv preprint arXiv:2607.12122*, 2026. URL https://arxiv.org/abs/2607.12122.

[222] M. Louis et al. Rebuttals move peer-review scores, but initial-review structure bounds the movement. *arXiv preprint arXiv:2606.22166*, 2026. URL https://arxiv.org/abs/2606.22166.

[223] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha. The AI scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024. URL https://arxiv.org/abs/2408.06292.

[224] C. Lu, C. Lu, R. T. Lange, Y. Yamada, S. Hu, J. Foerster, D. Ha, and J. Clune. Towards end-to-end automation of AI research. *Nature*, 651:914–919, 2026.

[225] S. Lu, C. Li, K. Luo, Z. Zhang, H. Wang, H. Xiao, L. Xiong, J. Wang, et al. AREX: Towards a recursively self-improving agent for deep research. *arXiv preprint arXiv:2607.21461*, 2026. URL https://arxiv.org/abs/2607.21461.

[226] J. Luo. XScientist: A git-like research protocol for long-running autonomous scientific discovery. *arXiv preprint arXiv:2607.12301*, 2026. URL https://arxiv.org/abs/2607.12301.

[227] Y. Luo, R. Wang, P. Gam, J. Cui, circlestarzero, S. Ni, J. Quanta, Q. Fu, and S. Hou. ChatPaper: Use LLM to summarize papers. https://github.com/kaixindelele/ChatPaper, 2023.

[228] Z. Luo, A. Kasirzadeh, and N. B. Shah. The more you automate, the less you see: Hidden pitfalls of AI scientist systems. *arXiv preprint arXiv:2509.08713*, 2025. URL https://arxiv.org/abs/2509.08713.

[229] Z. Luo, Z. Yang, Z. Xu, W. Yang, and X. Du. LLM4SR: A survey on large language models for scientific research. *arXiv preprint arXiv:2501.04306*, 2025. URL https://arxiv.org/abs/2501.04306.

[230] X. Lyu, H. Lu, X. Luo, and S. Yang. Beyond Instruction-Driven editing: Source-Grounded problem discovery with User-Governed repair for scientific posters. *arXiv preprint arXiv:2609.01813*, 2026. URL https://arxiv.org/abs/2609.01813.

[231] Y. Lyu, X. Zhang, X. Yi, Y. Zhao, S. Guo, W. Hu, J. Piotrowski, J. Kaliski, J. Urbani, Z. Meng, L. Zhou, and X. Yan. EvoScientist: Towards multi-agent evolving AI scientists for end-to-end scientific discovery. *arXiv preprint arXiv:2603.08127*, 2026. URL https://arxiv.org/abs/2603.08127.

[232] J. Lála, O. O'Donoghue, A. Shtedritski, S. Cox, S. G. Rodriques, and A. D. White. PaperQA: Retrieval-augmented generative agent for scientific research. *arXiv preprint arXiv:2312.07559*, 2023. URL https://arxiv.org/abs/2312.07559.

[233] M. M. K. M, V. N. Balasubramanian, and C. V. Jawahar. Unifying scientific communication: Fine-grained correspondence across scientific media. *arXiv preprint arXiv:2605.05831*, 2026. URL https://arxiv.org/abs/2605.05831.

[234] Q. Ma, C. Guo, Z. Tian, S. Wang, J. Xiao, Y. Yue, and Z. Zhang. Paper2Rebuttal: A multi-agent framework for transparent author response assistance. *arXiv preprint arXiv:2601.14171*, 2026. URL https://arxiv.org/abs/2601.14171.

[235] Y. Ma, Y. Zhao, S. Wu, Z. Chen, M. Patwardhan, and A. Cohan. ActReview: Rebuttal-Guided training data and rubric rewards for actionable peer review generation. *arXiv preprint arXiv:2609.09076*, 2026. URL https://arxiv.org/abs/2609.09076.

[236] Y. Ma, Y. Zhao, S. Wu, M. Patwardhan, and A. Cohan. IdeaAMBIG: Benchmarking Implementation-Critical gaps in Research-Idea specifications. *arXiv preprint arXiv:2609.10539*, 2026. URL https://arxiv.org/abs/2609.10539.

[237] B. P. Majumder, H. Surana, D. Agarwal, B. D. Mishra, A. Meena, A. Prakhar, T. Vora, T. Khot, A. Sabharwal, and P. Clark. DiscoveryBench: Towards data-driven discovery with large language models. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2407.01725.

[238] R. Meng, B. D. Mishra, J. Chen, C.-L. Li, P. Goyal, M. Parmar, Y. Song, Y. Song, R. Sinha, P. Ranganathan, B. Gokturk, J. Yoon, and T. Pfister. ScientistOne: Towards human-level autonomous research via chain-of-evidence. *arXiv preprint arXiv:2605.26340*, 2026. URL https://arxiv.org/abs/2605.26340.

[239] METR Team. Measuring AI ability to complete long tasks. https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks, 2025.

[240] J. Miao, J. R. Davis, Y. Zhang, J. K. Pritchard, and J. Zou. Paper2Agent: Reimagining research papers as interactive and reliable AI agents. *arXiv preprint arXiv:2509.06917*, 2025. URL https://arxiv.org/abs/2509.06917.

[241] S. Mishra, N. Rajeev, and T. Chakraborty. Tree-of-Concerns: Hierarchical multi-agent debate for unstated-limitation extraction in scientific critique. In *Findings of the Conference on Empirical Methods in Natural Language Processing*, 2026. URL https://arxiv.org/abs/2608.20777.

[242] L. Mitchener, A. Yiu, B. Chang, M. Bourdenx, T. Nadolski, A. Sulovari, E. C. Landsness, D. L. Barabasi, S. Narayanan, N. Evans, S. Reddy, M. Foiani, A. Kamal, L. P. Shriver, F. Cao, A. T. Wassie, J. M. Laurent, E. Melville-Green, M. Caldas, A. Bou, K. F. Roberts, S. Zagorac, T. C. Orr, M. E. Orr, K. J. Zwezdaryk, A. E. Ghareeb, L. McCoy, B. Gomes, E. A. Ashley, K. E. Duff, T. Buonassisi, T. Rainforth, R. J. Bateman, M. Skarlinski, S. G. Rodriques, M. M. Hinks, and A. D. White. Kosmos: An AI scientist for autonomous discovery. *arXiv preprint arXiv:2511.02824*, 2025. URL https://arxiv.org/abs/2511.02824.

[243] Y. Mo, T. Zheng, Y. Gao, R. Wang, N. N. K. H. Nam, K. K. W. Tam, J. Bai, Y. Song, G. Wong, and S. See. AgentIdeaBench: Benchmarking scientific ideation in the agent era. *arXiv preprint arXiv:2609.07611*, 2026. URL https://arxiv.org/abs/2609.07611.

[244] I. Mondal et al. A good talk does not look like a summary, it teaches you! measuring takeaways from paper-to-video talks. *arXiv preprint arXiv:2606.28531*, 2026. URL https://arxiv.org/abs/2606.28531.

[245] M. Moodi, X. Zhu, F. D. M. Silva, H. Chaput, and M. R. Taesiri. AutoWorldModel-Bench: A state-centric benchmark for automated world-model research. *arXiv preprint arXiv:2608.11216*, 2026. URL https://arxiv.org/abs/2608.11216.

[246] E. Moris and J. I. Orlando. Can coding agents build robust baselines? a skill-based approach for automating the medical imaging model-development pipeline. In *International Conference on Medical Image Computing and Computer-Assisted Intervention Workshops*, 2026. URL https://arxiv.org/abs/2608.23336.

[247] N. Morosin, A. A. Nadi, and M. P. Wallace. Generative AI use in statistical research: A literature review and code generation case study. *arXiv preprint arXiv:2608.11121*, 2026. URL https://arxiv.org/abs/2608.11121.

[248] J. Mun, C. Jung, X. Zhou, H. Kim, and M. Sap. GoodPoint: Learning constructive scientific paper feedback from author responses. *arXiv preprint arXiv:2604.11924*, 2026. URL https://arxiv.org/abs/2604.11924.

[249] M. Naddaf. More than half of researchers now use AI for peer review — often against guidance. *Nature News*, 649:273–274, 2026.

[250] J. Nam, J. Yoon, Y. Pan, Y. Wang, R. Meng, P. Ranganathan, and T. Pfister. ScientistTwo: Pioneering the human knowledge frontier with autonomous AI. *arXiv preprint arXiv:2609.19644*, 2026. URL https://arxiv.org/abs/2609.19644.

[251] D. Nathani, L. Madaan, N. Roberts, N. Bashlykov, A. Menon, V. Moens, A. Budhiraja, D. Magka, V. Vorotilov, G. Chaurasia, D. Hupkes, R. S. Cabral, T. Shavrina, J. Foerster, Y. Bachrach, W. Y. Wang, and R. Raileanu. MLGym: A new framework and benchmark for advancing AI research agents. *arXiv preprint arXiv:2502.14499*, 2025. URL https://arxiv.org/abs/2502.14499.

[252] B. Newman, Y. Lee, A. Naik, P. Siangliulue, R. Fok, J. Kim, D. S. Weld, J. C. Chang, and K. Lo. ArxivDIGESTables: Synthesizing scientific literature into tables using language models. In *Conference on Empirical Methods in Natural Language Processing*, pages 9612–9631, 2024. URL https://arxiv.org/abs/2410.22360.

[253] M.-A. Nguyen, M.-D. Nguyen, H. L. N.T., K. H. Dang, N. T. Dong, and D. D. Le. SurveyG: A multi-agent LLM framework with hierarchical citation graph for automated survey generation. *arXiv preprint arXiv:2510.07733*, 2025. URL https://arxiv.org/abs/2510.07733.

[254] S. Ni. ChatReviewer: ChatGPT-based paper reviewing and response generation. https://github.com/nishiwen1214/ChatReviewer, 2023.

[255] Y. Ni, P. Nie, K. Zou, X. Yue, and W. Chen. VisCoder: Fine-tuning LLMs for executable python visualization code generation. In *Conference on Empirical Methods in Natural Language Processing*, pages 2956–2983, 2025. URL https://arxiv.org/abs/2506.03930.

[256] J. Nie, Z. Yang, Z. Tang, Y. Zhang, X. Chu, X. Tian, and B. Han. DRNOISE: Benchmarking deep research agents in misleading evidence environments. *arXiv preprint arXiv:2607.17291*, 2026. URL https://arxiv.org/abs/2607.17291.

[257] J. Ning, X. Li, S. Zhong, J. Zeng, and G. Ke. Auto research for materials: Auditable AI-scientist workflows with held-out transfer. *arXiv preprint arXiv:2607.17100*, 2026. URL https://arxiv.org/abs/2607.17100.

[258] J. Ning et al. Closed-loop auto research for molecular property prediction: Discovering and certifying generalizable improvements. *arXiv preprint arXiv:2606.22731*, 2026. URL https://arxiv.org/abs/2606.22731.

[259] A. Novikov, N. Vu, M. Eisenberger, E. Dupont, P.-S. Huang, A. Z. Wagner, S. Shirobokov, B. Kozlovskii, F. J. R. Ruiz, A. Mehrabian, M. P. Kumar, A. See, S. Chaudhuri, G. Holland, A. Davies, S. Nowozin, P. Kohli, and

M. Balog. AlphaEvolve: A coding agent for scientific and algorithmic discovery. *arXiv preprint arXiv:2506.13131*, 2025. URL https://arxiv.org/abs/2506.13131.

[260] P. O. Oamen, O.-B. Osei, A. Mukherjee, C. Greisinger, S. Eger, P. Onobhayedo, and W. Zhao. How do VLMs behave when blind or misled? behavioral evaluation of VLMs on scientific figures. *arXiv preprint arXiv:2608.13267*, 2026. URL https://arxiv.org/abs/2608.13267.

[261] O. O'Donoghue, A. Shtedritski, J. Ginger, R. Abboud, A. E. Ghareeb, J. Booth, and S. G. Rodriques. BioPlanner: Automatic evaluation of LLMs on protocol planning. *arXiv preprint arXiv:2310.10632*, 2023. URL https://arxiv.org/abs/2310.10632.

[262] Y. Oh, Y.-J. Lee, J. Ahn, G. Kim, and D. Kang. Science or slop?: Benchmarking and mitigating scientific slop in AI-Generated papers. *arXiv preprint arXiv:2610.00531*, 2026. URL https://arxiv.org/abs/2610.00531.

[263] J. Ou, W. G. Walden, K. Sanders, Z. Jiang, K. Sun, J. Cheng, W. Jurayj, M. Wanner, S. Liang, C. Morgan, S. Han, W. Wang, C. May, H. Recknor, D. Khashabi, and B. V. Durme. ClaimCheck: How grounded are LLM critiques of scientific papers? In *Conference on Empirical Methods in Natural Language Processing*, pages 21712–21735, 2025. URL https://arxiv.org/abs/2503.21717.

[264] A. Ouyang, S. Guo, S. Arora, A. L. Zhang, W. Hu, C. Ré, and A. Mirhoseini. KernelBench: Can LLMs write efficient GPU kernels? In *International Conference on Machine Learning*, 2025. URL https://arxiv.org/abs/2502.10517.

[265] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. L. Wainwright, P. Mishkin, C. Zhang, S. Agarwal, K. Slama, A. Ray, J. Schulman, J. Hilton, F. Kelton, L. Miller, M. Simens, A. Askell, P. Welinder, P. Christiano, J. Leike, and R. Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022. URL https://arxiv.org/abs/2203.02155.

[266] B. Pan, Y. Fu, K. Wang, J. Lu, L. Pan, Z. Qian, Y. Chen, G. Wang, Y. Zhou, L. Zheng, Y. Tang, Z. Wen, Y. Wu, J. Lu, B. Zhu, M. Zhu, B. Zhang, and W. Chen. VIS-Shepherd: Constructing critic for LLM-based data visualization generation. *arXiv preprint arXiv:2506.13326*, 2025. URL https://arxiv.org/abs/2506.13326.

[267] W. Pang, K. Q. Lin, X. Jian, X. He, and P. Torr. Paper2Poster: Towards multimodal poster automation from scientific papers. In *Advances in Neural Information Processing Systems*, 2025. URL https://arxiv.org/abs/2505.21497.

[268] S. S. Panigrahi, J. Videnovic, and M. Brbic. HeurekaBench: A benchmarking framework for AI co-scientist. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2601.01678.

[269] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu. BLEU: A method for automatic evaluation of machine translation. In *Annual Meeting on Association for Computational Linguistics*, pages 311–318, 2002.

[270] L. Patel, N. Arabzadeh, H. Gupta, A. Sundar, I. Stoica, M. Zaharia, and C. Guestrin. DeepScholar-Bench: A live benchmark and automated evaluation for generative research synthesis. *arXiv preprint arXiv:2508.20033*, 2025. URL https://arxiv.org/abs/2508.20033.

[271] N. Patel, J. Crossman, E. Aggarwal, and E. Wenger. Are llms becoming similarly creative? evidence from three years of models. *arXiv preprint arXiv:2608.19437*, 2026. URL https://arxiv.org/abs/2608.19437.

[272] P. Pedinotti et al. STRUCTSURVEY: Structured agentic retrieval for automated survey paper generation. *arXiv preprint arXiv:2607.01243*, 2026. URL https://arxiv.org/abs/2607.01243.

[273] R. A. Poldrack. AI-Peer-Review: Multi-LLM peer review with meta-review. https://github.com/poldrack/ai-peer-review, 2024.

[274] V. Polshkov, M. Pitera, J. Yang, K. Priemko, M. Gaiduk, A. Nikolenko, D. Bykov, C. Southern, D. Yarats, and J. Ma. WANDR: A benchmark for wide and deep research. *arXiv preprint arXiv:2608.14747*, 2026. URL https://arxiv.org/abs/2608.14747.

[275] D. Pratt. Symposium: Trust via auditable records for communities of AI scientist agents. *arXiv preprint arXiv:2608.19511*, 2026. URL https://arxiv.org/abs/2608.19511.

[276] O. Press, A. Hochlehnert, A. Prabhu, V. Udandarao, O. Press, and M. Bethge. CiteME: Can language models accurately cite scientific claims? *arXiv preprint arXiv:2407.12861*, 2024. URL https://arxiv.org/abs/2407.12861.

[277] K. Pu, K. J. K. Feng, T. Grossman, T. Hope, B. D. Mishra, M. Latzke, J. Bragg, J. C. Chang, and P. Siangliulue. IdeaSynth: Iterative research idea development through evolving and composing idea facets with literature-

grounded feedback. In *CHI Conference on Human Factors in Computing Systems*, pages 1–31, 2025. URL https://arxiv.org/abs/2410.04025.

[278] Y. Pu, T. Lin, and H. Chen. PiFlow: Principle-aware scientific discovery with multi-agent collaboration. *arXiv preprint arXiv:2505.15047*, 2025. URL https://arxiv.org/abs/2505.15047.

[279] W. Qian, B. Xu, Z. Xie, B. Fan, G. Tang, J. Chen, X. Wu, M. Yang, C. Di, J. Li, L. Tung, P. Lai, Y. Xia, Z. Guo, Y. Xu, Y. Qin, S. Gan, X. Miao, and B. Cui. AutoSci: A memory-centric agentic system for the full scientific research lifecycle. *arXiv preprint arXiv:2605.31468*, 2026. URL https://arxiv.org/abs/2605.31468.

[280] Z. Qian, J. Lei, Y. Wang, and N. Cao. HypoForge: A self-improving multi-agent framework for automated hypothesis generation and testing via scientific skill learning. *arXiv preprint arXiv:2608.25770*, 2026. URL https://arxiv.org/abs/2608.25770.

[281] Z. Qian, B. Wu, Y. Wang, C. D. Yan, D. Dai, L. Zheng, J. Jiang, J. Zhang, and W. Wang. Spark-to-Paper: End-to-end research paper generation as a composable skill. *arXiv preprint arXiv:2608.11924*, 2026. URL https://arxiv.org/abs/2608.11924.

[282] Y. Qiu, H. Zhang, Z. Xu, M. Li, D. Song, Z. Wang, and K. Zhang. AI Idea Bench 2025: AI research idea generation benchmark. *arXiv preprint arXiv:2504.14191*, 2025. URL https://arxiv.org/abs/2504.14191.

[283] A. Qu, H. Zheng, Z. Zhou, Y. Yan, Y. Tang, S. Y. Ong, F. Hong, K. Zhou, C. Jiang, M. Kong, J. Zhu, X. Jiang, S. Li, C. Wu, B. K. H. Low, J. Zhao, and P. P. Liang. CORAL: Towards autonomous multi-agent evolution for open-ended discovery. *arXiv preprint arXiv:2604.01658*, 2026. URL https://arxiv.org/abs/2604.01658.

[284] Y. Qu, K. Huang, M. Yin, K. Zhan, D. Liu, D. Yin, H. C. Cousins, W. A. Johnson, X. Wang, M. Shah, R. B. Altman, D. Zhou, M. Wang, and L. Cong. CRISPR-GPT for agentic automation of gene-editing experiments. *arXiv preprint arXiv:2404.18021*, 2024. URL https://arxiv.org/abs/2404.18021.

[285] V. Raina, A. Liusie, and M. Gales. Is LLM-as-a-judge robust? investigating universal adversarial attacks on zero-shot LLM assessment. In *Conference on Empirical Methods in Natural Language Processing*, pages 7499–7517, 2024. URL https://arxiv.org/abs/2402.14016.

[286] S. Rajwal, S. Das, and T. Ghosal. Do llms know a good hypothesis when they see one? logit-based energy scoring outperforms prompted LLM-as-judge for scientific hypothesis ranking. *arXiv preprint arXiv:2608.17270*, 2026. URL https://arxiv.org/abs/2608.17270.

[287] B. Rank, H. Bhatnagar, A. Prabhu, S. Eisenberg, K. Nguyen, M. Bethge, and M. Andriushchenko. PostTrainBench: Can LLM agents automate LLM post-training? *arXiv preprint arXiv:2603.08640*, 2026. URL https://arxiv.org/abs/2603.08640.

[288] V. Rao, A. Kumar, H. Lakkaraju, and N. B. Shah. Detecting LLM-generated peer reviews. *arXiv preprint arXiv:2503.15772*, 2025. URL https://arxiv.org/abs/2503.15772.

[289] M. T. R. Ratul, Z. Chen, K. Fu, T. Ji, and L. Zhang. MasterSet: A large-scale benchmark for must-cite citation recommendation in the AI/ML literature. *arXiv preprint arXiv:2604.17680*, 2026. URL https://arxiv.org/abs/2604.17680.

[290] Y. Ren, X. Liu, Q. Sun, X. Zhang, J. Li, H. Zhang, and J. Wang. AutoResearch: Insight in, hallucination out. *arXiv preprint arXiv:2608.17906*, 2026. URL https://arxiv.org/abs/2608.17906.

[291] Z. Ren, R. Li, X. Zhang, Z. Pang, S. Ren, and J. Zhang. EviGraph: Evidence-guided autonomous research agents. *arXiv preprint arXiv:2608.04738*, 2026. URL https://arxiv.org/abs/2608.04738.

[292] J. A. Rodriguez, A. Puri, S. Agarwal, I. H. Laradji, P. Rodriguez, S. Rajeswar, D. Vazquez, C. Pal, and M. Pedersoli. StarVector: Generating scalable vector graphics code from images and text. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16175–16186, 2025. URL https://arxiv.org/abs/2312.11556.

[293] B. Romera-Paredes, M. Barekatain, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, P. Kohli, and A. Fawzi. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024.

[294] K. Ruan, X. Wang, J. Hong, P. Wang, Y. Liu, and H. Sun. LiveIdeaBench: Evaluating LLMs' scientific creativity and idea generation with minimal context. *arXiv preprint arXiv:2412.17596*, 2024. URL https://arxiv.org/abs/2412.17596.

[295] Q. Ruan and I. Gurevych. Author-in-the-loop response generation and evaluation: Integrating author expertise and intent in responses to peer review. *arXiv preprint arXiv:2602.11173*, 2026. URL https://arxiv.org/abs/2602.11173.

[296] M. Russinovich et al. Phantom references: Hallucinated citations that survive peer review at top-tier conferences. *arXiv preprint arXiv:2607.00738*, 2026. URL https://arxiv.org/abs/2607.00738.

[297] M. Russo, Y. Agarwal, T. Li, Z. Gu, M. Cafarella, O. Khattab, T. Kraska, and S. Madden. Carnot: Interpretable, interactive, and optimized execution of deep research queries. In *Proc. VLDB Endow.*, 2026. URL https://arxiv.org/abs/2608.09532.

[298] G. Russo Latona, M. H. Ribeiro, T. R. Davidson, V. Veselovsky, and R. West. The AI review lottery: Widespread AI-assisted peer reviews boost paper scores and acceptance rates. *arXiv preprint arXiv:2405.02150*, 2024. URL https://arxiv.org/abs/2405.02150.

[299] M. Ríos-García, N. Alampara, C. Gupta, I. Mandal, S. Mannan, A. A. Aghajani, N. M. A. Krishnan, and K. M. Jablonka. AI scientists produce results without reasoning scientifically. *arXiv preprint arXiv:2604.18805*, 2026. URL https://arxiv.org/abs/2604.18805.

[300] A. Safdar and M. Saadeldin. Long-horizon autonomous architecture research with a language-model agent: A behavioural case study. *arXiv preprint arXiv:2608.01995*, 2026. URL https://arxiv.org/abs/2608.01995.

[301] R. Saha, G. Juneja, D. Chaudhuri, N. Sajeevan, N. B. Shah, and D. Pruthi. Policies permitting LLM use for polishing peer reviews are currently not enforceable. *arXiv preprint arXiv:2603.20450*, 2026. URL https://arxiv.org/abs/2603.20450.

[302] D. Sahoo, M. Prasad, V. Majhi, J. Singh, V. Chamola, Y. Sinha, M. Mandal, and D. Kumar. When reject turns into accept: Quantifying the vulnerability of LLM-based scientific reviewers to indirect prompt injection. *arXiv preprint arXiv:2512.10449*, 2025. URL https://arxiv.org/abs/2512.10449.

[303] G. Sahu, H. Larochelle, L. Charlin, and C. Pal. ReviewerToo: Should AI join the program committee? a look at the future of peer review. *arXiv preprint arXiv:2510.08867*, 2025. URL https://arxiv.org/abs/2510.08867.

[304] R. Samanta, B. Saha, S. K. Ghosh, and R. Buyya. Agentr a stateful and recovery-aware software architecture for LLM-based auditable workflows. *arXiv preprint arXiv:2608.15264*, 2026. URL https://arxiv.org/abs/2608.15264.

[305] A. Sanyal, S. Schapiro, S. Shashidhar, R. Moon, L. R. Varshney, and D. Hakkani-Tur. Spark: A system for scientifically creative idea generation. In *International Conference on Computational Creativity*, 2025. URL https://arxiv.org/abs/2504.20090.

[306] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, volume 36, pages 68539–68551, 2023. URL https://arxiv.org/abs/2302.04761.

[307] S. Schmidgall and M. Moor. AgentRxiv: Towards collaborative autonomous research. *arXiv preprint arXiv:2503.18102*, 2025. URL https://arxiv.org/abs/2503.18102.

[308] S. Schmidgall, Y. Su, Z. Wang, X. Sun, J. Wu, X. Yu, J. Liu, M. Moor, Z. Liu, and E. Barsoum. Agent laboratory: Using LLM agents as research assistants. In *Conference on Empirical Methods in Natural Language Processing*, pages 5977–6043, 2025. URL https://arxiv.org/abs/2501.04227.

[309] S. Schmidgall, X. Zhu, M. Shaw, L. Yang, V. Liévin, J. Yang, Y. Zhuang, T. Strother, et al. Accelerating scientific research with gemini in the real-world. *arXiv preprint arXiv:2608.26701*, 2026. URL https://arxiv.org/abs/2608.26701.

[310] T. Schopf, T. Schreieder, and A. Aizawa. Think-Probe-Respond: Improving large language models as judges of research idea novelty. In *Findings of the Conference on Empirical Methods in Natural Language Processing*, 2026. URL https://arxiv.org/abs/2608.25660.

[311] N. Schuster, A. N. Salcedo, S. Bouchard, D. Frei, A. Pisani, J. E. Bautista, J. Zoubian, S. Escoffier, W. Liu, G. Valogiannis, and P. Zarrouk. Setting SAIL: Leveraging scientist-AI-loops for rigorous visualization tools. *arXiv preprint arXiv:2603.18145*, 2026. URL https://arxiv.org/abs/2603.18145.

[312] SciForge Team, Z. Gao, M. Fang, Y. Liu, H. Yang, X. Gu, S. Tang, S. Sun, et al. SciForge: An AI-native, multimodal workbench for scientific discovery. *arXiv preprint arXiv:2607.16038*, 2026. URL https://arxiv.org/abs/2607.16038.

[313] SenseTime. Office raccoon: Ai-powered office agent. https://www.sensetime.com/en/news-detail/51170569, 2026. News and Blog, March 4, 2026.

[314] M. Seo, J. Baek, S. Lee, and S. J. Hwang. Paper2Code: Automating code generation from scientific papers in machine learning. *arXiv preprint arXiv:2504.17192*, 2025. URL https://arxiv.org/abs/2504.17192.

[315] F. Shahhosseini, A. Marioriyad, A. Momen, M. S. Baghshah, M. H. Rohban, and S. H. Javanmard. Large language models for scientific idea generation: A creativity-centered survey. *arXiv preprint arXiv:2511.07448*, 2025. URL https://arxiv.org/abs/2511.07448.

[316] C. Shao, Y. Li, and F. Xu. SciNetBench: A relation-aware benchmark for scientific literature retrieval agents. *arXiv preprint arXiv:2601.03260*, 2026. URL https://arxiv.org/abs/2601.03260.

[317] E. Shao, Y. Wang, Y. Qian, Z. Pan, H. Liu, and D. Wang. SciSciGPT: Advancing human–AI collaboration in the science of science. *Nature Computational Science*, 6:301–315, 2026. URL https://arxiv.org/abs/2504.05559.

[318] Y. Shao, Y. Jiang, T. A. Kanell, P. Xu, O. Khattab, and M. S. Lam. Assisting in writing Wikipedia-like articles from scratch with large language models. *arXiv preprint arXiv:2402.14207*, 2024. URL https://arxiv.org/abs/2402.14207.

[319] V. Sharma, T. Joachims, and S. Dean. Do LLMs favor LLMs? quantifying interaction effects in peer review. *arXiv preprint arXiv:2601.20920*, 2026. URL https://arxiv.org/abs/2601.20920.

[320] M. Sharon and T. Hope. RATIO: A benchmark for retrieval across typed ideation operations in scientific literature. *arXiv preprint arXiv:2608.27394*, 2026. URL https://arxiv.org/abs/2608.27394.

[321] H. Shen, H. Yang, Z. Gu, and W. Han. ScholarGym: Benchmarking large language model capabilities in the information-gathering stage of deep research. *arXiv preprint arXiv:2601.21654*, 2026. URL https://arxiv.org/abs/2601.21654.

[322] M. Sheverev, D. Finkelstein, and S. Nikolenko. Beyond memory leaderboards: Evaluating scientific memory as budgeted context restoration. *arXiv preprint arXiv:2607.16848*, 2026. URL https://arxiv.org/abs/2607.16848.

[323] C. Shi, Q. Cai, Z. Chen, L. Zeng, Y. Zhao, J. Yu, J. Yu, and X. Li. APEX: Academic poster editing agentic expert. *arXiv preprint arXiv:2601.04794*, 2026. URL https://arxiv.org/abs/2601.04794.

[324] C. Shi, X. Ren, S. Song, H. Li, R. Sheng, and Y. Sun. Are LLMs ready for scientific discovery? a capability-oriented benchmark for AI scientists. *arXiv preprint arXiv:2607.11079*, 2026. URL https://arxiv.org/abs/2607.11079.

[325] J. Shi, Z. Zhang, B. Wu, Y. Liang, M. Fang, L. Chen, and Y. Zhao. PresentAgent: Multimodal agent for presentation video generation. In *Conference on Empirical Methods in Natural Language Processing*, pages 760–773, 2025. URL https://arxiv.org/abs/2507.04036.

[326] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 8634–8652, 2023. URL https://arxiv.org/abs/2303.11366.

[327] C. Si, D. Yang, and T. Hashimoto. Can LLMs generate novel research ideas? A large scale human study with 100+ NLP researchers. *arXiv preprint arXiv:2409.04109*, 2024. URL https://arxiv.org/abs/2409.04109.

[328] C. Si, T. Hashimoto, and D. Yang. The ideation-execution gap: Execution outcomes of LLM-generated versus human research ideas. *arXiv preprint arXiv:2506.20803*, 2025. URL https://arxiv.org/abs/2506.20803.

[329] C. Si, Z. Yang, Y. Choi, E. Candes, D. Yang, and T. Hashimoto. Towards execution-grounded automated AI research. *arXiv preprint arXiv:2601.14525*, 2026. URL https://arxiv.org/abs/2601.14525.

[330] M. Siddiqui, R. Pea, and H. Subramonyam. Script&shift: A layered interface paradigm for integrating content development and rhetorical strategy with LLM writing assistants. In *CHI Conference on Human Factors in Computing Systems*, pages 1–19, 2025. URL https://arxiv.org/abs/2502.10638.

[331] M. N. Siddiqui, V. Feliciano, R. Pea, and H. Subramonyam. AI in the writing process: How purposeful AI support fosters student writing. In *International Conference on Artificial Intelligence in Education*, pages 190–203, 2025. URL https://arxiv.org/abs/2506.20595.

[332] M. N. Siddiqui, N. Nasseri, A. Coscia, R. Pea, and H. Subramonyam. DraftMarks: Enhancing transparency in human-AI co-writing through interactive skeuomorphic process traces. In *arXiv preprint arXiv:2509.23505*, 2025. URL https://arxiv.org/abs/2509.23505.

[333] P. Q. D. Silva, S. Hazra, D. Lee, S. Kumar, and B. P. Majumder. Process-oriented evaluation of AI-assisted scientific writing. *arXiv preprint arXiv:2606.15583*, 2026. URL https://arxiv.org/abs/2606.15583.

[334] S. Sinhahajari, N. Majumder, and S. Poria. On the limits of LLM-as-judge for scientific novelty assessment. *arXiv preprint arXiv:2606.12071*, 2026. URL https://arxiv.org/abs/2606.12071.

[335] M. D. Skarlinski, S. Cox, J. M. Laurent, J. D. Braza, M. Hinks, M. J. Hammerling, M. Ponnapati, S. G. Rodriques, and A. D. White. Language agents achieve superhuman synthesis of scientific knowledge. *arXiv preprint arXiv:2409.13740*, 2024. URL https://arxiv.org/abs/2409.13740.

[336] Skywork AI. DeepResearchAgent: A hierarchical multi-agent system for deep research. https://github.com/SkyworkAI/DeepResearchAgent, 2025.

[337] H. Song, R. Zhang, J. Guo, X. Cheng, and M. d. Rijke. DeepRepro: State-aware subplanning for paper-to-code reproduction in evolving repositories. In *ACM International Conference on Information and Knowledge Management*, 2026. URL https://arxiv.org/abs/2608.26557.

[338] Y. Song, Y. Song, T. Pfister, and J. Yoon. PaperOrchestra: A multi-agent framework for automated AI research paper writing. *arXiv preprint arXiv:2604.05018*, 2026. URL https://arxiv.org/abs/2604.05018.

[339] H. Soudani, E. Lingg, F. Hasibi, and N. Rekabsaz. When deep research agents stagnate: Enhancing reasoning with retrieval-aware agent control. *arXiv preprint arXiv:2608.15191*, 2026. URL https://arxiv.org/abs/2608.15191.

[340] S. Sourav, S. Pal, M. J. Buehler, S. Das, F. Y. Wang, D. Soos, and T. Ghosal. Visualizing graph-to-answer mechanism recovery in materials-science hypothesis generation. *arXiv preprint arXiv:2608.04170*, 2026. URL https://arxiv.org/abs/2608.04170.

[341] D. Srikanth, B. Zhao, D. Xu, Y. Wu, and Z. Jiang. Recursive self-improvement of AI research agents. *arXiv preprint arXiv:2609.26457*, 2026. URL https://arxiv.org/abs/2609.26457.

[342] G. Starace, O. Jaffe, D. Sherburn, J. Aung, J. S. Chan, L. Maksin, R. Dias, E. Mays, B. Kinsella, W. Thompson, J. Heidecke, A. Glaese, and T. Patwardhan. PaperBench: Evaluating AI's ability to replicate AI research. In *International Conference on Machine Learning*, 2025. URL https://arxiv.org/abs/2504.01848.

[343] B. Su, P. Li, Y. Lu, and X. Chen. PEARL: Auditable repair for scientific reasoning graph extraction. *arXiv preprint arXiv:2607.17917*, 2026. URL https://arxiv.org/abs/2607.17917.

[344] H. Su, R. Chen, S. Tang, Z. Yin, X. Zheng, J. Li, B. Qi, Q. Wu, H. Li, W. Ouyang, P. Torr, B. Zhou, and N. Dong. Many heads are better than one: Improved scientific idea generation by a LLM-based multi-agent system. In *Annual Meeting of the Association for Computational Linguistics*, pages 28201–28240, 2025. URL https://arxiv.org/abs/2410.09403.

[345] T. Sun, E. Pan, Z. Yang, K. Sui, J. Shi, X. Cheng, T. Li, G. Zhang, W. Huang, J. Yang, and Z. Li. P2P: Automated paper-to-poster generation and fine-grained benchmark. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2505.17104.

[346] Y. Sun, C. Peng, Y. Yan, Z. Liu, S. Mei, B. Xu, X. Zhou, C. Chen, and M. Sun. HiEviDR-Bench: A benchmark for hierarchical evidence aggregation in deep research. *arXiv preprint arXiv:2607.25151*, 2026. URL https://arxiv.org/abs/2607.25151.

[347] Y. Sun, Z. Zeng, Y. Yang, C. Luo, W. Wang, Z. Liu, and J. Schmidhuber. SciDiagramEdit: Learning to edit scientific diagrams from paper revisions. *arXiv preprint arXiv:2607.15272*, 2026. URL https://arxiv.org/abs/2607.15272.

[348] A. Sundar, M. Chen, D. Tadimeti, G. Zhang, X. A. Li, N. B. Kumankumah, P. U. Ravva, S. Hasan, et al. Metag: A dataset to build agentic meta-reviewing capabilities. *arXiv preprint arXiv:2608.20488*, 2026. URL https://arxiv.org/abs/2608.20488.

[349] Y. Suzuki and F. Banaei-Kashani. TCA-SIR: Learning target-conditioned abstractions for scientific inspiration retrieval. *arXiv preprint arXiv:2607.28498*, 2026. URL https://arxiv.org/abs/2607.28498.

[350] SWE-bench Team. SWE-bench Leaderboards. https://www.swebench.com, 2026. Accessed: 2026-04-24.

[351] P. Taechoyotin and D. Acuna. REMOR: Automated peer review generation with LLM reasoning and multi-objective reinforcement learning. *arXiv preprint arXiv:2505.11718*, 2025. URL https://arxiv.org/abs/2505.11718.

[352] I. Takahara et al. Toward auditable AI scientists: A hypothesis evolution protocol for LLM agents. *arXiv preprint arXiv:2607.09195*, 2026. URL https://arxiv.org/abs/2607.09195.

[353] C. Tan, D. Lyu, S. Li, Z. Gao, J. Wei, S. Ma, Z. Liu, and S. Z. Li. Peer review as a multi-turn and long-context dialogue with role-based interactions. *arXiv preprint arXiv:2406.05688*, 2024. URL https://arxiv.org/abs/2406.05688.

[354] J. Tan and V. Syrgkanis. CausalForge: A formally grounded, self-improving agentic framework for automated research in causal inference. *arXiv preprint arXiv:2607.22511*, 2026. URL https://arxiv.org/abs/2607.22511.

[355] J. Tang, T. Fan, and C. Huang. AutoAgent: A fully-automated and zero-code framework for LLM agents. *arXiv preprint arXiv:2502.05957*, 2025. URL https://arxiv.org/abs/2502.05957.

[356] J. Tang, L. Xia, Z. Li, and C. Huang. AI-Researcher: Autonomous scientific innovation. *arXiv preprint arXiv:2505.18705*, 2025. URL https://arxiv.org/abs/2505.18705.

[357] X. Tang, X. Duan, and Z. G. Cai. Large language models for automated literature review. *arXiv preprint arXiv:2412.13612*, 2024. URL https://arxiv.org/abs/2412.13612.

[358] Y. Tang, Y. Fang, Y. Sun, W. Liu, W. Zhang, B. Liu, K. Liu, W. Zhang, and K. Chen. SciExplore: Evaluating autonomous agents from scientific navigation to information integration. *arXiv preprint arXiv:2607.20926*, 2026. URL https://arxiv.org/abs/2607.20926.

[359] M. V. T. Thai, T. Le, D. N. Manh, H. P. Nhat, and N. D. Q. Bui. SWE-EVO: Benchmarking coding agents in long-horizon software evolution scenarios. *arXiv preprint arXiv:2512.18470*, 2025. URL https://arxiv.org/abs/2512.18470.

[360] N. Thakkar, M. Yuksekgonul, J. Silberg, A. Garg, N. Peng, F. Sha, R. Yu, C. Vondrick, and J. Zou. A large-scale randomized study of large language model feedback in peer review. *Nature Machine Intelligence*, 8:326–336, 2026. URL https://arxiv.org/abs/2504.09737.

[361] M. Tian, L. Gao, S. D. Zhang, X. Chen, C. Fan, X. Guo, R. Haas, P. Ji, K. Krongchon, Y. Li, S. Liu, D. Luo, Y. Ma, H. Tong, K. Trinh, C. Tian, Z. Wang, B. Wu, Y. Xiong, S. Yin, M. Zhu, K. Lieret, Y. Lu, G. Liu, Y. Du, T. Tao, O. Press, J. Callan, E. Huerta, and H. Peng. SciCode: A research coding benchmark curated by scientists. *arXiv preprint arXiv:2407.13168*, 2024. URL https://arxiv.org/abs/2407.13168.

[362] Y. Tian, W. Cui, D. Deng, X. Yi, Y. Yang, H. Zhang, and Y. Wu. ChartGPT: Leveraging LLMs to generate charts from abstract natural language. *arXiv preprint arXiv:2311.01920*, 2023. URL https://arxiv.org/abs/2311.01920.

[363] G. Tie, P. Zhou, and L. Sun. A survey of AI scientists. *arXiv preprint arXiv:2510.23045*, 2025. URL https://arxiv.org/abs/2510.23045.

[364] K. Ueda, W. Hirota, T. Asakura, T. Omi, K. Takahashi, K. Arima, and T. Ishigaki. Exploring design of multi-agent LLM dialogues for research ideation. In *Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 322–337, 2025. URL https://arxiv.org/abs/2507.08350.

[365] R. van de Schoot, J. de Bruin, R. Schram, P. Zahedi, J. de Boer, F. Weijdema, B. Kramer, M. Huijts, M. Hoogerwerf, G. Ferdinands, A. Harkema, J. Willemsen, Y. Ma, Q. Fang, S. Hindriks, L. Tummers, and D. L. Oberski. An open source machine learning framework for efficient and transparent systematic reviews. *Nature Machine Intelligence*, 3:125–133, 2021.

[366] V. Vats, K. Sharma, S. Li, and S. Pei. AppliedScientist: Automated scientific revision through iterative AI reviewing. *arXiv preprint arXiv:2609.14738*, 2026. URL https://arxiv.org/abs/2609.14738.

[367] V. Vats, K. Sharma, and Y. Wang. SlideLab: Audience-Centered scientific slide generation and evaluation. *arXiv preprint arXiv:2609.30294*, 2026. URL https://arxiv.org/abs/2609.30294.

[368] A. Velasco, D. Rodriguez-Cardenas, D. Khati, D. N. Palacio, and D. Poshyvanyk. ECLAIR: A causally-grounded AI framework for scientific discovery in empirical software engineering. *arXiv preprint arXiv:2608.02323*, 2026. URL https://arxiv.org/abs/2608.02323.

[369] A. Vinaykumar, A. Li, S. Huang, and S. Liu. Any2Poster: Any-source poster generation across modalities and domains. *arXiv preprint arXiv:2606.02915*, 2026. URL https://arxiv.org/abs/2606.02915.

[370] C. Wang, H. Chen, H. Gao, H. Ding, Z. Liu, and Z. Tu. From simple QA to deep research: A verifiable benchmark constructed through iterative task evolution. *arXiv preprint arXiv:2608.02163*, 2026. URL https://arxiv.org/abs/2608.02163.

[371] C. Wang, M. Zhu, T. Huang, W. Li, Y. Zhang, Z. Zhu, Z. Zou, K. Yu, and L. Chen. LigBench: A unified and human-aligned benchmark for LLM-based research idea generation. *arXiv preprint arXiv:2608.13136*, 2026. URL https://arxiv.org/abs/2608.13136.

[372] H. Wang, P. Shojaee, K. Meidani, K. Sun, J. M. Hernández-Lobato, T. Head-Gordon, J. He, C. K. Reddy, C. Zhang, and Y. Du. Towards diverse scientific hypothesis search with large language models. *arXiv preprint arXiv:2606.10587*, 2026. URL https://arxiv.org/abs/2606.10587.

[373] J. Wang, W. Lv, B. Fu, J. Fu, J. Song, L. Zhang, L. Xue, L. Chen, Z. Xin, K. Li, and X. Cao. Act as a real researcher: A suite of benchmarks evaluating frontier LLMs and agentic harnesses in research lifecycle. *arXiv preprint arXiv:2606.07462*, 2026. URL https://arxiv.org/abs/2606.07462.

[374] M. Wang, J. Fang, S. Qiao, Z. Xu, H. Xu, H. Wang, S. Deng, L. Yang, et al. Mechanist: AI as a scientific instrument for discovering the mechanisms of intelligence. *arXiv preprint arXiv:2608.12036*, 2026. URL https://arxiv.org/abs/2608.12036.

[375] M. Wang, R. Lin, K. Hu, J. Jiao, N. Chowdhury, E. Chang, and T. Patwardhan. FrontierScience: Evaluating AI's ability to perform expert-level scientific tasks. *arXiv preprint arXiv:2601.21165*, 2026. URL https://arxiv.org/abs/2601.21165.

[376] Q. Wang, D. Downey, H. Ji, and T. Hope. SciMON: Scientific inspiration machines optimized for novelty. In *Annual Meeting of the Association for Computational Linguistics*, pages 279–299, 2024. URL https://arxiv.org/abs/2305.14259.

[377] Q. Wang, H. Wang, L. Chen, Z. Yang, G. Chen, H. Alinejad-Rokny, H. Li, Y. Lin, and M. Yang. FlowPIE: Test-time scientific idea evolution with flow-guided literature exploration. *arXiv preprint arXiv:2603.29557*, 2026. URL https://arxiv.org/abs/2603.29557.

[378] S. Wang, H. Chen, Y. Yin, S. Zhuang, B. Koopman, and G. Zuccon. Search, inspect, fetch: Exploiting structure-aware boolean retrieval for deep-research agents. *arXiv preprint arXiv:2608.02751*, 2026. URL https://arxiv.org/abs/2608.02751.

[379] S. Wang, X. Tian, P. Liu, and Y. Zhao. Workflow closure is not scientific closure in auto-research systems. *arXiv preprint arXiv:2605.26200*, 2026. URL https://arxiv.org/abs/2605.26200.

[380] W. Wang, L. Gu, L. Zhang, Y. Luo, Y. Dai, C. Shen, L. Xie, B. Lin, X. He, and J. Ye. SciPIP: An LLM-based scientific paper idea proposer. *arXiv preprint arXiv:2410.23166*, 2024. URL https://arxiv.org/abs/2410.23166.

[381] X. Wang, B. Li, Y. Song, F. F. Xu, X. Tang, M. Zhuge, J. Pan, Y. Song, B. Li, J. Singh, H. H. Tran, F. Li, R. Ma, M. Zheng, B. Qian, Y. Shao, N. Muennighoff, Y. Zhang, B. Hui, J. Lin, R. Brennan, H. Peng, H. Ji, and G. Neubig. OpenHands: An open platform for AI software developers as generalist agents. *arXiv preprint arXiv:2407.16741*, 2024. URL https://arxiv.org/abs/2407.16741.

[382] X. Wang, Y. Zhang, B. Xu, L. Hou, and J. Li. DeepWeaver: Bridging the evidence synthesis gap in open-ended question answering. *arXiv preprint arXiv:2608.18988*, 2026. URL https://arxiv.org/abs/2608.18988.

[383] Y. Wang, Y. Kordi, S. Mishra, A. Liu, N. A. Smith, D. Khashabi, and H. Hajishirzi. Self-Instruct: Aligning language models with self-generated instructions. In *Annual Meeting of the Association for Computational Linguistics*, pages 13484–13508, 2023. URL https://arxiv.org/abs/2212.10560.

[384] Y. Wang, Q. Guo, W. Yao, H. Zhang, X. Zhang, Z. Wu, M. Zhang, X. Dai, M. Zhang, Q. Wen, W. Ye, S. Zhang, and Y. Zhang. AutoSurvey: Large language models can automatically write surveys. *arXiv preprint arXiv:2406.10252*, 2024. URL https://arxiv.org/abs/2406.10252.

[385] Y. Wang, X. Ma, P. Nie, H. Zeng, Z. Lyu, Y. Zhang, B. Schneider, Y. Lu, X. Yue, and W. Chen. ScholarCopilot: Training large language models for academic writing with accurate citations. *arXiv preprint arXiv:2504.00824*, 2025. URL https://arxiv.org/abs/2504.00824.

[386] Y. Wang et al. NatureBench: Can coding agents match the published SOTA of nature-family papers? *arXiv preprint arXiv:2606.24530*, 2026. URL https://arxiv.org/abs/2606.24530.

[387] Z. Wang. Andy: A mathematical agent for rigorous proof and autonomous research. *arXiv preprint arXiv:2608.15052*, 2026. URL https://arxiv.org/abs/2608.15052.

[388] Z. Wang, H. Zhang, C.-L. Li, J. M. Eisenschlos, V. Perot, Z. Wang, L. Miculicich, Y. Fujii, J. Shang, C.-Y. Lee, and T. Pfister. Chain-of-table: Evolving tables in the reasoning chain for table understanding. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2401.04398.

[389] Z. Wang, A. Yuan, Y. Yao, L. Yao, H. Zuo, Z. Gong, Y. Liu, S. Li, et al. What proves you wrong: Benchmarking language models on falsifiable research ideation. *arXiv preprint arXiv:2608.22948*, 2026. URL https://arxiv.org/abs/2608.22948.

[390] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837, 2022. URL https://arxiv.org/abs/2201.11903.

[391] J. Wei, C. Tan, Q. Chen, G. Wu, S. Li, Z. Gao, L. Sun, B. Yu, and R. Guo. From words to structured visuals: A benchmark and framework for text-to-diagram generation and editing. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13315–13325, 2025. URL https://arxiv.org/abs/2411.11916.

[392] Y. Wei, F. Guo, Q. Zhou, Z. Fu, M. Ding, K. Yang, and Y. Zhang. Can retrievers find the same paper from different aspects? a multi-aspect full-paper scientific retrieval benchmark. *arXiv preprint arXiv:2608.15624*, 2026. URL https://arxiv.org/abs/2608.15624.

[393] Z. Wen, J. Cao, Z. Wang, B. Guo, R. Yang, and S. Liu. InteractiveSurvey: An LLM-based personalized and interactive survey paper generation system. *arXiv preprint arXiv:2504.08762*, 2025. URL https://arxiv.org/abs/2504.08762.

[394] Y. Weng, M. Zhu, G. Bao, H. Zhang, J. Wang, Y. Zhang, and L. Yang. CycleResearcher: Improving automated research via automated review. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2411.00816.

[395] H. Wijk, T. Lin, J. Becker, S. Jawhar, N. Parikh, T. Broadley, L. Chan, M. Chen, J. Clymer, J. Dhyani, E. Ericheva, K. Garcia, B. Goodrich, N. Jurkovic, H. Karnofsky, M. Kinniment, A. Lajko, S. Nix, L. Sato, W. Saunders, M. Taran, B. West, and E. Barnes. RE-Bench: Evaluating frontier AI R&D capabilities of language model agents against human experts. *arXiv preprint arXiv:2411.15114*, 2024. URL https://arxiv.org/abs/2411.15114.

[396] A. Wong, H. Cui, Y. Tan, X. Zhan, L. Lin, Y. Guo, Z. Dai, Z. Zeng, and W. Li. Eureka: Task-conditioned meta-agent orchestration for scientific discovery. *arXiv preprint arXiv:2608.19047*, 2026. URL https://arxiv.org/abs/2608.19047.

[397] Y. Woo, W. Lee, and J.-X. Huang. YouRA: A Persistent-State architecture for Evidence-Traceable autonomous research agents. *arXiv preprint arXiv:2610.01097*, 2026. URL https://arxiv.org/abs/2610.01097.

[398] H. Wu, B. Zheng, D. Song, Y. Jiang, J. Gao, L. Xing, L. Sun, and Y. Yuan. Towards a medical AI scientist. *arXiv preprint arXiv:2603.28589*, 2026. URL https://arxiv.org/abs/2603.28589.

[399] S. Wu, Y. Ma, Y. Zhao, T. Hu, O. Jiang, M. Patwardhan, and A. Cohan. RbtAct: Rebuttal as supervision for actionable review feedback generation. *arXiv preprint arXiv:2603.09723*, 2026. URL https://arxiv.org/abs/2603.09723.

[400] W. Wu, Z. Xu, Z. Zhang, Y. Zhao, and H. Tang. PresentAgent-2: Towards generalist multimodal presentation agents. *arXiv preprint arXiv:2605.11363*, 2026. URL https://arxiv.org/abs/2605.11363.

[401] Y. C. Wu, J. Jian, Z. Zhao, L. Cong, and M. Wang. LabClaw: Operating layer for labos. https://labclaw-ai.github.io, 2026.

[402] Y. Xiang, H. Yan, S. Ouyang, L. Gui, and Y. He. SciReplicate-Bench: Benchmarking LLMs in agent-driven algorithmic reproduction from research papers. *arXiv preprint arXiv:2504.00255*, 2025. URL https://arxiv.org/abs/2504.00255.

[403] J. Xiao, Y. Wang, C. Tang, P. Li, W. Ouyang, and S. Tang. LECTOR: Joint optimization of scientific reasoning graphs and introduction generation. *arXiv preprint arXiv:2605.25964*, 2026. URL https://arxiv.org/abs/2605.25964.

[404] L. Xiao et al. ResearchStudio-Reel: Automate the last mile of research from paper to poster, video, and blog. *arXiv preprint arXiv:2607.04438*, 2026. URL https://arxiv.org/abs/2607.04438.

[405] S. Xiao, S. Wang, X. Chen, K. Chao, M. Cui, F. Qian, C. Mei, F. Meng, L. Wang, Z. Yu, and J. Yin. Search shapes conclusions: Auditing evidence selection bias in deep research agents. *arXiv preprint arXiv:2609.39026*, 2026. URL https://arxiv.org/abs/2609.39026.

[406] A. Xie et al. RWGBench: Evaluating scholarly positioning in related work generation. *arXiv preprint arXiv:2606.24894*, 2026. URL https://arxiv.org/abs/2606.24894.

[407] X. Xie, Q. Long, M. Xiao, W. Ju, Y. Zhou, X. Wang, and H. Zhu. From Scientific Observations to Mechanisms: Benchmarking Hypothesis Generation by AI Scientists. *arXiv preprint arXiv:2610.05197*, 2026. URL https://arxiv.org/abs/2610.05197.

[408] A. Xin et al. EurekAgent: Agent environment engineering is all you need for autonomous scientific discovery. *arXiv preprint arXiv:2606.13662*, 2026. URL https://arxiv.org/abs/2606.13662.

[409] L. Xiong, K. Luo, Z. Xia, W. Zhang, J.-G. Yao, Z. Liu, J. Shao, J. Chen, H. Qian, X. Yang, Q. Yu, H. Li, C. Yue, X. Du, Y. Wang, Y. Liu, H. Xu, and Z. Dou. AutoResearchBench: Benchmarking AI agents on complex scientific literature discovery. *arXiv preprint arXiv:2604.25256*, 2026. URL https://arxiv.org/abs/2604.25256.

[410] C. Xu, Z. Deng, H. Liang, C. Yue, Z. Cui, P. Ye, and W. Cai. SciFigAlign: Scoring scientific figures by fine-tuned alignment of visuals with manuscript evidence. *arXiv preprint arXiv:2607.27066*, 2026. URL https://arxiv.org/abs/2607.27066.

[411] H. Xu, R. Xu, S. Ren, and J. Zhang. Project2Task: Graph-guided project-level planning for autonomous research. *arXiv preprint arXiv:2608.05225*, 2026. URL https://arxiv.org/abs/2608.05225.

[412] R. Xu, Y. Wang, L. Tao, W. Xuan, and Y. Luo. Diversifying personalized research ideation against AI-induced homogenization. *arXiv preprint arXiv:2607.28087*, 2026. URL https://arxiv.org/abs/2607.28087.

[413] W. Xu, Y. Zhou, Y. Zhou, Q. Cao, S. Li, J. Bu, B. Liu, Y. Chen, X. He, X. Zhao, X. Zhuang, F. Wang, Z. Zhou, Q. Feng, W. Huang, J. Wei, H. Wu, Y. Yang, G. Wang, S. Xu, Z. Huang, X. Liu, J. Liu, C. Tang, W. Li, Y. Chen, J. Ning, P. Jiang, C. Ma, Y. Du, C. Ji, H. Xu, M. Hu, J. Zheng, X. Chen, Y. Wu, F. Jiang, X. Chen, X. Tang, Y. Fu, Y. Lu, Y. Zhang, L. Sun, C. Li, J. Ma, W. Liu, Y. Liu, K.-C. Wu, S. Chai, Y. Wang, O. Zhangjin, C. Tang, S. Zhang, W. Cao, J. Ren, T. Cui, Z. Yao, J. Deng, Y. Sun, F. Liu, W. Wei, J. Xu, Z. Li, J. Gong, Z. Guo, Z. Yao, Z. Chen, T. Peng, F. Yu, B. Zhang, D. Zhou, S. Tang, J. Liu, F. Ling, Y. Lu, Y. Ren, B. Fei, Z. Zhao, X. Gu, R. Su, X.-M. Wu, W. Si, Y. Liu, H. Chen, X. Yan, X. Yang, J. Yan, J. Wu, Q. Zheng, C. Li, Z. Gao, H. Kong, J. He, M. Su, T. Fu, P. Ye, C. Song, N. Dong, Y. Li, H. Fu, S. Sun, L. Cheng, J. Lin, W. Ouyang, B. Zhou, W. Zhang, and L. Bai. Probing scientific general intelligence of LLMs with scientist-aligned workflows. *arXiv preprint arXiv:2512.16969*, 2025. URL https://arxiv.org/abs/2512.16969.

[414] W. Xu, S. Li, T. Ye, Q. Cao, Y. Chen, H. Gao, Y. Wang, Q. Li, K. Li, S. Xu, S. Chai, F. Yu, X. Zhao, Z. Zhao, W. Ma, Z. Guo, K. Wu, H. Zhou, H. Yin, L. Cheng, C. Hu, H. Li, L. Mi, X. Xie, Y. Zhou, R. Chen, Z. Zhou, X. Guo, Y. Zhou, X. He, S. Xu, X. Gu, J. Wu, M. Liu, C. Song, F. Ling, D. Zhou, S. Tang, Y. Li, M. Su, P. Ye, S. Sun, B. Wang, X. Yang, Z. Yin, T. Fu, G. Zhai, W. Ouyang, B. Zhang, L. Bai, and W. Zhang. ResearchClawBench: A benchmark for end-to-end autonomous scientific research. *arXiv preprint arXiv:2606.07591*, 2026. URL https://arxiv.org/abs/2606.07591.

[415] W. Xu, T. Mi, Y. Liu, Y. Nan, Z. Zhou, L. Ye, L. Zhang, Y. Qiao, and P. Liu. ASI-Evolve: AI accelerates AI. https://github.com/GAIR-NLP/ASI-Evolve, 2026.

[416] X. Xu, C. Feng, C. Zha, W. He, M. He, B. Xiao, and X. Gao. ProteinMCP: An agentic AI framework for autonomous protein engineering. *Protein Science*, 35(4):e70547, 2026.

[417] Y. Xu, K. Zhang, Y. Yu, M. Zhang, S. Meng, P. Chu, and Z. Tu. Scaling scientific discovery environments for turn-level agentic RL. *arXiv preprint arXiv:2607.28990*, 2026. URL https://arxiv.org/abs/2607.28990.

[418] Z. Xu, Z. Xue, T. Hu, Y. Wang, Y. Liu, and J. Zhang. Deep academic survey: Stateful agentic closed-loop paradigm for academic survey automation. *arXiv preprint arXiv:2608.18034*, 2026. URL https://arxiv.org/abs/2608.18034.

[419] Y. Yamada, R. T. Lange, C. Lu, S. Hu, C. Lu, J. Foerster, J. Clune, and D. Ha. The AI Scientist-v2: Workshop-level automated scientific discovery via agentic tree search. *arXiv preprint arXiv:2504.08066*, 2025. URL https://arxiv.org/abs/2504.08066.

[420] G. Yan, S. Li, P. Ma, Q. Zhang, P. Ma, J. Wang, M.-H. Hsieh, and Y. Du. OmniQEC: discovering practical quantum error-correcting codes by an AI scientist. *arXiv preprint arXiv:2607.25865*, 2026. URL https://arxiv.org/abs/2607.25865.

[421] X. Yan, S. Feng, J. Yuan, R. Xia, B. Wang, B. Zhang, and L. Bai. SurveyForge: On the outline heuristics, memory-driven generation, and multi-dimensional evaluation for automated survey writing. *arXiv preprint arXiv:2503.04629*, 2025. URL https://arxiv.org/abs/2503.04629.

[422] H. Yang and Y. Cao. Efficiency matters in autonomous research. *arXiv preprint arXiv:2607.24647*, 2026. URL https://arxiv.org/abs/2607.24647.

[423] J. Yang, C. E. Jimenez, A. Wettig, K. Lieret, S. Yao, K. Narasimhan, and O. Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems*, volume 37, pages 50528–50652, 2024. URL https://arxiv.org/abs/2405.15793.

[424] M. Yang. ResearchClaw: Personal AI research assistant with extensible skills. https://github.com/ymx10086/ResearchClaw, 2025.

[425] R. Yang, Y. Li, and S. Li. ARIS: Fully autonomous research via adversarial multi-agent collaboration. https://github.com/wanshuiyin/Auto-claude-code-research-in-sleep, 2026.

[426] X. Yang, X. Yang, S. Fang, Y. Zhang, J. Wang, B. Xian, Q. Li, J. Li, M. Xu, Y. Li, H. Pan, Y. Zhang, W. Liu, Y. Shen, W. Chen, and J. Bian. R&D-Agent: An LLM-agent framework towards autonomous data science. *arXiv preprint arXiv:2505.14738*, 2025. URL https://arxiv.org/abs/2505.14738.

[427] X. Yang, S. Sarwar, J. Cheng, Z. Shi, D. Li, H. Chen, H. Zhang, C. Guo, J. He, and Z. Liao. Scaling automatic research agents via world models. *arXiv preprint arXiv:2608.12564*, 2026. URL https://arxiv.org/abs/2608.12564.

[428] X. Yang, Z. Sha, J. Li, J. Yu, Y. Sun, M. Zhao, J. Fang, X. Guo, Y. Wu, X. Hu, Y. Luo, Q. Liu, and Z. Wang. No hidden prompts needed! you can game AI peer review with presentation-only revisions. *arXiv preprint arXiv:2606.13044*, 2026. URL https://arxiv.org/abs/2606.13044.

[429] Y. Yang, W. Jiang, Y. Wang, Y. Song, Y. Wang, and C. Zhang. Auto-Slides: An interactive multi-agent system for creating and customizing research presentations. *arXiv preprint arXiv:2509.11062*, 2025. URL https://arxiv.org/abs/2509.11062.

[430] Y. Yang, C. Lan, J. Sang, and Y. Lu. CSPO: Alleviating reward ambiguity for structured table-to-LaTeX generation. *arXiv preprint arXiv:2604.10918*, 2026. URL https://arxiv.org/abs/2604.10918.

[431] Z. Yang, Z. Zhou, S. Wang, X. Cong, X. Han, Y. Yan, Z. Liu, Z. Tan, P. Liu, D. Yu, Z. Liu, X. Shi, and M. Sun. MatPlotAgent: Method and evaluation for LLM-based agentic scientific data visualization. *arXiv preprint arXiv:2402.11453*, 2024. URL https://arxiv.org/abs/2402.11453.

[432] Z. Yang, W. Liu, B. Gao, Y. Liu, W. Li, T. Xie, L. Bing, W. Ouyang, E. Cambria, and D. Zhou. MOOSE-Chem2: Exploring LLM limits in fine-grained scientific hypothesis discovery via hierarchical search. In *Advances in Neural Information Processing Systems*, 2025. URL https://arxiv.org/abs/2505.19209.

[433] Z. Yang, W. Liu, B. Gao, T. Xie, Y. Li, W. Ouyang, S. Poria, E. Cambria, and D. Zhou. MOOSE-Chem: Large language models for rediscovering unseen chemistry scientific hypotheses. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2410.07076.

[434] Z. Yang, F. Liu, Y. Ning, and H. Liu. EvoDS: Self-evolving autonomous data science agent with skill learning and context management. *arXiv preprint arXiv:2606.03841*, 2026. URL https://arxiv.org/abs/2606.03841.

[435] Z. Yang, X. Liu, and X. Xu. SciIntegrity-Bench: A benchmark for evaluating academic integrity in AI scientist systems. *arXiv preprint arXiv:2605.10246*, 2026. URL https://arxiv.org/abs/2605.10246.

[436] Z. Yang, D. Ma, H. Li, Z. Wang, T. Huang, Z. Hu, C. Wang, Y. Zhang, X. Wu, K. Yu, and L. Chen. DeepSurvey: Enhancing analytical depth and citation reliability in automated survey generation. *arXiv preprint arXiv:2605.29522*, 2026. URL https://arxiv.org/abs/2605.29522.

[437] Z. Yang, Y. Zhao, W. Liu, and X. Li. MERIT: Matching expertise via rubric-informed training for reviewer assignment. *arXiv preprint arXiv:2605.27865*, 2026. URL https://arxiv.org/abs/2605.27865.

[438] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.03629.

[439] Y. Yao, H. Zhu, P. Wang, J. Ren, X. Yang, Q. Chen, X. Li, D. Shi, J. Li, Q. Wang, S. Wang, X. Liu, J. Wu, M. Liu, and W. Zhou. O-Researcher: An open ended deep research model via multi-agent distillation and agentic RL. *arXiv preprint arXiv:2601.03743*, 2026. URL https://arxiv.org/abs/2601.03743.

[440] J. Ye, Y. Wang, Y. Huang, D. Chen, Q. Zhang, N. Moniz, T. Gao, W. Geyer, C. Huang, P.-Y. Chen, N. V. Chawla, and X. Zhang. Justice or prejudice? Quantifying biases in LLM-as-a-Judge. *arXiv preprint arXiv:2410.02736*, 2024. URL https://arxiv.org/abs/2410.02736.

[441] R. Ye, X. Pang, J. Chai, J. Chen, Z. Yin, Z. Xiang, X. Dong, J. Shao, and S. Chen. Are we there yet? revealing the risks of utilizing large language models in scholarly peer review. *arXiv preprint arXiv:2412.01708*, 2024. URL https://arxiv.org/abs/2412.01708.

[442] S. Yoon and D. Lee. Personalized deep research query refinement with graph-scaffolded evidence grounding. *arXiv preprint arXiv:2608.05876*, 2026. URL https://arxiv.org/abs/2608.05876.

[443] H. Yu, Z. Hong, Z. Cheng, K. Zhu, K. Xuan, J. Yao, T. Feng, and J. You. ResearchTown: Simulator of human research community. In *International Conference on Machine Learning*, pages 73051–73096. PMLR, 2025. URL https://arxiv.org/abs/2412.17767.

[444] L. Yu, X. Xu, Y. Zhou, S. He, and A. Pan. Beyond execution: Auditing experimental fidelity in LLM-driven scientific research. *arXiv preprint arXiv:2608.26753*, 2026. URL https://arxiv.org/abs/2608.26753.

[445] S. Yu, M. Luo, A. Madasu, V. Lal, and P. Howard. Is your paper being reviewed by an LLM? benchmarking AI text detection in peer review. *arXiv preprint arXiv:2502.19614*, 2025. URL https://arxiv.org/abs/2502.19614.

[446] T. Yu, C. Tao, H. Shen, H. Li, R. Chen, L. Teng, L. Wang, Y. Li, Q. Chen, C. Xu, and L. Wang. Closed-loop AI achieves certifiable engineering design. *arXiv preprint arXiv:2608.21976*, 2026. URL https://arxiv.org/abs/2608.21976.

[447] X. Yu, J. Han, Z. Chen, C. Zhang, X. Fu, J. Tan, Z. You, Y. Jian, Y.-P. Wang, and C. Chai. DiagramRAG: A lightweight framework to retrieve scientific diagram for figure generation. *arXiv preprint arXiv:2605.27931*, 2026. URL https://arxiv.org/abs/2605.27931.

[448] J. Yuan, X. Yan, B. Zhang, T. Chen, B. Shi, W. Ouyang, Y. Qiao, L. Bai, and B. Zhou. Dolphin: Moving towards closed-loop auto-research through thinking, practice, and feedback. In *Annual Meeting of the Association for Computational Linguistics*, pages 21768–21789, 2025. URL https://arxiv.org/abs/2501.03916.

[449] R. Yuan, Y. Li, Y. Du, F. Cai, Y. Liu, H. P. Chan, C. Lin, Y. Chen, J. Yang, B. Dai, P. Lu, and C. Xiao. AutoDataBench: A data-centric testbed for accelerating auto research. *arXiv preprint arXiv:2609.40097*, 2026. URL https://arxiv.org/abs/2609.40097.

[450] M. Yuksekgonul, D. Koceja, X. Li, F. Bianchi, J. McCaleb, X. Wang, J. Kautz, Y. Choi, J. Zou, C. Guestrin, and Y. Sun. Learning to discover at test time. *arXiv preprint arXiv:2601.16175*, 2026. URL https://arxiv.org/abs/2601.16175.

[451] S. Zeng, K. Tian, K. Zhang, Y. wang, J. Gao, R. Liu, S. Yang, J. Li, X. Long, J. Ma, B. Qi, and B. Zhou. ReviewRL: Towards automated scientific review with RL. In *Conference on Empirical Methods in Natural Language Processing*, pages 16931–16943, 2025. URL https://arxiv.org/abs/2508.10308.

[452] W. Zeng, M. Ouyang, L. Cui, and H. T. Ng. SlideTailor: Personalized presentation slide generation for scientific papers. In *AAAI Conference on Artificial Intelligence*, volume 40, pages 34584–34592, 2026. URL https://arxiv.org/abs/2512.20292.

[453] B. Zhang, S. Feng, X. Yan, J. Yuan, R. Ma, Y. Hu, Z. Yu, X. He, S. Huang, S. Hou, Z. Nie, Z. Wang, J. Liu, T. Peng, P. Ye, D. Zhou, S. Zhang, X. Wang, Y. Zhang, M. Li, Z. Tu, X. Yue, W. Ouyang, B. Zhou, and L. Bai. InternAgent: When agent becomes the scientist – building closed-loop system from hypothesis to verification. *arXiv preprint arXiv:2505.16938*, 2025. URL https://arxiv.org/abs/2505.16938.

[454] D. Zhang, Z. Bao, S. Du, Z. Zhao, K. Zhang, D. Bao, and Y. Yang. Re2: A consistency-ensured dataset for full-stage peer review and multi-turn rebuttal discussions. *arXiv preprint arXiv:2505.07920*, 2025. URL https://arxiv.org/abs/2505.07920.

[455] H. Zhang, H. Cui, Y. Wang, Y. Tian, Q. Guo, C. Wang, J. Wu, C. Song, and Y. Zhang. IterSurvey: Deep literature survey automation with an iterative workflow. *arXiv preprint arXiv:2510.21900*, 2025. URL https://arxiv.org/abs/2510.21900.

[456] J. Zhang, S. Hu, C. Lu, R. Lange, and J. Clune. Darwin godel machine: Open-ended evolution of self-improving agents. *arXiv preprint arXiv:2505.22954*, 2025. URL https://arxiv.org/abs/2505.22954.

[457] J. Zhang, J. Zhang, Z. Cui, J. Yang, L. Zhang, B. Hui, Q. Liu, Z. Wang, L. Wang, and J. Lin. PlotCraft: Pushing the limits of LLMs for complex and interactive data visualization. *arXiv preprint arXiv:2511.00010*, 2025. URL https://arxiv.org/abs/2511.00010.

[458] K. Zhang, Y. Sun, X. Ren, C. Yi, and H. Yang. AutoSR: Automatic symbolic regression by searching research states. *arXiv preprint arXiv:2608.16876*, 2026. URL https://arxiv.org/abs/2608.16876.

[459] L. Zhang, Y. Zhao, Y. Cao, R. Chen, M. Jing, J. Lai, Z. Tang, J. Xie, et al. Benchmarking agents for proving theorems in quantum algorithms and quantum information. *arXiv preprint arXiv:2607.21533*, 2026. URL https://arxiv.org/abs/2607.21533.

[460] T. Zhang and H. Sun. SciNav: A general agent framework for scientific coding tasks. *arXiv preprint arXiv:2603.20256*, 2026. URL https://arxiv.org/abs/2603.20256.

[461] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations*, 2020. URL https://arxiv.org/abs/1904.09675.

[462] T. Zhang, H. Lin, Z. Liu, C. Chen, and W. Zhang. SciFlow-Bench: Evaluating structure-aware scientific diagram generation via inverse parsing. *arXiv preprint arXiv:2602.09809*, 2026. URL https://arxiv.org/abs/2602.09809.

[463] X. Zhang, Z. Zhang, Z. Fu, B. Lin, and W. Wang. From inertia to objectivity: Improving deep research agents with noise isolation. In *Conference on Empirical Methods in Natural Language Processing*, 2026. URL https://arxiv.org/abs/2608.23045.

[464] Y. Zhang, Y. Wang, F. Zhang, R. Zhao, J. Li, J. Tang, and J. Zhang. SurveyReview: A reviewer-aligned benchmark for survey evaluators. *arXiv preprint arXiv:2608.07641*, 2026. URL https://arxiv.org/abs/2608.07641.

[465] Y. Zhang, E. Zeng, J. Lee, and T. Y. Yu. An AI scientist that doesn't drift: Taste, structure, and falsifiable findings in a quadruped navigation research loop. *arXiv preprint arXiv:2608.07542*, 2026. URL https://arxiv.org/abs/2608.07542.

[466] Z. Zhang, X. Zhang, J. Wei, Y. Xu, and C. You. PosterGen: Aesthetic-aware multi-modal paper-to-poster generation via multi-agent LLMs. *arXiv preprint arXiv:2508.17188*, 2025. URL https://arxiv.org/abs/2508.17188.

[467] Z. Zhang, J. Ruan, X. Gao, R. Zhou, L. Meng, L. Hu, T. Liu, and Y. Fu. Edit2TikZ: A comprehensive and challenging benchmark for scientific figure editing with tikz. *arXiv preprint arXiv:2608.13441*, 2026. URL https://arxiv.org/abs/2608.13441.

[468] Z. Zhang, N. Wang, S. Galhotra, and C. Cardie. How far are we from true auto-research? *arXiv preprint arXiv:2605.19156*, 2026. URL https://arxiv.org/abs/2605.19156.

[469] B. Zhao, J. Zhang, C. Whitehouse, M. Jiang, M. Shvartsman, A. Charnalia, D. Magka, T. Shavrina, D. Dunfield, O. M. Aodha, and Y. Bachrach. APRES: An agentic paper revision and evaluation system. *arXiv preprint arXiv:2603.03142*, 2026. URL https://arxiv.org/abs/2603.03142.

[470] B. Zhao et al. Autonomous scientific discovery via iterative meta-reflection. *arXiv preprint arXiv:2607.01131*, 2026. URL https://arxiv.org/abs/2607.01131.

[471] H. Zhao, S. Si, Z. Wang, Z. Wang, L. Chen, X. Li, Z. Liang, M. Sun, and M. Zhang. Crafter: A multi-agent harness for editable scientific figure generation from diverse inputs. *arXiv preprint arXiv:2605.30611*, 2026. URL https://arxiv.org/abs/2605.30611.

[472] K. Zhao, W. Lin, Q. Zheng, F. Xu, and Y. Li. Deep ideation: Designing LLM agents to generate novel research ideas on scientific concept network. *arXiv preprint arXiv:2511.02238*, 2025. URL https://arxiv.org/abs/2511.02238.

[473] M. Zhao, J. Dong, K. Xu, Z. Hasan, C. Fan, S. Jiang, S. Mao, T. Lingya, et al. ScienceFlow: A long-horizon agent for ML research, scientific discovery and beyond. *arXiv preprint arXiv:2608.14354*, 2026. URL https://arxiv.org/abs/2608.14354.

[474] X. Zhao, Z. Sang, Y. Li, Q. Shi, W. Zhao, S. Wang, D. Zhang, X. Han, Z. Liu, and M. Sun. AutoReproduce: Automatic AI experiment reproduction with paper lineage. *arXiv preprint arXiv:2505.20662*, 2025. URL https://arxiv.org/abs/2505.20662.

[475] Y. Zhao, W. Chen, Z. Xu, M. Patwardhan, C. Wang, Y. Liu, L. Vig, and A. Cohan. AbGen: Evaluating large language models in ablation study design and evaluation for scientific research. In *Annual Meeting of the Association for Computational Linguistics*, pages 12479–12491, 2025. URL https://arxiv.org/abs/2507.13300.

[476] Y. Zhao, T. Wei, J. Zou, X. Ning, Y. Bei, L. Chen, S. Rana, W. H. Yang, H. Tong, and J. He. PaperMind: Benchmarking agentic reasoning and critique over scientific papers in multimodal LLMs. *arXiv preprint arXiv:2604.21304*, 2026. URL https://arxiv.org/abs/2604.21304.

[477] H. Zheng, X. Guan, H. Kong, W. Zhang, J. Zheng, W. Zhou, H. Lin, Y. Lu, X. Han, and L. Sun. PPTAgent: Generating and evaluating presentations beyond text-to-slides. In *Conference on Empirical Methods in Natural Language Processing*, pages 14402–14418, 2025. URL https://arxiv.org/abs/2501.03936.

[478] H. Zheng, G. Mo, X. Yan, Q. Yuan, W. Zhang, X. Chen, Y. Lu, H. Lin, X. Han, and L. Sun. DeepPresenter: Environment-grounded reflection for agentic presentation generation. *arXiv preprint arXiv:2602.22839*, 2026. URL https://arxiv.org/abs/2602.22839.

[479] T. Zheng, Z. Deng, H. T. Tsang, W. Wang, J. Bai, Z. Wang, and Y. Song. From automation to autonomy: A survey on large language models in scientific discovery. *arXiv preprint arXiv:2505.13259*, 2025. URL https://arxiv.org/abs/2505.13259.

[480] T. Zheng, H. Liu, C. Huang, H. Bao, S. Zhang, R. Liu, R. Dai, R. Chen, C. Liu, T. Xiong, X. Wu, H. Zhang, and H. Huang. LLMs improving LLMs: Agentic discovery for test-time scaling. *arXiv preprint arXiv:2605.08083*, 2026. URL https://arxiv.org/abs/2605.08083.

[481] Y. Zheng, S. Sun, L. Qiu, D. Ru, C. Jiayang, X. Li, J. Lin, B. Wang, Y. Luo, R. Pan, Y. Xu, Q. Min, Z. Zhang, Y. Wang, W. Li, and P. Liu. OpenResearcher: Unleashing AI for accelerated scientific research. In *Conference on Empirical Methods in Natural Language Processing*, pages 209–218, 2024. URL https://arxiv.org/abs/2408.06941.

[482] J. Zhou, Z. Fan, X. Wu, T. Yu, F. Zhang, and L. Wang. Deep research pretraining via predictive navigation. *arXiv preprint arXiv:2608.00432*, 2026. URL https://arxiv.org/abs/2608.00432.

[483] Q. Zhou, Z. Zhang, Z. Li, and L. Sun. "Give a positive review only": An early investigation into in-paper prompt injection attacks and defenses for AI reviewers. *arXiv preprint arXiv:2511.01287*, 2025. URL https://arxiv.org/abs/2511.01287.

[484] C. Zhu, J. Xiong, R. Ma, Z. Lu, Y. Liu, and L. Li. When your reviewer is an LLM: Biases, divergence, and prompt injection risks in peer review. *arXiv preprint arXiv:2509.09912*, 2025. URL https://arxiv.org/abs/2509.09912.

[485] D. Zhu, R. Meng, Y. Song, X. Wei, S. Li, T. Pfister, and J. Yoon. PaperBanana: Automating academic illustration for AI scientists. *arXiv preprint arXiv:2601.23265*, 2026. URL https://arxiv.org/abs/2601.23265.

[486] M. Zhu, Y. Weng, L. Yang, and Y. Zhang. DeepReview: Improving LLM-based paper review with human-like deep thinking process. *arXiv preprint arXiv:2503.08569*, 2025. URL https://arxiv.org/abs/2503.08569.

[487] M. Zhu, Z. Lin, Y. Weng, P. Lu, Q. Xie, Y. Wei, S. Liu, Q. Sun, and Y. Zhang. AutoFigure: Generating and refining publication-ready scientific illustrations. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2602.03828.

[488] P. Zhu, L. Li, L. Yang, S. Su, and J. Shao. Is deep research reliable? misleading knowledge induces false conclusions. *arXiv preprint arXiv:2607.20891*, 2026. URL https://arxiv.org/abs/2607.20891.

[489] X. Zhu, Y. Cai, Z. Liu, C. Wang, F. Li, W. Jin, W. Liu, Z. Bing, B. Zheng, J. Chai, S. Tang, R. Ye, Y. Du, X. Pang, Y. Du, T. Miao, Y. Zhang, R. Liao, Z. Ding, L. Zhang, Y. Wang, W. E, and S. Chen. EvoMaster: A foundational evolving agent framework for agentic science at scale. *arXiv preprint arXiv:2604.17406*, 2026. URL https://arxiv.org/abs/2604.17406.

[490] Z. Zhu, K. Q. Lin, and M. Z. Shou. Paper2Video: Automatic video generation from scientific papers. *arXiv preprint arXiv:2510.05096*, 2025. URL https://arxiv.org/abs/2510.05096.

[491] Z. Zhuang, J. Chen, H. Xu, Y. Jiang, and J. Lin. Large language models for automated scholarly paper review: A survey. *Information Fusion*, 124:103332, 2025. URL https://arxiv.org/abs/2501.10326.

[492] Z. Zhuang, C. Lao, P. Xu, H. Xu, R. Yang, Y. He, P. Zhang, J. Cao, et al. From trajectories to evidence: Auditable experimental records for industrial research agents. *arXiv preprint arXiv:2608.05235*, 2026. URL https://arxiv.org/abs/2608.05235.